TRAJECTORY OPTIMIZATION

-LECTURE NOTE

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1. A Brief History of Optimal Control

To be updated later

Simple Optimization Problems
- The shortest path joining two points: a straight line segment.
- Isoperimetric problem (i.e., Dido's problem) to find the plane curve of a given length that encloses the largest possible area : the curve is a circle.

1638 Galileo posed two shape problems;
- The shape of a heavy chain suspended between two points (the catenary).
- The shape of a wire such that a bead sliding along its end-part in minimum time (the brachystochrone).

1662 Fermat found that Light chooses the path in minimum time

1685 Newton was the first to solve the shape of a body with minimum drag. He did not publish the result until 1694.

1696 Johann Bernoulli (the Netherlands, professor of univ. of Basel) Challenged the Brachystochrone problem

6 people; Johann Bernoulli, Jakob Bernoulli, Leibniz, l'Hopital, Tschirnhaus, Newton.

1744 Euler(1707-1783) published the collected works (when he was 13)

"Nothing at all takes places in the universe in which some rule of maximum or minimum does not appear."

Euler formulated the problem in general terms.

; Finding the curve *q(t)* over the internal $a \le t \le b$ with given values *q(a)*, *q(b)* which minimize

$$J = \int_a^b L(t, q(t), \dot{q}(t))dt \qquad (1)$$

For some given function $L(t, q(t), \dot{q}(t))$

a necessary condition of optimality for the curve q

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}} = \frac{\partial L}{\partial q} \tag{2}$$

→ solution technique; essential geometric

To compute left side of (2), one first evaluate $\frac{\partial L}{\partial \dot{q}}$ treating $\dot{q}$ as an independent variable (not $\dot{q}(t)$, foundation of time), then plugs in q(t) and $\dot{q}(t)$ for q and $\dot{q}$, and finally differentiates with respest to t.

$$\frac{d}{dt}\left[\frac{\partial L}{\partial u}(q(t),\dot{q}(t),t)\right] = \frac{\partial L}{\partial q}(q(t),\dot{q}(t),t) \tag{2'}$$

where $L(q,u,t)$ is a function of $q \in \mathbf{R}^n, u \in \mathbf{R}^n, t \in \mathbf{R}^n$ (i.e. u $\equiv \dot{q}$ (t)).

1755 Lagrange(1736-1813) described an analytical approach (when he was 19)

→ based on "variations" of the optimal curve and using his "undetermined multipliers"

→ which led directly to Euler's necessary condition Euler adopted this approach and rename the subject "the calculus of variations"

Euler - Lagrange equation based on first - order variations, yields only a stationary condition

1786 Legendre(1752-1833) studied the second variation and produced a second - order necessary condition of optimality(minimum).

$$\frac{\partial^2 L}{\partial \dot{q}^2}\left(t, q(t), \dot{q}(t)\right) \geq 0 \tag{3}$$

Legendre deribed it for the scalar case, but it was later extended to the vector case by Clebsch(1833-1872).

→ now known as Legendre – Clebsch condition.

$$\frac{\partial^2 L}{\partial \dot{q}_i \partial \dot{q}_j} \geq 0 \quad , \quad 1 \leq i, j < n \qquad \text{; the Hessian matrix}$$

<Motivation> The Legendre condition is clearly the second order necessary condition for a minimum of a function, but eq(2) does not look at all like the first order condition for a minimum of that same function.

=> It is natural to ask whether there might be a way to relate the two conditions. Is it possible that both can be expressed as necessary conditions for a minimum of one and the same function?
=> The maximum principle.

[(Solution)
Let us look at another way of writing (2)

$$\mathbf{H}(q,u,p,t) \equiv \langle p,u \rangle - L(q,u,t) \tag{i}$$

$$\dot{q}(t) \equiv u$$

Define $p(t) \equiv \frac{\partial L}{\partial u}(q(t),\dot{q}(t),t)$ (ii)

$$\frac{\partial \mathbf{H}}{\partial p} = u$$

$$\dot{q}(t) = u = \frac{\partial \mathbf{H}}{\partial p} \quad \Rightarrow \dot{q}(t) = \frac{\partial \mathbf{H}}{\partial p} \tag{iii}$$

Eq(2′) $\frac{d}{dt}\left[\frac{\partial L}{\partial u}\right] = \frac{\partial L}{\partial q}$, $\frac{\mathrm{d}}{\mathrm{dt}}[\mathrm{p}] = \frac{\partial \mathrm{L}}{\partial \mathrm{q}}$, $\frac{\partial \mathbf{H}}{\partial \mathrm{q}} = -\frac{\partial \mathrm{L}}{\partial \mathrm{q}}$

$$\frac{dp}{dt} = -\frac{\partial \mathbf{H}}{\partial q} \tag{iv}$$

We know $\frac{\partial \mathbf{H}}{\partial u} = p - \frac{\partial L}{\partial u}$

So, equation (ii) gives

$$\frac{\partial \mathbf{H}}{\partial u} = 0 \tag{v}$$

The system of eqs usually written more concisely as

$$\frac{dq}{dt} = \frac{\partial \mathbf{H}}{\partial p}, \quad \frac{dp}{dt} = -\frac{\partial \mathbf{H}}{\partial q}, \quad \frac{\partial \mathbf{H}}{\partial u} = 0 \tag{vi}$$

is exactly equivalent to (2) under completely general condition if **H** is defined as (i).

H is called $\boxed{\text{control Hamitonian}}$

Since $\mathbf{H} = \textit{-L} + \textit{a linear function of } u$, eq (3) is completely equivalent to

$$\frac{\partial^2 \mathbf{H}}{\partial u^2} \le 0 \qquad \left(\textit{i.e.,} \ \frac{\partial^2 \mathbf{H}}{\partial \dot{q}^2} \le 0 \right) \tag{vii}$$

⇨ check eq(v) and eq(vii)

$$\frac{\partial \mathbf{H}}{\partial u} = 0, \frac{\partial^2 \mathbf{H}}{\partial u^2} \le 0$$

H must have a maximum as a function of u.

End]

Before 1838, William Roman Hamilton(1805-1865) had been reformulating the equations of mechanics as a variational principle (least action).

He introduced the function (now known as "Hamiltonian function")

$$\mathrm{H}(\mathrm{t},\mathrm{p},\mathrm{q}) \equiv \langle \mathrm{p},\dot{\mathrm{q}}\rangle - \mathrm{L}(\mathrm{q},\dot{\mathrm{q}},\mathrm{t}) \tag{4}$$

$$\text{where } \mathrm{p} = \frac{\partial \mathrm{L}}{\partial \dot{\mathrm{q}}}(\mathrm{t},\mathrm{q},\dot{\mathrm{q}}) \tag{5}$$

treat $\dot{q}$ not as an independent variable, but as a function of t, q, p.

$$\rightarrow \dot{\mathrm{q}}(\mathrm{t}) = \frac{\partial \mathrm{H}}{\partial \mathrm{p}}(\mathrm{t},\mathrm{p}(\mathrm{t}),\mathrm{q}(\mathrm{t})), \quad \dot{\mathrm{p}}(\mathrm{t}) = -\frac{\partial \mathrm{H}}{\partial \mathrm{q}}(\mathrm{t},\mathrm{p}(\mathrm{t}),\mathrm{q}(\mathrm{t})) \tag{6}$$

if and only if the Euler-Lagrange equation(2) is satisfied.

Eq(6) resembles eq(vi), but is not all the same. The difference is that in Hamilton's definition, $\dot{q}$ is supposed to be treated not as an independent variable but as a function of q, p, t, defined implicitly by the eq (5).

It is easy to see that, if the map (q, $\dot{q}$,t) -> (q,p,t) defined by (5) can be inverted, i.e. if we can solve (5) for $\dot{q}$ as a function of p, q, t, then (6) is equivalent to (vi).

Indeed, it is clear that

$$H(q,p,t) \equiv \mathbf{H}(\ q,u(q,p,t),t)$$

$$\frac{\partial H}{\partial q} = \frac{\partial \mathbf{H}}{\partial q} + \frac{\partial \mathbf{H}}{\partial u}\frac{\partial u}{\partial q} \tag{6'}$$

Since $\frac{\partial \mathbf{H}}{\partial u} = 0 \ for\ u = u(q,p,t) \Rightarrow \frac{\partial H}{\partial q} = \frac{\partial \mathbf{H}}{\partial q}$

Note: It should be clear that the Hamiltonian reformulation of the Euler-

Lagrange equations in terms of the "control Hamiltonian" is at least as natural as the classical one, and perhaps even simpler. Moreover, the control formulation has at least one obvious advantage, namely,
(A1) the control version of the Hamilton equations is equivalent to the Euler-Lagrange system under completely general conditions, whereas the classical version only makes sense when the transformation eq(5) can be inverted, at least locally, to solve for $\dot{q}$ as a function of p, q, t.

Eqs(v) and (vii) suggest something. Clearly, H must have a maximum as a function of u. So we state this as a conjecture.

<u>Conjecture M1</u> ; Besides eq(vi), an additional necessary condition for optimality is that **H** , as a function of u, have a maximum at $\dot{q}(t)$ for each t.

1838 Jacobi(1804-1851) showed that it could be more compactly written in terms of what is now known as the Hamilton-Jacobi equation (basis of "dynamic programming" developed by Bellman over 100 years later).

$$\phi_t(t,q) + H\big(t, \phi_q(t,q), q(t)\big) = 0 \qquad (7)$$

If $\phi(t,q)$ is a twice continuously differentiable solution of this equation, then equation (6) defined a regular optimal trajectory, provided that

(a) the strict Legendre-Clebsch condition is satisfied

$$H_{pp} < 0$$

(b) there are no points conjugate to a along trajectory(the Jacobi condition)

Weierstrauss (1815-1897) considered strong variations (δq small, but no restriction on $\delta\dot{q}$). He considered the problem of minimizing an integral I of the form

$I = \int_a^b L(q(s), \dot{q}(s))ds$ for Lagrangian such that $L(q,\dot{q})$ is positively homogeneous with respect to the velocity $\dot{q}$ (that is, $L(q, \alpha\dot{q}) = \alpha L(q,\dot{q})$ for all $q, \dot{q}$ all $\alpha(\geq 0)$ and does not depend on time)

He introduced the “excess function”

$$E\left(q,u,\bar{u}\right)=L\left(q,\bar{u}\right)-\frac{\partial L}{\partial u}(q,u)\cdot\bar{u} \tag{8}$$

Depending on three sets of independent variables q,u *and* $\bar{u}$.
He then proved his side condition:

For a curve $s\to q^*(s)$ to be solution of the minimum problem, The function E has to be $E\geq 0$ when evaluated for $q\to q^*(s)$, $u\to\dot{q}^*(s)$ and a completely arbitrary $\bar{u}$ (is arbitrary value of u).

Note : Homogeneity property $\Rightarrow$ $L(q,u)=\dfrac{\partial L}{\partial u}(q,u)\cdot u$

So, the “excess function” equally can be written

$$E\left(q,u,\bar{u}\right)=L\left(q,\bar{u}\right)-\frac{\partial L}{\partial u}(q,u)\cdot\bar{u}-\left[L(q,u)-\frac{\partial L}{\partial u}(q,u)\cdot u\right]$$

$$=\left[L\left(q,\bar{u}\right)-\langle p,\bar{u}\rangle\right]-\left[L(q,u)-\langle p,u\rangle\right]$$

$$=-\mathbf{H}\left(q,\bar{u},p\right)-\left(-\mathbf{H}\left(q,u,p\right)\right)=\mathbf{H}\left(q,u,p\right)-\mathbf{H}\left(q,\bar{u},p\right)$$

He esseintially discovered “maximum principle”, however he did it using a obscured procedure and then missed some profound implication of his discovery. Weierstrass’ condition, expressed in terms of the control Hamiltonian, simply says that
(MAX) along an optimal curve t ->q(t), if we define p(t) via (ii), then for every t, the vaule u=$\dot{q}(t)$ must maximize the (control) Hamiltonian **H**(q(t),u,p(t),t) as a function of u.

(Max) can be simplified considerably. Indeed, the requirement that p(t) be defined via (ii) is now redundant: if H(q(t),u,p(t),t), regarded as a function of u, has a maximum at u=$\dot{q}(t)$, then $\dfrac{\partial H}{\partial u}(q(t),\dot{q}(t),p(t),t)=0$, so p(t) has to be given by (ii). Moreover, the vanishing of $\dfrac{\partial H}{\partial u}(q(t),\dot{q}(t),p(t),t)$ is also one of the conditions of (vi). So we can state (vi) and (MAX) together.
Then,

(Necessary Condition for Optimality: NCO) If a curve $q(t)$is a solution of the maximization problem, then there has to exist a function $p(t)$ such that the following three conditions hold for all t

$$\dot{q}(t) = \frac{\partial \mathbf{H}}{\partial p}\left(q(t), \dot{q}(t), p(t), t\right)$$

$$\dot{p}(t) = -\frac{\partial \mathbf{H}}{\partial q}\left(q(t), \dot{q}(t), p(t), t\right)$$

$$\mathbf{H}\left(q(t), \dot{q}(t), p(t), t\right) = \max_{u} \mathbf{H}\left(q(t), u, p(t), t\right) \qquad (9)$$

H must have a maximum as a function of u.

NCO encapsulates in one single statement the combined power of the Euler-Lagrange necessary conditions and Weierstraa side condition as well as the Legendre condition.

With control Hamiltonian, Weierstrass's side condition becomes much simpler. The control Hamiltonian does not require the introduction of an "excess function". One does not need to include a formula specifiying how p(t) is defined, since eq (9) does this automatically.

If you compare (NCO) with all the other necessary conditions that we had written earlier, quite amazingly the derivatives with respect to the u variable are gone.

Principles of Mathematical Guessing[1] suggest that the existence of the $\frac{\partial L}{\partial u}$ is not needed. → Then there is no longer any reason to insist that the range of values of u be the whole space; any subset of $\mathbf{R}^n$ would do, since the minimization that occurs in eq(9) makes sense over any set.

Conjecture M2 ;

(NCO) should still be a necessary condition for optimality even for problems where u=$\dot{q}(t)$ is restricted to belong to some <u>subset</u> U of $\mathbf{R}^n$, and L(q,u,t) is <u>not</u> required to be differentiable with respect to u.

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[1] Principles of Mathematical Guessing

: If a statement is proved under some specific restrictions but turns out not to involve these restrictions at all, chances are that the restrictions are not needed and the statement is valid even without them.

Now that we have liberated ourselves from the constraint that L be differentiable with respect to u.

u (i.e., $\dot{q}(t)$) can be anything?

→ Instead of letting $\dot{q}$ be u, allow $\dot{q}(t)$ to be even more arbitrary. (It ought to be possible for u (i.e., $\dot{q}$) to be anything, and NCO will still work.)

→ After applying the Principle of Mathematical Guessing again, we can write $\underbrace{\dot{q} = f(q,u,t)}_{a \quad general \quad function}$

→ $\langle p,u \rangle = \langle p, f(q,u,t) \rangle$

Conjecture M3 ;

(NCO) should still be a necessary condition for optimality even for problems where q is restricted to satisfy a differential equation $\dot{q} = f(q,u,t)$, with the "control function" t -> u(t) taking values in some set U and allowed to be a "completely arbitrary" U-valued function of t, and the Hamiltonian H now being defined by

$$H(q,u,p,t) = \langle p, f(q,u,t) \rangle - L(q,u,t) \tag{10}$$

A minor modification is needed to make Conjecture M3 to be the Pontryagin maximum principle.

Consider the restriction of the class of admissible function $\dot{q}(t)$ to a subset of $\mathbf{R}^n$, specially so that they also satisfy the set of equations.

$$g\left(t,q(t),\dot{q}(t)\right) = 0, \quad t \in [a,b] \tag{11}$$

⇒ a general set of differential-algebraic equations

⇒ sufficient conditions were imposed to ensure that

$\dot{q} = f\left(t,q(t)\right)$ satisfying equation (11) : problem of Lagrange since it was solved by the use of Lagrange multipliers.

⇒ The complete solution set the scene for parameterizing the degrees of freedom implicit in the Lagrange problem by considering constraints of the form

$$\dot{q} = f\left(t, q\left(t\right), u\left(t\right)\right) \tag{12}$$

↓u(t); "controls", which can be chosen at each $t \in \left(a,b\right)$ yielding the optimal control problem.

optimal control problem :

Find $u(t)$ on *(a, b)* to minimize

$$J = \int_a^b L\left(t, q\left(t\right), \dot{q}\left(t\right)\right) dt$$

Subject to $\dot{q} = f\left(t, q\left(t\right), u\left(t\right)\right)$

$$t \in \left(a,b\right)$$
$$u\left(t\right) \in \Omega \subset R^m$$
$$q\left(a\right),\ q\left(b\right) \quad given$$

1962 Pontryagin

established the necessary conditions in his famous "<u>maximum principle</u>"

$$\dot{q}\left(t\right) = \frac{\partial H}{\partial p} \quad , \qquad \dot{p}\left(t\right) = -\frac{\partial H}{\partial q}$$
$$H\left(t, p, q, \dot{q}\right) = \max_{u \in \Omega} \boldsymbol{H}\left(t, p, q, u\right) \tag{13}$$

$$u^* = \max_u \mathbf{H}\left(t, p, q, u\right)$$

2. Introduction to Optimal Control

- Optimal Control Problem
  - Find the best strategy (control) to maximize/minimize a performance index (objective function, cost function)
  - Compatible with a given dynamic system (equations of motion)
  - Possibly satisfying a set of equality/inequality constraints

- Mathematical Statement
  - Maximize/Minimize a performance index

    $$J\left(\vec{x}(t)\right) = \varphi\left(\vec{x}\left(t_f\right), t_f\right) + \int_{t_0}^{t_f} L\left(\vec{x}(t), \vec{u}(t), t\right) dt$$

  - Subject to a dynamical system (nonholonomic constraints)

    $$\dot{\vec{x}}(t) = f\left(\vec{x}(t), \vec{u}(t), t\right)$$

  - Possibly satisfying a set of equality/inequality state/control constraints (holonomic constraints)

    $$\vec{0} = g\left(\vec{x}(t), \vec{u}(t), t\right)$$
    $$\vec{0} \le h\left(\vec{x}(t), \vec{u}(t), t\right)$$

  - Simple Example

    Minimize

    $$J = \tfrac{1}{2}\int_{t_0}^{t_f} \left(x^2(t) + \alpha u^2(t)\right) dt, \quad \alpha > 0 \text{ is constant}$$

    subject to

    $$\dot{x}(t) = f\left(x(t), u(t)\right)$$

- (A variety of) Performance Index
  - Mayer Cost : $J = \varphi\left(\vec{x}\left(t_f\right), t_f\right)$
  - Lagrange Cost : $J = \int_{t_0}^{t_f} L\left(\vec{x}, \vec{u}, t\right) dt$

- Bolza Cost : $J = \varphi\left(\vec{x}\left(t_f\right), t_f\right) + \int_{t_0}^{t_f} L\left(\vec{x}, \vec{u}, t\right) dt$

- Minimum Fuel : $J = \int_{t_0}^{t_f} \left|\vec{u}\right| dt$, $J = \int_{t_0}^{t_f} \vec{u} \cdot \vec{u}\, dt$

  Depends on the Propulsion System and its Rocket Equation

- Minimum Time : $J = \int_{t_0}^{t_f} 1 dt = t_f - t_0$

  or $J = \int_{t_0}^{t_f} 100\, dt = 100(t_f - t_0)$

  Same meaning mathematically

  Numerical optimization may lead to different results (scaling issue)

- Quadratic Cost : $J = \vec{x}^T\left(t_f\right) Q_f \vec{x}\left(t_f\right) + \int_{t_0}^{t_f} \left(\vec{x}^T Q \vec{x} + u^T R u\right) dt$

- Example: Linear Quadratic Regulator

  - Minimize $J = \frac{1}{2}\int_0^\infty \left(x^2 + u^2\right) dt$

    subject to $\left.\begin{matrix} \dot{x} = u,\ x(0) = 1 \\ u = -kx \end{matrix}\right\} \Rightarrow \dot{x} = -kx,\ x(0) = 1$

    In this simple problem, the task is to <u>find k</u> to minimize the performance index, while satisfying the equations of motion and the initial condition

  - Solution

$$\dot{x} = -kx \Rightarrow x\left(t\right) = x\left(0\right) e^{-kt}$$

$$\begin{aligned} J\left(k\right) &= \frac{1}{2}\int_0^\infty \left(x^2\left(0\right) e^{-2kt} + k^2 x^2\left(0\right) e^{-2kt}\right) dt \\ &= \frac{1}{2} x^2\left(0\right) \int_0^\infty e^{-2kt}\left(1 + k^2\right) dt \\ &= \frac{1}{2} x^2\left(0\right)\left(1 + k^2\right) \frac{1}{-2k}\left[e^{-2kt}\right]_0^\infty \\ &= \frac{1}{4}\frac{\left(1 + k^2\right)}{k} \end{aligned}$$

$\frac{\partial J}{\partial k} = 0$ (necessary condition for a minimum = stationarity condition)

$$\Rightarrow \frac{1}{4}\frac{2k^2-\left(1+k^2\right)}{k^2}=0$$
$$\Rightarrow k^2=1$$
$$\Rightarrow k=\pm 1$$

$\frac{\partial^2 J}{\partial k^2}>0$ (sufficient condition for a minimum = convexity condition)

$$\Rightarrow \frac{\partial^2 J}{\partial k^2}=\frac{\partial}{\partial k}\left(\frac{-1}{4k^2}+\frac{1}{4}\right)=\frac{1}{2k^3}>0$$
$$\Rightarrow k^*=1$$
$$\Rightarrow J^*=\frac{1}{2}$$

k is not a function of time $\Rightarrow$ parameter optimization

- Transformation into a classical form of Calculus of Variations:
  When u is not used and $\dot{x}$ is not restricted (i.e., no equations of motion such as $\dot{x}=-kx$), the above problem becomes:
  Minimize $J=\frac{1}{2}\int_0^\infty \left(x^2+\dot{x}^2\right)dt,\ \ x\left(0\right)=1$
  This is a standard form of Calculus of Variations that does not treat control variables u.

- Procedure to Solving Optimal Control Problem
  - For a given physical optimal control problem, mathematically formulate an optimal control problem, i.e., define
    - Performance Index
    - Dynamical System
    - Equality/Inequality Constraints
  - Determine a control function that minimizes or maximizes the performance index, while satisfying the associated dynamical system and constraints
    - Minimum (or Maximum) Principle by Pontryagin
      - $\Rightarrow$ Nonlinear two-point boundary-value problem (TPBVP)
      - $\Rightarrow$ Open Loop Solution in general, except LQR

- Dynamic programming by Bellman
  - ⇒ Hamiltonian-Jacobi-Bellman Equation (HJBE)
  - ⇒ Closed-Loop Solution in general, i.e., Solution in Feedback Form

  In the frame of Dynamic Programming, an optimal policy is found by employing the intuitively appealing concept called the Principle of Optimality

- Principle of Optimality: Whatever the initial state and initial decision are, the remaining decision must constitute an optimal policy with regard to the state resulting from the first decision

  Ex) If a-b-e is the optimal path from a to e, b-e is the optimal path from b to e.

- Many complex aerospace problems that are not amenable to classical techniques have been solved by using optimal control theory
- The optimal control law, if it can be obtained, usually requires a digital computer for implementation (except for the linear quadratic regulator problem); it is a full state feedback (vs. output feedback) control law, and thus all of the states must be available for feedback to the controller computer for implementation (except for the linear regulator problem)

3. The Calculus of Variations

3.1. Fundamental Concepts

3.1.1. Functionals

- A function ***f*** is a rule of correspondence that assigns each <u>element</u> $\vec{q}$ in a certain set **D** to a unique element in a set **R**

  D: domain of ***f***
  R: range of ***f***

  Ex) $f(\vec{q}) = \sqrt{q_1^2 + q_2^2 + \cdots + q_n^2}$

  $q_1, \cdots q_n$; coordinate of a point in *n*-dimentional Euclidean space.
  $f(\vec{q})$ is the real number assigned by *f* and represents the distance of the point $\vec{q}$ from the origin.

- A functional ***J*** is a rule of correspondence that assigns each <u>function</u> $\vec{x}$ in a certain set **Ω** to a unique real number **S**

  Ω: domain of ***J***
  S: range of ***J***

  A functional is a "function of a function"

  Ex) $\underbrace{J(x)}_{functional} = \int_{t_0}^{t_f} \underbrace{x(t)}_{function}\, dt$

  J is a unique real number.
  J is the area under the $x(t)$ curve.

3.1.2 Linearity of Functional

- *J* is a linear functional if and only if it satisfies:
  - Principle of Homogeneity
    $J(\alpha\vec{x}) = \alpha J(\vec{x})$ for all $\vec{x} \in \Omega$, real number α, $\alpha\vec{x} \in \Omega$

■ Principle of Additivity

$J(\vec{x}_1+\vec{x}_2)=J(\vec{x}_1)+J(\vec{x}_2)$ for $\vec{x}_1, \vec{x}_2, and\ \vec{x}_1+\vec{x}_2 \in \Omega$

In other words, *J* is a linear functional if and only if it belongs to the Linear Vector Space

■ Example

◆ $J(x)=\int_{t_0}^{t_f} x(t)dt$ $\Rightarrow$ Linear Functional

Homogeneity

$$J(\alpha x)=\int_{t_0}^{t_f}\alpha x(t)\,dt=\alpha\int_{t_0}^{t_f}x(t)\,dt \Rightarrow J(\alpha x)=\alpha J(x)$$

Additivity

$$J(x_1)=\int_{t_0}^{t_f}x_1(t)\,dt,\ J(x_2)=\int_{t_0}^{t_f}x_2(t)\,dt$$

$$J(x_1+x_2)=\int_{t_0}^{t_f}[x_1(t)+x_2(t)]\,dt=\int_{t_0}^{t_f}x_1(t)\,dt+\int_{t_0}^{t_f}x_2(t)\,dt=J(x_1)+J(x_2)$$

◆ $J(x)=\int_{t_0}^{t_f} x^2(t)dt$ $\Rightarrow$ Non-Linear Functional

Homogeneity?

$$\left.\begin{aligned} J(\alpha x) &= \int_{t_0}^{t_f}\alpha^2 x^2(t)\,dt \\ &= \alpha^2\int_{t_0}^{t_f}x^2(t)\,dt \\ &= \alpha^2 J(x) \end{aligned}\right\} \Rightarrow J(\alpha x)\neq \alpha J(x)$$

3.1.3 Increment of Functional

- Increment of Function *f*

  Let $\vec{q}$, $\vec{q}+\Delta\vec{q}$ be the elements of *f*

  Then, increment of *f* is defined as $\Delta f \equiv f(\vec{q}+\Delta\vec{q})-f(\vec{q})$

  Δf depends on both $\vec{q}$ and $\Delta\vec{q}$ $\Rightarrow$ $\Delta f(\vec{q},\Delta\vec{q})$

- Increment of Functional J

  Let $\vec{x}$, $\vec{x}+\delta\vec{x}$ be the functions for J where $\delta\vec{x}$ is the variation of the function $\vec{x}$

  Then, increment of J is defined as $\Delta J \equiv J(\vec{x}+\delta\vec{x}) - J(\vec{x})$

  ΔJ depends on both $\vec{x}$ and $\delta\vec{x}$ $\Rightarrow$ $\Delta J(\vec{x}, \delta\vec{x})$

  Example

$$J(x) = \int_{t_0}^{t_f} x^2(t)dt$$

$$\begin{aligned} \Delta J &= J(x+\delta x) - J(x) \\ &= \int_{t_0}^{t_f} (x+\delta x)^2 dt - \int_{t_0}^{t_f} x^2 dt \\ &= \int_{t_0}^{t_f} \left[2x\delta x + (\delta x)^2 \right] dt \end{aligned}$$

3.1.4 Variation of Functional

- The increment of a function f of n variables can be written as

$$\Delta f(\vec{q}, \Delta\vec{q}) = \underbrace{df(\vec{q}, \Delta\vec{q})}_{a\ linear\ function\ of\ \Delta\vec{q}} + g(\vec{q}, \Delta\vec{q}) \cdot \|\Delta\vec{q}\|$$

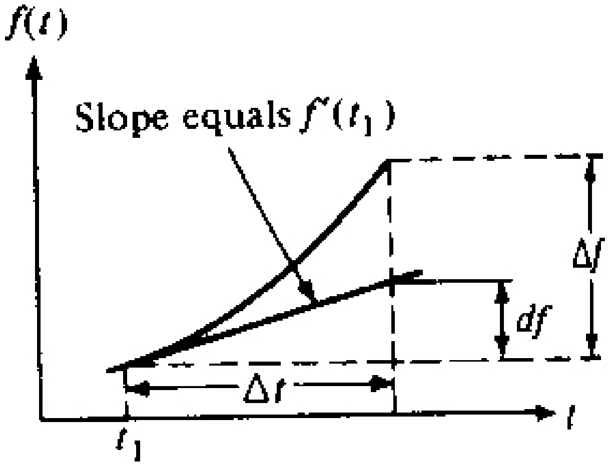
  If $\lim_{\|\Delta\vec{q}\| \to 0} g(\vec{q}, \Delta\vec{q}) = 0$

  Then, f is said to be differentiable at $\vec{q}$ and df is a <u>differential</u> of f at the point $\vec{q}$.

  If f is a differentiable function of one variable t, its differential can be written

  $df(t, \Delta t) = f'(t)\Delta t$ first-order approximation to Δf

  where $f'(t)$ is the derivative of f at t



- The increment of a functional (⇒Total Variation of Functional)

$$\Delta J(\vec{x},\delta\vec{x}) = \underbrace{\delta J(\vec{x},\delta\vec{x})}_{\text{linear term in } \delta\vec{x}} + g(\vec{x},\delta\vec{x})\cdot\|\delta\vec{x}\|$$

If $\lim\limits_{\|\delta\vec{x}\|\to 0}\{g(\vec{x},\delta\vec{x})\} = 0$

Then, J is said to be differentiable on $\vec{x}$ and δJ is the variation of J evaluated for the function $\vec{x}$

δJ is the linear approximation to the difference in the functional J caused by $\vec{x}$, $\vec{x}+\delta\vec{x}$

3.1.5 Maxima and Minima of Functional

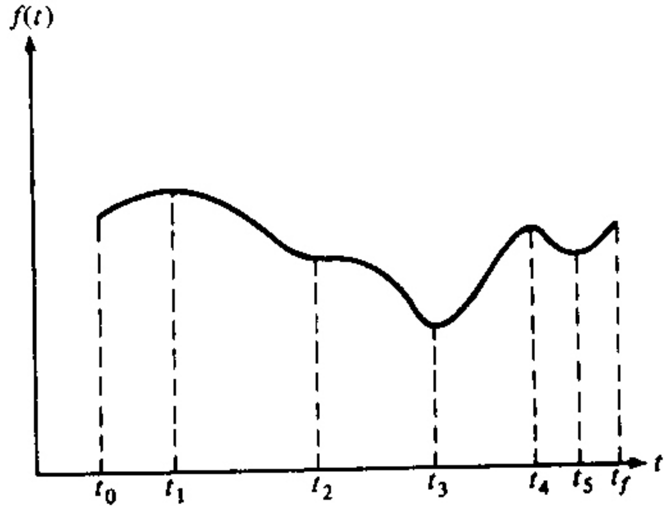
- A function f with its domain $\boldsymbol{D}$ has a relative extremum at the point $\vec{q}^*$,

  if there exists $\varepsilon > 0$ such that $\|\vec{q}-\vec{q}^*\| < \varepsilon$ for $\forall\,\vec{q}\in D$ and the increment of f has the same sign:

  $f(\vec{q}^*)$ is a relative minimum, if $\Delta f = f(\vec{q}) - f(\vec{q}^*) > 0$ (a)

  $f(\vec{q}^*)$ is a relative maximum, if $\Delta f = f(\vec{q}) - f(\vec{q}^*) < 0$ (b)

If equation (a)/(b) are satisfied for arbitrarily large ε, then $f(\vec{q}^*)$ is a global or absolute minimum/maximum



stationary points:	t_1, t_2, t_3, t_4, t_5
extreme points	t_1, t_3, t_4, t_5
relative minima	t_0, t_3, t_5
relative maxima	t_1, t_4, t_f
global minima	t_3
global maxima	t_1

- A functional J with domain Ω has a relative <u>extremum</u> at $\vec{x}^*$ (<u>extremal</u>) if there exists an $\varepsilon > 0$ such that $\left\| \vec{X} - \vec{X}^* \right\| < \varepsilon$ for $\forall \vec{X} \in \Omega$ and the increment of J has the same sign:

  $J(\vec{x}^*)$ is a relative minimum if $\Delta J = J(\vec{x}) - J(\vec{x}^*) > 0$ (c)

  $J(\vec{x}^*)$ is a relative maximum if $\Delta J = J(\vec{x}) - J(\vec{x}^*) < 0$ (d)

  If equation (c)/(d) is satisfied for arbitrarily large ε, then $J(\vec{x}^*)$ is a global or absolute minimum/maximum

- Let $\vec{x}^*$ be the <u>extremal</u> and $J(\vec{x}^*)$ be the <u>extremum</u>.
  Then, $\Delta J(\vec{x}^*)$ must have <u>the same sign</u> for every $\vec{x}$ near $\vec{x}^*$, i.e., every $\delta\vec{x}$

If $\delta\vec{x}$ is sufficiently small, then sign ΔJ = sign δJ

ΔJ : Total Variation δJ First Variation

$$\Delta J(\vec{x},\delta\vec{x}) = \underbrace{\delta J(\vec{x},\delta\vec{x})}_{linear\ term\ in\ \delta\vec{x}} + g(\vec{x},\delta\vec{x})\cdot\|\delta\vec{x}\|$$

If $\lim_{\|\delta\vec{x}\|\to 0}\{g(\vec{x},\delta\vec{x})\} = 0$, then $\Delta J \simeq \delta J$

$$\begin{aligned}
\Delta J(\vec{x}^*,\delta\vec{x}) &= J(\vec{x}^* + \delta\vec{x}) - J(\vec{x}^*) \\
&= J(\vec{x}^*) + \frac{\partial J}{\partial\vec{x}}\Big|_{\vec{x}^*}\delta\vec{x} + \frac{1}{2}\frac{\partial^2 J}{\partial\vec{x}^2}\Big|_{\vec{x}^*}(\delta\vec{x})^2 + \cdots - J(\vec{x}^*) \\
&= \frac{\partial J}{\partial\vec{x}}\Big|_{\vec{x}^*}\delta\vec{x} + \frac{1}{2}\frac{\partial^2 J}{\partial\vec{x}^2}\Big|_{\vec{x}^*}(\delta\vec{x})^2 + H.O.T
\end{aligned}$$

$$where\ \frac{\partial J}{\partial\vec{x}}\Big|_{\vec{x}^*}\delta\vec{x} \equiv \delta J(\vec{x}^*,\delta\vec{x});\ first\ variation$$

$$\frac{1}{2}\frac{\partial^2 J}{\partial\vec{x}^2}\Big|_{\vec{x}^*}(\delta\vec{x})^2 \equiv \delta^2 J;\ second\ variation$$

δJ is a linear function of $\delta\vec{x}$

3.1.6 **Fundamental Theorem of the Calculus of Variation**

- Fundamental Theorem (Necessary Condition) used in finding extreme values of functions
  - the differential must be zero at an extreme point (except extrema at the boundaries of closed region)

- Analogous Theorem exists in Variational Problems
  - the variation must be zero on an exteremal curve, if there are no bounds imposed on the curves

- **Fundamental Theorem of the Calculus of Variations:**

  Let $\vec{x}(t): R \to \Omega \subset R^n$ and $J(\vec{x}): R^n \to R$ (that is, let $\vec{x}$ be a vector function of t in the class Ω) and $J(\vec{x})$ be a differentiable function of $\vec{x}$).

If $\vec{x}^*$ is an extremal and $J(\vec{x})$ is a smooth (differentiable) functional, $\vec{x}$ is not constrained in the interior and by any boundaries, the first variation of J must vanish on $\vec{x}^*$, that is, $\underline{\delta J(\vec{x}^*, \delta\vec{x}) = 0}$ for all admissible $\delta\vec{x}$.

- Proof by Contradiction

  Assume that $\vec{x}^*$ is an extremal and that $\delta J(\vec{x}^*, \delta\vec{x}) \neq 0$.

  Then, the increment ΔJ can be made to change sign in an arbitrarily small neighborhood of $\vec{x}^*$, which contradicts the relative extremum for J at $\vec{x}^*$, i.e., that the increment of J ($= \Delta J$) has the same sign at extremal $\vec{x}^*$.

  Recall

  - If $\vec{x}^*$ be the extremal and $J(\vec{x}^*)$ be the extremum, then $\Delta J(\vec{x}^*)$ must have the same sign for every $\vec{x}$ near $\vec{x}^*$.
  - If $\delta\vec{x}$ is sufficiently small, sign ΔJ = sign δJ

- Proof in detail

  The increment is

$$\Delta J(\vec{x}^*, \delta\vec{x}) = J(\vec{x}^* + \delta\vec{x}) - J(\vec{x}^*) = \delta J(\vec{x}^*, \delta\vec{x}) + g(\vec{x}^*, \delta\vec{x}) \cdot \|\delta\vec{x}\|$$

  Then,

$$\lim_{\|\delta\vec{x}\| \to 0} g(\vec{x}^*, \delta\vec{x}) = 0 \Rightarrow \Delta J \approx \delta J$$

I. Let us select a variation

$$\delta\vec{x}(t) = \alpha\delta\vec{x}^{(1)}(t) \text{ such that } \alpha > 0 \text{ and } \left\|\alpha\delta\vec{x}^{(1)}\right\| < \varepsilon$$

   Suppose that

$$\delta J(\vec{x}^*, \alpha\delta\vec{x}^{(1)}) < 0 \quad (\text{Note } \delta J(\vec{x}^*, \delta\vec{x}) = \left.\frac{\partial J}{\partial \vec{x}}\right|_{\vec{x}^*} \delta\vec{x})$$

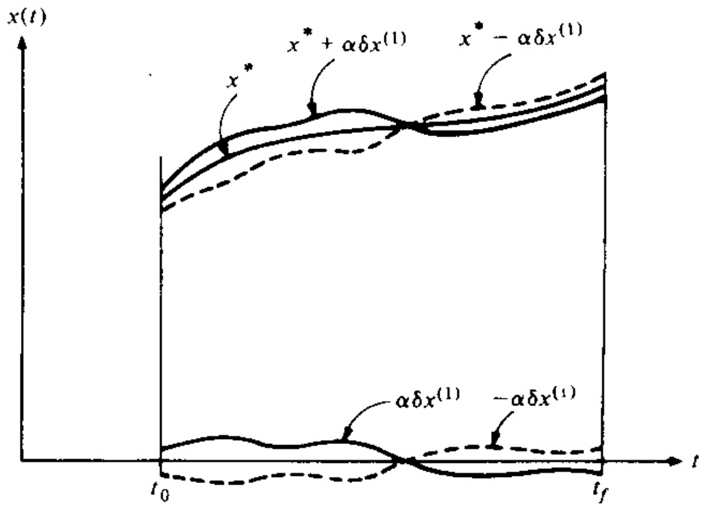
   Since δJ is a linear functional of $\delta\vec{x}$, the principle of homogeneity gives

$$\delta J\left(\vec{x}^*,\alpha\delta\vec{x}^{(1)}\right)=\left.\frac{\partial J}{\partial\vec{x}}\right|_{\vec{x}^*}\alpha\delta\vec{x}^{(1)}=\alpha\delta J\left(\vec{x}^*,\delta\vec{x}^{(1)}\right)<0$$

$$\Rightarrow \delta J\left(\vec{x}^*,\delta\vec{x}^{(1)}\right)<0$$

$$\Rightarrow \delta J\left(\vec{x}^*,\alpha\delta\vec{x}^{(1)}\right)=\delta J\left(\vec{x}^*,\delta\vec{x}\right)<0 \quad (\text{note } \delta\vec{x}=\alpha\delta\vec{x}^{(1)})$$

$$\Rightarrow \Delta J\left(\vec{x}^*,\delta\vec{x}\right)<0$$



II. Let us select another variation

$$\delta\vec{x}(t)=-\alpha\delta\vec{x}^{(1)}(t) \text{ such that } \alpha>0, \ \left\|-\alpha\delta\vec{x}^{(1)}\right\|<\varepsilon$$

Similarly, the principle of homogeneity gives

$$\delta J\left(\vec{x}^*,-\alpha\delta\vec{x}^{(1)}\right)=-\alpha\delta J\left(\vec{x}^*,\delta\vec{x}^{(1)}\right)>0 \quad (\text{since } \delta J\left(\vec{x}^*,\delta\vec{x}^{(1)}\right)<0)$$

$$\Rightarrow \delta J\left(\vec{x}^*,-\alpha\delta\vec{x}^{(1)}\right)=\delta J\left(\vec{x}^*,\delta\vec{x}\right)>0 \quad (\text{note } \delta\vec{x}=-\alpha\delta\vec{x}^{(1)})$$

$$\Rightarrow \Delta J\left(\vec{x}^*,\delta\vec{x}\right)>0$$

III. From I and II,
if $\delta J\left(\vec{x}^*,\delta\vec{x}\right)\neq 0$, then

$$\Delta J\left(\vec{x}^*,\alpha\delta\vec{x}^{(1)}\right)=\Delta J\left(\vec{x}^*,\delta\vec{x}\right)<0$$

$$\text{where } \delta\vec{x}=\alpha\delta\vec{x}^{(1)}, \ \text{because } \delta\vec{x} \text{ can be arbitrary near } \vec{x}^*$$

$$\Delta J\left(\vec{x}^*,-\alpha\delta\vec{x}^{(1)}\right)=\Delta J\left(\vec{x}^*,\delta\vec{x}\right)>0$$

where $\delta\vec{x}=-\alpha\delta\vec{x}^{(1)}$, *because* $\delta\vec{x}$ *can be arbitrary near* $\vec{x}^*$,

which contradicts the assumption that if $\vec{x}^*$ is extremal, then ΔJ has the same sign. These arguments conclude that if $\vec{x}$ is an extremal, it is necessary that $\delta J\left(\vec{x}^*,\delta\vec{x}\right)=0$ for arbitrary $\delta\vec{x}$.

3.2 Functionals of a Single Function

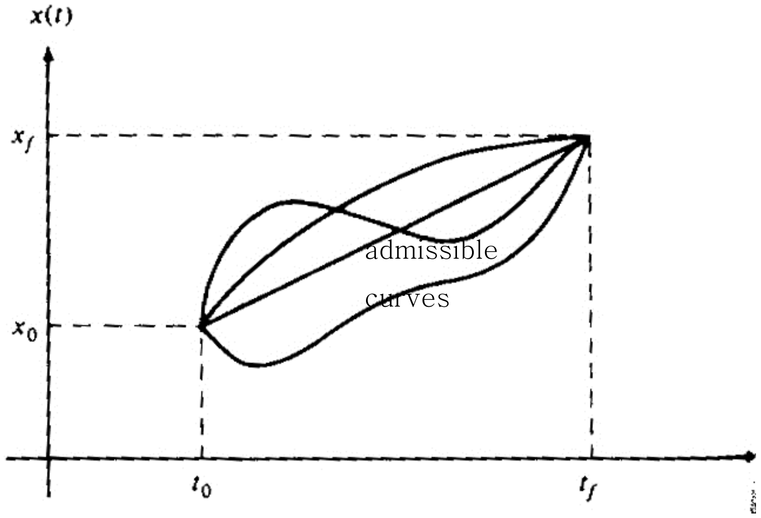
3.2.1 The simplest Variation Problem

Class Ω		Type of Problem
x must be continuous	C	$J=\int g(x,t)dt$
x must be continuous with continuous first derivatives $(x\subset D^1)$	D^1	$J=\int g(x,\dot{x},t)dt$
x must be continuous with continuous first and second derivatives $(x\subset D^2)$	D^2	$J=\int g(x,\dot{x},\ddot{x},t)dt$

- Find a function *x*(t)* (extremal) for which the following functional *J(x)* has a relative extremum

  $$J(x)=\int_{t_0}^{t_f} g(x(t),\dot{x}(t),t)dt$$

  t_0 and t_f are fixed, $x(t_0)$ and $x(t_f)$ are given



The <u>admissible</u> solutions must be curves $x \in \Omega$ which also satisfy the end conditions

The search begins by finding the curves that satisfy the <u>fundamental theorem</u>

$$\Delta J(x^*,\dot{x}^*,t;\delta x,\delta\dot{x}) = J(x^*+\delta x,\dot{x}^*+\delta\dot{x},t) - J(x^*,\dot{x}^*,t)$$

$$= \int_{t_0}^{t_f} \underbrace{g(x,\dot{x},t)}_{\substack{\text{Taylor series}\\ \text{expansion}\\ \text{about } x^*,\dot{x}^*}} dt - \int_{t_0}^{t_f} g(x^*,\dot{x}^*,t)\,dt$$

$$= \int_{t_0}^{t_f} \left[g(x^*,\dot{x}^*,t) + \left.\frac{\partial g}{\partial x}\right|_{x^*,\dot{x}^*} \delta x + \left.\frac{\partial g}{\partial \dot{x}}\right|_{x^*,\dot{x}^*} \delta\dot{x} + \underbrace{H.O.T.}_{neglect} \right] dt - \int_{t_0}^{t_f} g(x^*,\dot{x}^*,t)\,dt,$$

$$H.O.T. = \frac{1}{2}\left[\left.\frac{\partial^2 g}{\partial x^2}\right|_{x^*,\dot{x}^*} (\delta x)^2 + 2\left.\frac{\partial g}{\partial x \partial \dot{x}}\right|_{x^*,\dot{x}^*} \delta x\delta\dot{x} + \left.\frac{\partial^2 g}{\partial \dot{x}^2}\right|_{x^*,\dot{x}^*} (\delta\dot{x})^2 \right] + \theta\left[(\delta x)^2,(\delta\dot{x})^2\right]$$

Extract the first variation in ΔJ that are linear in δx and $\delta\dot{x}$

$$\Delta J \simeq \delta J\left(x,\delta x\right)$$

$$= \int_{t_0}^{t_f}\left[\left.\frac{\partial g}{\partial x}\right|_{x^*,\dot{x}^*}\delta x + \left.\frac{\partial g}{\partial \dot{x}}\right|_{x^*,\dot{x}^*}\delta \dot{x}\right]dt$$

$$\left(\Uparrow \int_{t_0}^{t_f} u\dot{v}dt = uv\Big|_{t_0}^{t_f} - \int_{t_0}^{t_f}\dot{u}vdt\right)$$

$$= \frac{\partial g}{\partial \dot{x}}\delta x\Big|_{t_0}^{t_f} + \int_{t_0}^{t_f}\left[\frac{\partial g}{\partial x}\delta x - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right)\delta x\right]dt$$

$$\delta x(t_0) = 0,\ \delta x(t_f) = 0\ \ fixed\ (natural\ bounday\ condition)$$

$$= \int_{t_0}^{t_f}\left[\frac{\partial g}{\partial x} - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right)\right]\delta x(t)dt \underset{\substack{fundamental\\ theorem}}{=} 0$$

Since $\delta x(t)$ is arbitrary,

$$\frac{\partial g}{\partial x}\left(x^*,\dot{x}^*,t\right) - \frac{d}{dt}\left[\frac{\partial g}{\partial \dot{x}}\left(x^*,\dot{x}^*,t\right)\right] = 0$$

⇒ **Euler-Lagrange equation**

I. A <u>necessary</u> condition for x* to be an extremal

II. In general, nonlinear, time-varying, second-order, ordinary differential equation

III. The boundary condition are split

⇒ $x(t_0)$, $x(t_f)$ are given, instead of $[x(t_0),\dot{x}(t_0)]$ or $[x(t_f),\dot{x}(t_f)]$

IV. Nonlinear, <u>Two-Point-Boundary-Value-Problem (TPBVP)</u>

⇒ Extremely Dfficult

⇒ Usually cannot be solved analytically

⇒ Numerical integration is used

- Find a function *x*(t)* (extremal) for which the following functional *J(x)* has a relative extremum

$$J(x) = \int_{t_0}^{t_f} g\left(x(t),\dot{x}(t),\ddot{x}(t),t\right)dt$$

t_0 and t_f are fixed, $x(t_0)$, $x(t_f)$, $\dot{x}(t_0)$ and $\dot{x}(t_f)$ are given

$$\delta J = \int_{t_0}^{t_f}\left[\frac{\partial g}{\partial x}\delta x + \frac{\partial g}{\partial \dot{x}}\delta\dot{x} + \frac{\partial g}{\partial \ddot{x}}\delta\ddot{x}\right]dt$$

$$\left(\begin{array}{l} \Uparrow \int_{t_0}^{t_f}\left(\frac{\partial g}{\partial \dot{x}}\delta\dot{x}\right)dt = \frac{\partial g}{\partial \dot{x}}\delta x\Big|_{t_0}^{t_f} - \int_{t_0}^{t_f}\frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right)\delta x dt \\ \quad \int_{t_0}^{t_f}\left(\frac{\partial g}{\partial \ddot{x}}\delta\ddot{x}\right)dt = \frac{\partial g}{\partial \ddot{x}}\delta\dot{x}\Big|_{t_0}^{t_f} - \int_{t_0}^{t_f}\frac{d}{dt}\left(\frac{\partial g}{\partial \ddot{x}}\right)\delta\dot{x} dt \end{array}\right)$$

$$= \frac{\partial g}{\partial \ddot{x}}\delta\dot{x}\Big|_{t_0}^{t_f} + \frac{\partial g}{\partial \dot{x}}\delta x\Big|_{t_0}^{t_f} + \int_{t_0}^{t_f}\left[\frac{\partial g}{\partial x}\delta x - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right)\delta x - \frac{d}{dt}\left(\frac{\partial g}{\partial \ddot{x}}\right)\delta\dot{x}\right]dt$$

$$\left(\Uparrow \int_{t_0}^{t_f}\left(\frac{d}{dt}\left(\frac{\partial g}{\partial \ddot{x}}\right)\delta\dot{x}\right)dt = \frac{d}{dt}\left(\frac{\partial g}{\partial \ddot{x}}\right)\delta x\Big|_{t_0}^{t_f} - \int_{t_0}^{t_f}\frac{d^2}{dt^2}\left(\frac{\partial g}{\partial \ddot{x}}\right)\delta x dt\right)$$

$$= -\frac{d}{dt}\left(\frac{\partial g}{\partial \ddot{x}}\right)\delta x\Big|_{t_0}^{t_f} + \int_{t_0}^{t_f}\left[\frac{\partial g}{\partial x}\delta x - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right)\delta x + \frac{d^2}{dt^2}\left(\frac{\partial g}{\partial \ddot{x}}\right)\delta x\right]dt$$

$$= \int_{t_0}^{t_f}\left[\frac{d^2}{dt^2}\left(\frac{\partial g}{\partial \ddot{x}}\right) - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right) + \frac{\partial g}{\partial x}\right]\delta x\, dt$$

$$= 0$$

Since $\delta x(t)$ is arbitrary,

$$\frac{d^2}{dt^2}\left(\frac{\partial g}{\partial \ddot{x}}\right) - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right) + \frac{\partial g}{\partial x} = 0,$$ Euler-Lagrange equation

- Find a function *x*(t)* (extremal) for which the following functional *J(x)* has a relative extremum

$$J(x) = \int_{t_0}^{t_f} g\left(x(t), \dot{x}(t), \ddot{x}(t), \dddot{x}(t), t\right)dt$$

t_0 and t_f are fixed, x, $\dot{x}$ and $\ddot{x}$ are given at both end points

Similar derivation yields

$$-\frac{d^3}{dt^3}\left(\frac{\partial g}{\partial \dddot{x}}\right) + \frac{d^2}{dt^2}\left(\frac{\partial g}{\partial \ddot{x}}\right) - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right) + \frac{\partial g}{\partial x} = 0$$

Generalization
$$(-1)^n\frac{d^n}{dt^n}\left(\frac{\partial g}{\partial \overset{n}{x}}\right) + (-1)^{n-1}\frac{d^{n-1}}{dt^{n-1}}\left(\frac{\partial g}{\partial \overset{n-1}{x}}\right) + \cdots + \frac{\partial g}{\partial x} = 0$$

- Example (2-Dimensional Low-Thrust Orbit Transfer)

I. Governing Equations of Motion

$$m\ddot{\vec{r}} = -m\nabla\phi + m\vec{u}$$

$$\vec{r} = r\hat{e}_r \ (\,polar\ coordinate\,)$$

$$\vec{u}\ : thrust\ acceleration\ (control)$$

$$\phi = \phi(\vec{r}\,) : gravitational\ \ potential,\ \ \nabla\phi(\vec{r}\,) = \frac{\partial\phi(\vec{r}\,)}{\partial\vec{r}} = \frac{\partial\phi}{\partial r}\hat{e}_r + \frac{1}{r}\frac{\partial\phi}{\partial\theta}\hat{e}_\theta$$

II. (Minimum Control-Effort Problem) Find $\vec{u}$ to minimize

$$J = \frac{1}{2}\int_0^T \left(\vec{u}^T\vec{u}\right)dt,\quad T,\ r(0)\ and\ r(T)\ are\ specified$$

$$J\left(\vec{r},\ddot{\vec{r}}\right) = \frac{1}{2}\int_0^T \left(\ddot{\vec{r}} + \nabla\phi\right)^T\left(\ddot{\vec{r}} + \nabla\phi\right)dt \ \Rightarrow\ J(\vec{r},\ddot{\vec{r}}\,) = \int_0^T g(\vec{r},\ddot{\vec{r}}\,)dt$$

Necessary Condition

$$\frac{d^2}{dt^2}\left(\frac{\partial g}{\partial\ddot{\vec{r}}}\right) - \frac{d}{dt}\left(\frac{\partial g}{\partial\dot{\vec{r}}}\right) + \frac{\partial g}{\partial\vec{r}} = 0$$

$$\frac{\partial g}{\partial\vec{r}} = \frac{\partial}{\partial\vec{r}}\left[\left(\ddot{\vec{r}} + \nabla\phi\right)^T\left(\ddot{\vec{r}} + \nabla\phi\right)\right] = 2\left(\ddot{\vec{r}} + \nabla\phi\right)\frac{\partial}{\partial\vec{r}}\nabla\phi(\vec{r}\,) = 2\left(\ddot{\vec{r}} + \nabla\phi\right)\nabla^2\phi(\vec{r}\,)$$

$$\frac{\partial g}{\partial\dot{\vec{r}}} = 0$$

$$\frac{\partial g}{\partial\ddot{\vec{r}}} = \frac{\partial}{\partial\ddot{\vec{r}}}\left[\left(\ddot{\vec{r}} + \nabla\phi\right)^T\left(\ddot{\vec{r}} + \nabla\phi\right)\right] = 2\left(\ddot{\vec{r}} + \nabla\phi\right)\frac{\partial}{\partial\ddot{\vec{r}}}\left(\ddot{\vec{r}} + \nabla\phi\right) = 2\left(\ddot{\vec{r}} + \nabla\phi\right)$$

$$\Rightarrow \frac{d^2}{dt^2}\left(\ddot{\vec{r}} + \nabla\phi\right) + \nabla^2\phi(\vec{r}\,)\left(\ddot{\vec{r}} + \nabla\phi\right) = 0$$

$$\Rightarrow \ddot{\vec{u}} + \nabla^2\phi\vec{u} = 0,\quad \vec{r}(0),\ \vec{r}(T),\ \dot{\vec{r}}(0),\ \dot{\vec{r}}(T),\ T\ \ are\ \ specified$$

$$\Rightarrow\ Two-Point-Boundary-Value-Problem\ \ (TPBVP)$$

$$\Rightarrow\ \vec{u}^*(t)$$

$$\Rightarrow m\ddot{\vec{r}} = -m\nabla\phi + m\vec{u}^*(t)$$

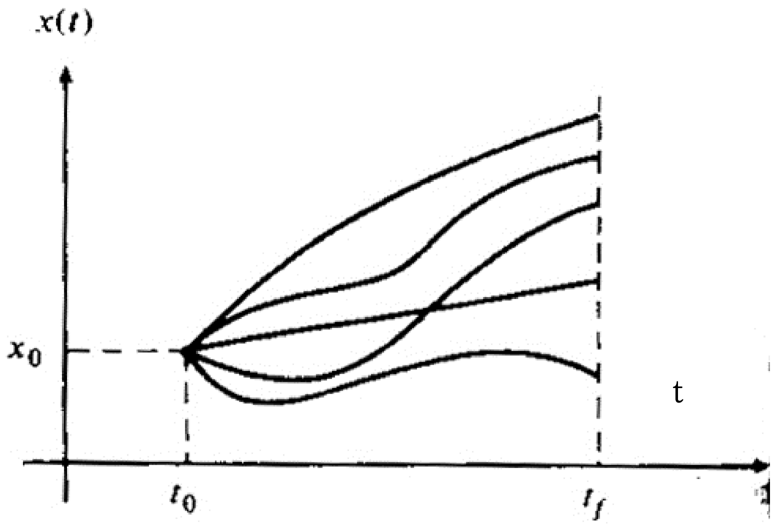
$$\Rightarrow \vec{r}^*(t)$$

3.2.2 Final Time Specified, $x(t_f)$ Free

- Find a function $x^*(t)$ (extremal) for which the following functional $J(x)$ has a relative extremum

$$J(x)=\int_{t_0}^{t_f} g\left(x(t),\dot{x}(t),t\right)dt$$

t_0 and t_f are specified, $x(t_0)$ is given, $x(t_f)$ is free



$$\delta J=\int_{t_0}^{t_f}\left[\frac{\partial g}{\partial x}\delta x+\frac{\partial g}{\partial \dot{x}}\delta\dot{x}\right]dt$$

$$=\frac{\partial g}{\partial \dot{x}}\delta x\Big|_{t_0}^{t_f}+\int_{t_0}^{t_f}\left[\frac{\partial g}{\partial x}-\frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right)\right]\delta x\,dt$$

$$\left(\Uparrow\quad \frac{\partial g}{\partial \dot{x}}\delta x\Big|_{t_0}^{t_f}=\frac{\partial g}{\partial \dot{x}}\delta x(t_f)-\frac{\partial g}{\partial \dot{x}}\delta x(t_0)=\frac{\partial g}{\partial \dot{x}}\delta x(t_f)\right)$$

$$=\frac{\partial g}{\partial \dot{x}}\delta x\Big|_{t=t_f}\quad\delta x(t_f)+\int_{t_0}^{t_f}\left[\frac{\partial g}{\partial x}-\frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right)\right]\delta x dt$$

$$=0$$

- Since $\delta J\left(x^*,\delta x\right)=0$ and δx is arbitrary, the following conditions must hold:

$$\frac{\partial g}{\partial x}(x,\dot{x},t)-\frac{d}{dt}\left[\frac{\partial g}{\partial \dot{x}}(x,\dot{x},t)\right]=0$$

$$\left.\frac{\partial g}{\partial \dot{x}}\right|_{t=t_f}=0$$

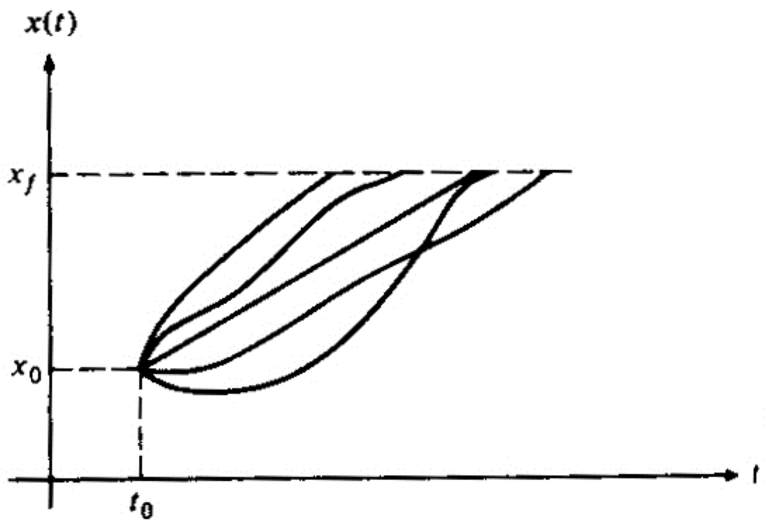
The first condition is the Euler-Lagrange equation, and the second condition at the boundary is called Transversality Condition (Natural Boundary Condition)

3.2.3 Final Time Free, *$x(t_f)$* Specified

- Find a function *$x^*(t)$* (extremal) for which the following functional *$J(x)$* has a relative extremum

  $$J(x)=\int_{t_0}^{t_f} g(x(t),\dot{x}(t),t)dt$$

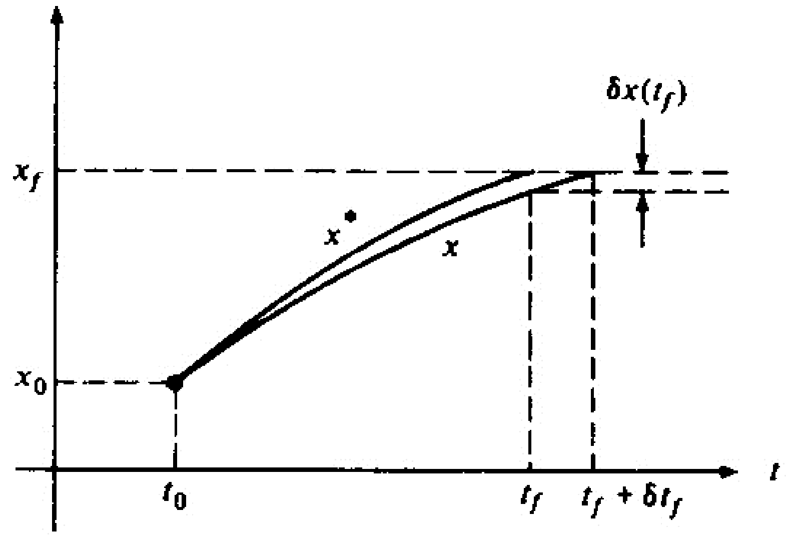
  t_0 is fixed, t_f is free, $x(t_0)$ and $x(t_f)$ are given



$$
\begin{aligned}
\Delta J &= \int_{t_0}^{t_f+\delta t_f} g(x,\dot{x},t)dt - \int_{t_0}^{t_f} g\left(x^*,\dot{x}^*,t\right)dt \\
&= \int_{t_0}^{t_f} g(x,\dot{x},t)dt + \int_{t_f}^{t_f+\delta t_f} g(x,\dot{x},t)dt - \int_{t_0}^{t_f} g\left(x^*,\dot{x}^*,t\right)dt \\
&= \int_{t_0}^{t_f} \left[g(x,\dot{x},t) - g\left(x^*,\dot{x}^*,t\right)\right]dt + \int_{t_f}^{t_f+\delta t_f} g(x,\dot{x},t)dt \\
&= \int_{t_0}^{t_f} \left[\left.\frac{\partial g}{\partial x}\right|_{x^*,\dot{x}^*} \delta x + \left.\frac{\partial g}{\partial \dot{x}}\right|_{x^*,\dot{x}^*} \delta \dot{x} + H.O.T. \right]dt + g\left(x\left(t_f\right),\dot{x}\left(t_f\right),t_f\right)\delta t_f \\
&= \int_{t_0}^{t_f} \left[\left.\frac{\partial g}{\partial x}\right|_{x^*,\dot{x}^*} \delta x + \left.\frac{\partial g}{\partial \dot{x}}\right|_{x^*,\dot{x}^*} \delta \dot{x} + H.O.T. \right]dt \\
&\quad + \left[g\left(x^*\left(t_f\right),\dot{x}^*\left(t_f\right),t_f\right) + \left.\frac{\partial g}{\partial \dot{x}}\right|_{x^*,\dot{x}^*} \delta x\left(t_f\right) + \left.\frac{\partial g}{\partial \dot{x}}\right|_{x^*,\dot{x}^*} \delta \dot{x}\left(t_f\right) + H.O.T. \right]\delta t_f
\end{aligned}
$$

$$
\left(\Uparrow \; \delta x(t)\delta t_f \in \theta\left[(\delta x)^2,\left(\delta t_f\right)^2\right],\; \delta\dot{x}\left(t_f\right)\delta t_f \in \theta\left[(\delta \dot{x})^2,\left(\delta t_f\right)^2\right]\right)
$$

$$
\begin{aligned}
\delta J &= \int_{t_0}^{t_f} \left[\left.\frac{\partial g}{\partial x}\right|_{x^*,\dot{x}^*} \delta x + \left.\frac{\partial g}{\partial \dot{x}}\right|_{x^*,\dot{x}^*} \delta \dot{x} \right]dt + g\left(x^*\left(t_f\right),\dot{x}^*\left(t_f\right),t_f\right)\delta t_f \\
&= \left.\frac{\partial g}{\partial \dot{x}}\delta x\right|_{t_0}^{t_f} + \int_{t_0}^{t_f}\left[\frac{\partial g}{\partial x} - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right)\right]\delta x\, dt + g\left(x^*,\dot{x}^*,t_f\right)\delta t_f
\end{aligned}
$$



$$-\delta x(t_f) \simeq \dot{x}(t_f)\,\delta t_f$$

$$\Uparrow \ \text{assume } \dot{x}(t_f) = \dot{x}^*(t_f)$$

$$\Rightarrow -\delta x(t_f) \simeq \dot{x}^*(t_f)\,\delta t_f$$

$$\Rightarrow \quad \delta x(t_f) \simeq -\dot{x}^*(t_f)\,\delta t_f$$

Note that $\delta x(t_f)$ depends on δt_f and is neither zero nor free.

The dependence of $\delta x(t_f)$ on δt_f must be linearly approximated because δJ consists of the first-order terms in the increment of ΔJ

$$\delta J = \left.\frac{\partial g}{\partial \dot{x}}\right|_{x^*,\dot{x}^*} \delta x(t_f) - \left.\frac{\partial g}{\partial \dot{x}}\right|_{x^*,\dot{x}^*} \delta x(t_0) + g(x^*,\dot{x}^*,t_f)\,\delta t_f + \int_{t_0}^{t_f}\left[\frac{\partial g}{\partial x} - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right)\right]_{x^*,\dot{x}^*}\delta x(t)\,dt$$

$$\left(\Uparrow \ \delta x(t_f) \simeq -\dot{x}^*(t_f)\,\delta t_f\right)$$

$$= \left[-\frac{\partial g}{\partial x}\dot{x}^*(t_f) + g\right]\delta t_f + \int_{t_0}^{t_f}\left[\frac{\partial g}{\partial x} - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right)\right]_{x^*,\dot{x}^*}\delta x(t)\,dt$$

$$= 0$$

$$\left(\Uparrow \ \delta x(t_f)\, and\ \delta t_f\ are\ arbitrary\right)$$

$$\frac{\partial g}{\partial x} - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right) = 0 \qquad \text{Euler-Lagrange equation}$$

$$g(x^*,\dot{x}^*,t_f) - \frac{\partial g}{\partial \dot{x}}(x^*,\dot{x}^*,t_f)\,\dot{x}^*(t_f) = 0 \qquad \text{Transversality Condition}$$

- Example

$$J = \int_0^{t_f}(\dot{x}^2 + x)\,dt, \quad x(0) = 0,\ x(t_f) = 1,\ t_f \text{ is free}$$

$$\Rightarrow g(x,\dot{x},t_f) = (\dot{x}^2 + x)$$

I. Euler-Lagrange equation

$$\frac{\partial g}{\partial x}-\frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right)=0$$
$$\Rightarrow -2\ddot{x}+1=0 \Rightarrow \ddot{x}=\tfrac{1}{2}$$
$$\Rightarrow x=\tfrac{1}{4}t^2+Bt+C$$

$$x(0)=0 \Rightarrow C=0$$
$$x(t_f)=\tfrac{1}{4}t_f^2+Bt_f=1$$

II. Transversality condition

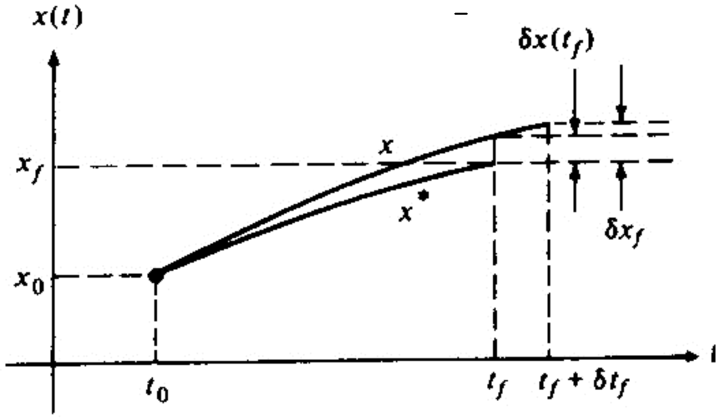
$$g-\frac{\partial g}{\partial \dot{x}}\dot{x}=0 \quad \text{at} \quad t=t_f$$
$$\Rightarrow \dot{x}^2+x-2\dot{x}^2=0 \Rightarrow x-\dot{x}^2=0$$
$$\left(\Uparrow \quad \dot{x}=\tfrac{1}{2}t+B \Rightarrow \dot{x}^2=\tfrac{1}{4}t^2+Bt+B^2\right)$$
$$\Rightarrow \left[x=\dot{x}^2\right]_{t=t_f}$$
$$\Rightarrow \tfrac{1}{4}t_f^{\,2}+Bt_f=\tfrac{1}{4}t_f^{\,2}+Bt_f+B^2$$
$$\Rightarrow B^2=0 \Rightarrow B=0$$
$$\Rightarrow \begin{cases} x^*(t)=\tfrac{1}{4}t^2 \\ x(t_f)=\tfrac{1}{4}t_f^{\,2}=1 \Rightarrow t_f=\pm 2 \Rightarrow t_f=2 \end{cases}$$

3.2.4 Problems with both the final time t_f and $x(t_f)$ free

- Find a function $x^*(t)$ (extremal) for which the following functional $J(x)$ has a relative extremum

$$J(x)=\int_{t_0}^{t_f} g(x(t),\dot{x}(t),t)\,dt$$

t_0 and $x(t_0)$ are specified, t_f and $x(t_f)$ are free



$\delta x(t_f)$: the difference at $t = t_f$

the state variation at $t = t_f$

δx_f : the difference at the end point of the two curves.

the state variation when t_f and $x(t_f)$ are free.

$\delta x(t_f) \neq \delta x_f$, in general.

$$\delta x_f \simeq \delta x(t_f) + \dot{x}(t_f)\delta t_f$$
$$\Uparrow \dot{x}(t_f) \simeq \dot{x}^*(t_f)$$
$$\delta x(t_f) \simeq \delta x_f - \dot{x}^*(t_f)\delta t_f$$

$$\begin{aligned}\Delta J &= \int_{t_0}^{t_f+\delta t_f} g(x,\dot{x},t)dt - \int_{t_0}^{t_f} g(x^*,\dot{x}^*,t)\,dt \\ &= \int_{t_0}^{t_f}\left[g(x,\dot{x},t) - g(x^*,\dot{x}^*,t)\right]dt + \int_{t_f}^{t_f+\delta t_f} g(x,\dot{x},t)dt\end{aligned}$$

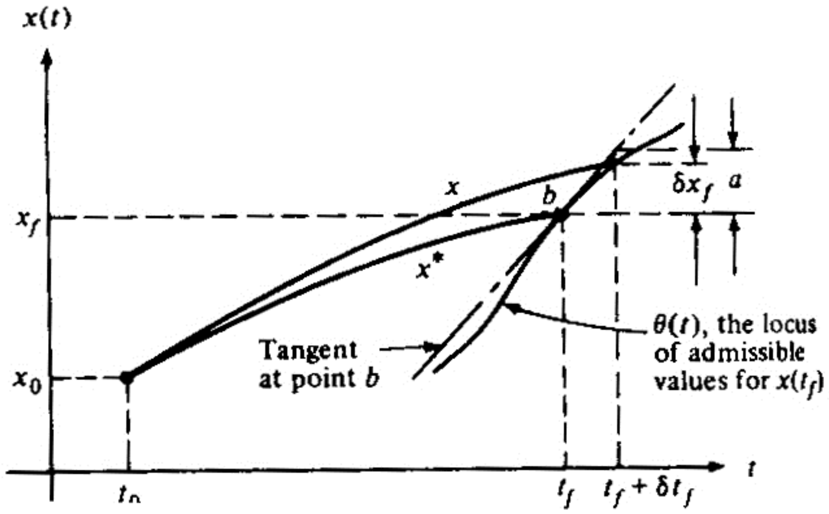
$$
\begin{aligned}
\delta J &= \int_{t_0}^{t_f}\left[\frac{\partial g}{\partial x}\delta x + \frac{\partial g}{\partial \dot{x}}\delta \dot{x}\right]dt + g\left(x^*\left(t_f\right),\dot{x}^*\left(t_f\right),t_f\right)\delta t_f \\
&= \frac{\partial g}{\partial \dot{x}}\delta x\Big|_{t_0}^{t_f} + g\left(x^*,\dot{x}^*,t_f\right)\delta t_f + \int_{t_0}^{t_f}\left[\frac{\partial g}{\partial x} - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right)\right]\delta x dt \\
&\quad \left(\Uparrow \frac{\partial g}{\partial \dot{x}}\delta x\Big|_{t_0}^{t_f} = \frac{\partial g}{\partial \dot{x}}\delta x\left(t_f\right) - \frac{\partial g}{\partial \dot{x}}\delta x\left(t_0\right) = \frac{\partial g}{\partial \dot{x}}\left(\delta x_f - \dot{x}^*\delta t_f\right)\right) \\
&= \frac{\partial g}{\partial \dot{x}}\delta x_f + \left[g - \frac{\partial g}{\partial \dot{x}}\dot{x}^*\right]\delta t_f + \int_{t_0}^{t_f}\left[\frac{\partial g}{\partial x} - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right)\right]\delta x dt \\
&= 0
\end{aligned}
$$

I. $x^*\left(t_f\right)$ and t_f^* are not related $\Rightarrow \delta x\left(t_f\right)$ and δx_f are arbitrary

$$\frac{\partial g}{\partial x} - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right) = 0 \qquad \text{Euler-Lagrange equation}$$

$$\left.\begin{aligned}
&\left.\frac{\partial g}{\partial \dot{x}}\right|_{t=t_f} = 0 \\
&\left(g - \frac{\partial g}{\partial \dot{x}}\dot{x}^*\right)_{t=t_f} = 0
\end{aligned}\right\} \Rightarrow g\left(x^*,\dot{x}^*,t_f\right) = 0 \qquad \text{Transversality Condition}$$

II. $x^*\left(t_f\right)$ and t_f^* are related by $x^*\left(t_f\right) = \theta\left(t_f\right)$



$$\left(\Downarrow \quad \dot{\theta}\left(t_f\right) \simeq \frac{\delta x_f}{\delta t_f} \Rightarrow \delta x_f \simeq \dot{\theta}\left(t_f\right)\delta t_f \right)$$

$$\begin{aligned}
\delta J &= \frac{\partial g}{\partial \dot{x}}\dot{\theta}\left(t_f\right)\delta t_f + \left[g - \frac{\partial g}{\partial \dot{x}}\dot{x}^*\left(t_f\right) \right]\delta t_f + \int_{t_0}^{t_f}\left[\frac{\partial g}{\partial x} - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right) \right]\delta x dt \\
&= \left[g - \frac{\partial g}{\partial \dot{x}}\left(\dot{x}^* - \dot{\theta}\left(t_f\right)\right) \right]\delta t_f + \int_{t_0}^{t_f}\left[\frac{\partial g}{\partial x} - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right) \right]\delta x dt \\
&= 0
\end{aligned}$$

$$\frac{\partial g}{\partial x} - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right) = 0 \qquad \text{Euler-Lagrange equation}$$

$$g + \frac{\partial g}{\partial \dot{x}}\left[\dot{\theta} - \dot{x}^*\right]\Bigg|_{x^*\left(t_f\right),\dot{x}^*\left(t_f\right),t_f} = 0 \qquad \text{Transversality Condition}$$

$$\Rightarrow g\left(x^*\left(t_f\right),\dot{x}^*\left(t_f\right),t_f\right) + \frac{\partial g}{\partial \dot{x}}\left(x^*\left(t_f\right),\dot{x}^*\left(t_f\right),t_f\right)\left[\dot{\theta}\left(t_f\right) - \dot{x}^*\left(t_f\right)\right] = 0$$

- Example (Problem 4-13, Kirk, p.181): Determine an extremal for the functional

$$J\left(x\right) = \int_0^{t_f} \sqrt{1 + \dot{x}^2\left(t\right)}dt$$

$x\left(0\right) = 2$, t_f and $x\left(t_f\right)$ are free on the curve $\theta(t_f) = -4t_f + 5$

The Euler-Lagrange equation is derived as

$$\frac{\partial g}{\partial x}-\frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right)=0, \quad g=\sqrt{1+\dot{x}^2(t)}$$

$$\Rightarrow \frac{d}{dt}\left(\frac{\dot{x}}{\left(1+\dot{x}^2\right)^{1/2}}\right)=0$$

$$\Rightarrow \frac{\ddot{x}}{\left(1+\dot{x}^2\right)^{1/2}}-\frac{1}{2}\dot{x}\frac{2\dot{x}\ddot{x}}{\left(1+\dot{x}^2\right)^{3/2}}=0$$

$$\Rightarrow \frac{\ddot{x}}{\left(1+\dot{x}^2\right)^{1/2}}\left[1-\frac{\dot{x}^2}{\left(1+\dot{x}^2\right)}\right]=0$$

$$\Rightarrow \ddot{x}(t)=0 \quad \left(\because 1+\dot{x}^2>0 \text{ and } 1-\frac{\dot{x}^2}{\left(1+\dot{x}^2\right)}\neq 0\right)$$

$$\Rightarrow x^*(t)=C_1 t+C_2 \qquad \text{(i)}$$

From the initial boundary condition,

$$x(0)=2 \quad \rightarrow \quad C_2=2$$

The Transversality Condition is

$$g+\frac{\partial g}{\partial \dot{x}}\left(\dot{\theta}-\dot{x}\right)=0$$

$$\Rightarrow \left(1+\dot{x}^2\right)^{1/2}+\frac{\dot{x}}{\left(1+\dot{x}^2\right)^{1/2}}\left[-4-\dot{x}\right]=0$$

$$\Rightarrow 1+\dot{x}^2-4\dot{x}-\dot{x}^2=0 \qquad \text{at } t=t_f \qquad \text{(ii)}$$

$$\Rightarrow -4\dot{x}+1=0$$

$$\Rightarrow \dot{x}=\frac{1}{4}$$

From (i) & (ii)

$$\dot{x}\left(t_f\right)=C_1=\tfrac{1}{4}$$

The free final time t_f is determined from

$$x\left(t_f\right)=\theta\left(t_f\right)$$

$$\Rightarrow \tfrac{1}{4}t_f+2=-4t_f+5$$

$$\Rightarrow t_f=\tfrac{12}{17}$$

Summarized, the solution is

$$x^*(t) = \tfrac{1}{4}t + 2,\ t_f = \tfrac{12}{17}$$

Summary of 3.1 and 3.2

- Definition
  I. Functional
  II. Variation
  III. Extremum/Extremal
  IV. Relative/Absolute Minimum/Maximum
  sign ΔJ = sign δJ

- Fundamental Theorem of the Calculus of Variation

  $\delta J\left(\vec{x}^*, \delta x\right) = 0$ for extremum

- Necessary Conditions for Extremum

  $$J(x) = \int_{t_0}^{t_f} g\big(x(t), \dot{x}(t), t\big)dt$$

  t_0 and $x(t_0)$ are given, and the admissible curve $x(t)$ are smooth

  $$\delta J\left(x^*, \delta x\right) = \frac{\partial g}{\partial x}\delta x_f + \left[g - \frac{\partial g}{\partial \dot{x}}\dot{x}\right]\delta t_f + \int_{t_0}^{t_f}\left[\frac{\partial g}{\partial x} - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right)\right]\delta x dt = 0$$

  Whatever type of the boundary condition is, the Euler-Lagrange equation must be satisfied:

  $$\frac{\partial g}{\partial x}\left(x^*(t), \dot{x}^*(t), t\right) - \frac{d}{dt}\left[\frac{\partial g}{\partial \dot{x}}\left(x^*(t), \dot{x}^*(t), t\right)\right] = 0$$

  Transversality Conditions

  I. t_f and $x(t_f)$ are specified:

$$\delta t_f = 0 \text{ and } \delta x_f = \delta x\left(t_f\right) = 0$$

II. t_f is specified, $x\left(t_f\right)$ is free:

$$\delta t_f = 0 \text{ and } \delta x_f = \delta x(t_f) \neq 0$$

$$\left.\frac{\partial g}{\partial \dot{x}}\right|_{t=t_f} = 0$$

III. t_f is free, $x\left(t_f\right)$ is specified:

$$\delta t_f \neq 0 \text{ and } \delta x_f = 0$$

$$\left(g - \frac{\partial g}{\partial x}\dot{x}^*\right)_{t=t_f} = 0$$

IV. t_f and $x\left(t_f\right)$ are free:

$$\delta t_f \neq 0 , \delta x_f \neq 0$$

I. $x^*\left(t_f\right)$ and $t_f^{\ *}$ are not related

$$\left.\begin{array}{l} \left.\dfrac{\partial g}{\partial \dot{x}}\right|_{t=t_f} = 0 \\ \left(g - \dfrac{\partial g}{\partial \dot{x}}\dot{x}^*\right)_{t=t_f} = 0 \end{array}\right\} \Rightarrow \left. g\right|_{t=t_f} = 0$$

II. $x^*\left(t_f\right) = \theta\left(t_f\right)$

$$\delta x_f \simeq \dot{\theta}\delta t_f$$

$$\left[g + \frac{\partial g}{\partial \dot{x}}\left(\dot{\theta} - \dot{x}^*\right)\right]_{t=t_f} = 0$$

3.3 Functionals involving Several Independent Functions

- Results so far are for the functionals of a single function x and its first derivative $\dot{x}$

- Generalization to include functionals of several independent functions and their first-derivatives results in matrix form:
  I. Regardless of the boundary conditions, the Euler-Lagrange equations are necessary conditions to be satisfied:
$$\frac{\partial g}{\partial \vec{x}}\left(\vec{x}^*(t),\dot{\vec{x}}^*(t),t\right)-\frac{d}{dt}\left[\frac{\partial g}{\partial \dot{\vec{x}}}\left(\vec{x}^*(t),\dot{\vec{x}}^*(t),t\right)\right]=\vec{0}$$
  II. The required boundary conditions are found from the following equation by making the appropriate substitutions for $\delta\vec{x}_f$ and δt_f.

$$\boxed{\left[\frac{\partial g}{\partial \dot{\vec{x}}}\left(\vec{x}^*(t),\dot{\vec{x}}^*(t),t\right)\right]^T_{t=t_f}\delta\vec{x}_f+\left[g\left(\vec{x}^*(t),\dot{\vec{x}}^*(t),t\right)-\left\{\frac{\partial g}{\partial \dot{\vec{x}}}\left(\vec{x}^*(t),\dot{\vec{x}}^*(t),t\right)\right\}^T\dot{\vec{x}}(t)\right]_{t=t_f}\delta t_f=0}$$

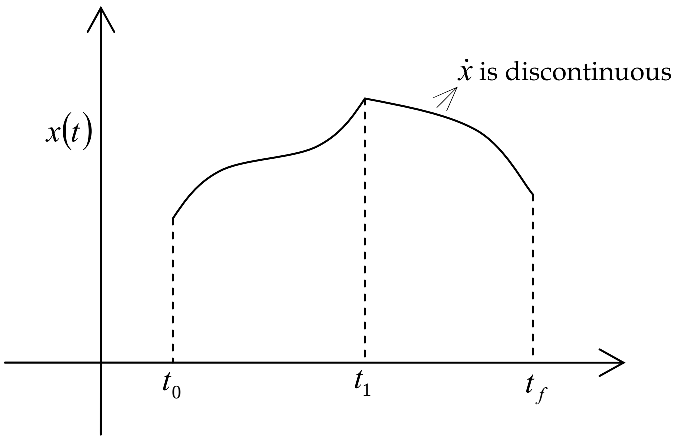
Table 4-1 DETERMINATION OF BOUNDARY-VALUE RELATIONSHIPS

Problem description	*Substitution*	*Boundary conditions*	*Remarks*
1. $\mathbf{x}(t_f)$, t_f both specified (*Problem 1*)	$\delta\mathbf{x}_f=\delta\mathbf{x}(t_f)=\mathbf{0}$ $\delta t_f=0$	$\mathbf{x}^*(t_0)=\mathbf{x}_0$ $\mathbf{x}^*(t_f)=\mathbf{x}_f$	$2n$ equations to determine $2n$ constants of integration
2. $\mathbf{x}(t_f)$ free; t_f specified (*Problem 2*)	$\delta\mathbf{x}_f=\delta\mathbf{x}(t_f)$ $\delta t_f=0$	$\mathbf{x}^*(t_0)=\mathbf{x}_0$ $\frac{\partial g}{\partial\dot{\mathbf{x}}}(\mathbf{x}^*(t_f),\dot{\mathbf{x}}^*(t_f),t_f)=\mathbf{0}$	$2n$ equations to determine $2n$ constants of integration
3. t_f free; $\mathbf{x}(t_f)$ specified (*Problem 3*)	$\delta\mathbf{x}_f=\mathbf{0}$	$\mathbf{x}^*(t_0)=\mathbf{x}_0$ $\mathbf{x}^*(t_f)=\mathbf{x}_f$ $g(\mathbf{x}^*(t_f),\dot{\mathbf{x}}^*(t_f),t_f)$ $-\left[\frac{\partial g}{\partial\dot{\mathbf{x}}}(\mathbf{x}^*(t_f),\dot{\mathbf{x}}^*(t_f),t_f)\right]^T\dot{\mathbf{x}}^*(t_f)=0$	$(2n+1)$ equations to determine $2n$ constants of integration and t_f
4. t_f, $\mathbf{x}(t_f)$ free and independent (*Problem 4*)	—	$\mathbf{x}^*(t_0)=\mathbf{x}_0$ $\frac{\partial g}{\partial\dot{\mathbf{x}}}(\mathbf{x}^*(t_f),\dot{\mathbf{x}}^*(t_f),t_f)=\mathbf{0}$ $g(\mathbf{x}^*(t_f),\dot{\mathbf{x}}^*(t_f),t_f)=0$	$(2n+1)$ equations to determine $2n$ constants of integration and t_f
5. t_f, $\mathbf{x}(t_f)$ free but related by $\mathbf{x}(t_f)=\boldsymbol{\theta}(t_f)$ (*Problem 4*)	$\delta\mathbf{x}_f=\frac{d\boldsymbol{\theta}}{dt}(t_f)\,\delta t_f$†	$\mathbf{x}^*(t_0)=\mathbf{x}_0$ $\mathbf{x}^*(t_f)=\boldsymbol{\theta}(t_f)$ $g(\mathbf{x}^*(t_f),\dot{\mathbf{x}}^*(t_f),t_f)$ $+\left[\frac{\partial g}{\partial\dot{\mathbf{x}}}(\mathbf{x}^*(t_f),\dot{\mathbf{x}}^*(t_f),t_f)\right]^T\left[\frac{d\boldsymbol{\theta}}{dt}(t_f)-\dot{\mathbf{x}}^*(t_f)\right]=0$†	$(2n+1)$ equations to determine $2n$ constants of integration and t_f

† $\frac{d\boldsymbol{\theta}}{dt}$ denotes the $n\times 1$ column vector $\left[\frac{d\theta_1}{dt}\ \ \frac{d\theta_2}{dt}\ \ \cdots\ \ \frac{d\theta_n}{dt}\right]^T$.

3.4 Piecewise-Smooth Extremals

- Piecewise Smoothness
  - When $\dot{\vec{x}}(t)$ is discontinuous, $\vec{x}(t)$ is said to have <u>corners</u>
  - While $\dot{\vec{x}}(t)$ is discontinuous at corners for a finite number of times in the interval $[t_0, t_f]$, $\vec{x}(t)$ is continuous but not smooth



$$\dot{x}^*(t_1^-) \neq \dot{x}^*(t_1^+) \text{ but } x^*(t_1^-) = x^*(t_1^+)$$

  - t_1^- and t_1^+ indicates the moment just before and after the discontinuity of $\dot{x}$, i.e., just before and after the corner

- Find a function *x*(t)* (extremal) for which the following functional *J(x)* has a relative extremum (which allows corner extremals =broken extremals)

  $$J(x) = \int_{t_0}^{t_f} g(x(t), \dot{x}(t), t)dt$$

  t_0 and t_f are fixed, $x(t_0)$ and $x(t_f)$ are given

  $$\delta J = \int_{t_0}^{t_1} \left[\frac{\partial g}{\partial x} \delta x + \frac{\partial g}{\partial \dot{x}} \delta \dot{x} \right] dt + \int_{t_1}^{t_f} \left[\frac{\partial g}{\partial x} \delta x + \frac{\partial g}{\partial \dot{x}} \delta \dot{x} \right] dt$$

  - Assume that t_1 and $x(t_1)$ are not related

$$t_0 \quad \to t_1^-, \qquad t_1^+ \quad \to t_f$$

Can apply the same method as that for the case of free t_f and $x(t_f)$:

$$\delta J = \int_{t_0}^{t_1}\left[\frac{\partial g}{\partial x}\delta x + \frac{\partial g}{\partial \dot{x}}\delta \dot{x}\right]dt - \int_{t_f}^{t_1}\left[\frac{\partial g}{\partial x}\delta x + \frac{\partial g}{\partial \dot{x}}\delta \dot{x}\right]dt$$

$\delta x_{t_1} = \delta x(t_1) + \dot{x}^*(t_1)\delta t_1$ and δt_1 are free

Recall that in the case of free t_f and $x(t_f)$,

$$\delta J = \frac{\partial g}{\partial \dot{x}}\delta x_f + \left[g - \frac{\partial g}{\partial \dot{x}}\dot{x}^*\right]\delta t_f + \int_{t_0}^{t_f}\left[\frac{\partial g}{\partial x} - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right)\right]\delta x dt = 0$$

Similary, we have

$$\begin{aligned}
\delta J &= \left[\frac{\partial g}{\partial \dot{x}}\delta x_{t_1} + \left(g - \frac{\partial g}{\partial \dot{x}}\dot{x}\right)\delta t_1\right]_{t=t_1^-} + \int_{t_0}^{t_1}\left[\frac{\partial g}{\partial x} - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right)\right]\delta x\, dt \\
&\quad - \left[\frac{\partial g}{\partial \dot{x}}\delta x_{t_1} + \left(g - \frac{\partial g}{\partial \dot{x}}\dot{x}\right)\delta t_1\right]_{t=t_1^+} + \int_{t_1}^{t_f}\left[\frac{\partial g}{\partial x} - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right)\right]\delta x\ dt \\
&= \left[\left.\frac{\partial g}{\partial \dot{x}}\right|_{t=t_1^-} - \left.\frac{\partial g}{\partial \dot{x}}\right|_{t=t_1^+}\right]\delta x_{t_1} + \left[\left(g - \frac{\partial g}{\partial \dot{x}}\dot{x}\right)_{t=t_1^-} - \left(g - \frac{\partial g}{\partial \dot{x}}\dot{x}\right)_{t=t_1^+}\right]\delta t_1 \\
&\quad + \int_{t_0}^{t_1}\left[\frac{\partial g}{\partial x} - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right)\right]\delta x\, dt + \int_{t_1}^{t_f}\left[\frac{\partial g}{\partial x} - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right)\right]\delta x\ dt \\
&= 0
\end{aligned}$$

Since t_1 and $x(t_1)$ are assumed to be not related. $\delta x(t_1)$ and δt_1 are independent and arbitrary, so each of their coefficients must vanish:

I. Euler-Lagrange equation

II. $\left.\frac{\partial g}{\partial \dot{x}}\right|_{t=t_1^-} = \left.\frac{\partial g}{\partial \dot{x}}\right|_{t=t_1^+}$

III. $g - \frac{\partial g}{\partial \dot{x}}\dot{x}\,\Big|_{t=t_1^-} = g - \frac{\partial g}{\partial \dot{x}}\dot{x}\,\Big|_{t=t_1^+}$

II and III are called Weierstrass-Erdman Corner conditions, which must hold for an extremal at t_1, i.e., at a corner

If there exists multiple corners at $t_1, t_2, \ldots, t_n$, then the conditions II and

III must be satisfied at each corner

■ Assume that $x(t_1)$ and t_1 are related by $x(t_1)=\theta(t_1)$ $\Rightarrow$

$$\delta x_{t_1} \simeq \dot{\theta}(t_1)\,\delta t_1$$

$$\begin{aligned}
\delta J &= \left[\frac{\partial g}{\partial \dot{x}}\delta x_{t_1} + \left(g - \frac{\partial g}{\partial \dot{x}}\dot{x}\right)\delta t_1\right]_{t=t_1^-} + \int_{t_0}^{t_1}\left[\frac{\partial g}{\partial x} - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right)\right]\delta x\,dt \\
&\quad - \left[\frac{\partial g}{\partial \dot{x}}\delta x_{t_1} + \left(g - \frac{\partial g}{\partial \dot{x}}\dot{x}\right)\delta t_1\right]_{t=t_1^+} + \int_{t_1}^{t_f}\left[\frac{\partial g}{\partial x} - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right)\right]\delta x\,dt \\
&\qquad \left(\Uparrow \quad \delta x_{t_1} \simeq \dot{\theta}(t_1)\,\delta t_1\right) \\
&= \left[\frac{\partial g}{\partial \dot{x}}\dot{\theta}(t_1)\delta t_1 + \left(g - \frac{\partial g}{\partial \dot{x}}\dot{x}\right)\delta t_1\right]_{t=t_1^-} + \int_{t_0}^{t_1}\left[\frac{\partial g}{\partial x} - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right)\right]\delta x\,dt \\
&\quad - \left[\frac{\partial g}{\partial \dot{x}}\dot{\theta}(t_1)\delta t_1 + \left(g - \frac{\partial g}{\partial \dot{x}}\dot{x}\right)\delta t_1\right]_{t=t_1^+} + \int_{t_1}^{t_f}\left[\frac{\partial g}{\partial x} - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right)\right]\delta x\,dt \\
&= \left[\frac{\partial g}{\partial \dot{x}}\dot{\theta}\Big|_{t=t_1^-} + \left(g - \frac{\partial g}{\partial \dot{x}}\dot{x}\right)_{t=t_1^-} - \frac{\partial g}{\partial \dot{x}}\dot{\theta}\Big|_{t=t_1^+} - \left(g - \frac{\partial g}{\partial \dot{x}}\dot{x}\right)_{t=t_1^+}\right]\delta t_1 \\
&\quad + \int_{t_0}^{t_1}\left[\frac{\partial g}{\partial x} - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right)\right]\delta x\,dt + \int_{t_1}^{t_f}\left[\frac{\partial g}{\partial x} - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right)\right]\delta x\,dt \\
&= 0
\end{aligned}$$

The Weierstrass-Erdman condition is

$$\left[\frac{\partial g}{\partial \dot{x}}\left(\dot{\theta} - \dot{x}\right) + g\right]_{t=t_1^-} = \left[\frac{\partial g}{\partial \dot{x}}\left(\dot{\theta} - \dot{x}\right) + g\right]_{t=t_1^+}$$

Extension into the case of functionals of several independent functions is straightforward

- Theorem: Condition for No Corner

If $g_{\dot{x}\dot{x}}(x,\dot{x},t) \neq 0, \forall\, x,\dot{x},t$ in the domain of *g*, then any $x(t)$ with a corner can not extremize *J*.

Proof is given by using Mean Value Theorem[2]

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[2] Mean Value Theorem:
If $f(x)$ is smooth and its derivative exists in the interval (a,b), then there exists a number

Since $g_{\dot{x}\dot{x}}\left(x,\dot{x},t\right)$ exists, (i.e., $g_{\dot{x}}(x,\dot{x},t)$ is differentiable), $g_{\dot{x}}(x,\dot{x},t)$ is continuous with respect to $\dot{x}$, i.e.,

$$g_{\dot{x}}(x,\dot{x}(t_1^+),t_1^+) - g_{\dot{x}}(x,\dot{x}(t_1^-),t_1^-) = 0$$

Let $f \equiv g_{\dot{x}}$, $\mathrm{a} = \dot{x}(t_1^-)$, $b = \dot{x}(t_1^+)$

Then,

$$f\left(b\right) = g_{\dot{x}}(x,\dot{x}(t_1^+),t_1^+)$$
$$f\left(\mathrm{a}\right) = g_{\dot{x}}(x,\dot{x}(t_1^-),t_1^-)$$

$$f\,'(\cdot) = g_{\dot{x}\dot{x}}$$

By the Mean Value Theorem, there exists $\xi \equiv \dot{x}^*\left(t_1^-\right) + \alpha\left[\dot{x}^*\left(t_1^+\right) - \dot{x}^*\left(t_1^-\right)\right]$ such that

$$f(b) - f(a) = f\,'(\xi)(b-a)$$
$$\Rightarrow g_{\dot{x}}(x,\dot{x}(t_1^+),t_1^+) - g_{\dot{x}}(x,\dot{x}(t_1^-),t_1^-) = g_{\dot{x}\dot{x}}\left(x^*\left(t_1\right),\xi,t_1\right)\left[\dot{x}^*\left(t_1^+\right) - \dot{x}^*\left(t_1^-\right)\right]$$

If $g_{\dot{x}\dot{x}} \neq 0$ at any value of $x,\dot{x},t$, then $\dot{x}^*\left(t_1^+\right) - \dot{x}^*\left(t_1^-\right) = 0$,

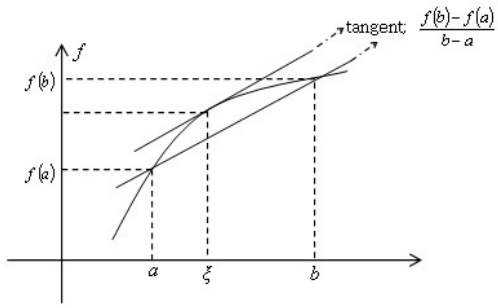
Which concludes that there exists no corner

- Example: Given

$$J = \int_{t_0}^{t_f}\left[a\dot{x}^2 + bx\dot{x} + cx^2\right]dt \qquad a \neq 0, \quad x(0) = x_0, \quad x(t_f) = x_f,$$

ξ in (a,b) such that

$$f(b) = f(a) + f'(\xi)(b-a)$$
$$where\ \xi = a + (b-a)\alpha, \qquad 0 < \alpha < 1$$



show that $x^*(t)$ can have no corner

$$\frac{\partial^2 g}{\partial \dot{x}^2} = \frac{\partial}{\partial \dot{x}}\left[2a\dot{x} + bx\right] = 2a \neq 0 \qquad (\because a \neq 0) \ \Rightarrow \ \text{no corner!}$$

3.5 Constrained Extrema

3.5.1 Point/Path Constraints

- Determine a set of necessary conditions for a function $x^*(t)$ to be an extremal for a functional

  $$J\left(x(t)\right) = \int_{t_0}^{t_f} g\left(x(t), \dot{x}(t), t\right) dt$$

  subject to $f(x(t), t) = 0$ (Holonomic Constraint)

  $x(t_0)$ and $x(t_f)$ are (assumed to be) given for simplicity of analysis

  The path constraint would be present if the admissible curves were required to lie on a specified surface

  Form the augmented functional by using the Lagrange Multiplier to adjoin the path constraints:

  $$J_{\mathrm{a}}\left(x(t)\right) = \int_{t_0}^{t_f} \left(g(x(t), \dot{x}(t), t) + \Lambda(t) f(x(t), t)\right) dt$$

  Λ; optimal weight function $\Lambda(t)$

  $f = 0 \Rightarrow J_a = J$

$$\begin{aligned}\delta J_a(x) &= \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x}\delta x + \frac{\partial g}{\partial \dot{x}}\delta \dot{x} + \Lambda \frac{\partial f}{\partial x}\delta x + \delta\Lambda \cdot f\right] dt \\ &= \frac{\partial g}{\partial \dot{x}}\delta x\Big|_{t_0}^{t_f} + \int_{t_0}^{t_f}\left[\left[\frac{\partial g}{\partial x} - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right) + \Lambda\frac{\partial f}{\partial x}\right]\delta x + \delta\Lambda \cdot f\right] dt \\ &= 0\end{aligned}$$

$$\frac{\partial g}{\partial x} - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right) + \Lambda\frac{\partial f}{\partial x} = 0 \qquad \text{Euler-Lagrange equation}$$

$f(x(t),t)=0$ (Given) Path Constraints

3.5.2 Differential Equation Constraints (Nonholonomic)

- Determine a set of necessary conditions for a function $x^*(t)$ to be an extremal for a functional

$$J(x)=\int_{t_0}^{t_f} g(x,\dot{x},t)dt$$

subject to $f(x(t),\dot{x}(t),t)=0$ (Nonholonomic Constraint)

$x(t_0)$ and $x(t_f)$ are given

Similarly,

$$J_{\mathrm{a}}(x(t))=\int_{t_0}^{t_f}\left(g(x(t),\dot{x}(t),t)+\Lambda(t)f(x(t),\dot{x}(t),t)\right)dt$$

$\Lambda(t)$; optimal weight function

$$\delta J_a(x)=\int_{t_0}^{t_f}\left[\frac{\partial g}{\partial x}\delta x+\frac{\partial g}{\partial \dot{x}}\delta\dot{x}+\Lambda\frac{\partial f}{\partial x}\delta x+\Lambda\frac{\partial f}{\partial \dot{x}}\delta\dot{x}+\delta\Lambda\cdot f\right]dt$$

$$=\frac{\partial g}{\partial \dot{x}}\delta x\Big|_{t_0}^{t_f}+\Lambda\frac{\partial f}{\partial \dot{x}}\delta x\Big|_{t_0}^{t_f}+\int_{t_0}^{t_f}\left\{\left[\frac{\partial g}{\partial x}-\frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right)+\Lambda\frac{\partial f}{\partial x}-\frac{d}{dt}\left(\Lambda\frac{\partial f}{\partial \dot{x}}\right)\right]\delta x+\delta\Lambda\cdot f\right\}dt$$

$$=0$$

$$\frac{\partial g}{\partial x}-\frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}+\Lambda\frac{\partial f}{\partial \dot{x}}\right)+\Lambda\frac{\partial f}{\partial x}=0$$ Euler-Lagrange equation

$f(x(t),\dot{x}(t),t)=0$ (Given) diff. equation constraints

3.5.3 Isoperimetric Constraints

- Determine a set of necessary conditions for a function $x^*(t)$ to be an extremal for a functional

$$J=\int_{t_0}^{t_f} g(x,\dot{x},t)dt$$

subject to $\int_{t_0}^{t_f} F(x(t),\dot{x}(t),t)dt = c$

$x(t_0)$ and $x(t_f)$ are given

c is a specified constant, which can represent total fuel or energy available in control problems

The isoperimetric constraint can be put into the form of differential equation constraints by defining a new variable

$$z(t) = \int_{t_0}^{t} F\left(x(t),\dot{x}(t),t\right)dt$$

$z(t_0) = 0,\quad z(t_f) = c$ $\qquad z$; additional state variable

$$\dot{z}(t) = F(x(t),\dot{x}(t),t)$$

$F(x(t),\dot{x}(t),t) - \dot{z}(t) \equiv f(x(t),\dot{x}(t),t,\dot{z}) = 0$, f; a new differential constraint

Augmented functional

$$J_a(x) = \int_{t_0}^{t_f} (g + \Lambda \cdot f)dt$$

$$\begin{aligned}
\delta J_a(x) &= \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x}\delta x + \frac{\partial g}{\partial \dot{x}}\delta \dot{x} + \Lambda \frac{\partial f}{\partial x}\delta x + \Lambda \frac{\partial f}{\partial \dot{x}}\delta \dot{x} + \Lambda \frac{\partial f}{\partial \dot{z}}\delta \dot{z} + \delta\Lambda \cdot f \right] dt \\
&= \frac{\partial g}{\partial \dot{x}}\delta x\Big|_{t_0}^{t_f} + \Lambda \frac{\partial f}{\partial \dot{x}}\delta x\Big|_{t_0}^{t_f} + \Lambda \frac{\partial f}{\partial \dot{z}}\delta z\Big|_{t_0}^{t_f} \\
&\quad + \int_{t_0}^{t_f} \left\{ \left[\frac{\partial g}{\partial x} - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}} + \Lambda \frac{\partial f}{\partial \dot{x}} \right) + \Lambda \frac{\partial f}{\partial x} \right] \delta x - \frac{d}{dt}\left(\Lambda \frac{\partial f}{\partial \dot{z}} \right)\delta z + \delta\Lambda \cdot f \right\} dt \\
&= 0
\end{aligned}$$

$$\frac{\partial g}{\partial x} - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}} + \Lambda \frac{\partial f}{\partial \dot{x}} \right) + \Lambda \frac{\partial f}{\partial x} = 0 \qquad \text{Euler-Lagrange equation}$$

$$\frac{d}{dt}\left(\Lambda \frac{\partial f}{\partial \dot{z}} \right) = 0 \xrightarrow[f = F - \dot{z} \to f_{\dot{z}} = -1]{} \frac{d}{dt}(-\Lambda) = 0 \quad \Rightarrow \Lambda(t) = const.$$

$\Rightarrow$ Lagrange multiplier is constant

$$f = 0 \quad \Rightarrow \dot{z}(t) = F\left(x(t),\dot{x}(t),t\right),\quad z(t_0) = 0,\quad z(t_f) = c$$

3.6 Second Variation and Sufficient Conditions

- The increment (total variation) of a functional

$$\Delta J \equiv J(\vec{x}+\delta\vec{x}) - J(\vec{x})$$

$$\Delta J(\vec{x}, \delta\vec{x}) = \underbrace{\delta J(\vec{x}, \delta\vec{x})}_{\text{linear term in } \delta\vec{x}} + g_1(\vec{x}, \delta\vec{x}) \cdot \|\delta\vec{x}\|$$

If $\lim_{\|\delta\vec{x}\|\to 0} \{g_1(\vec{x}, \delta\vec{x})\} = 0,$

then, J is said to be differentiable on $\vec{x}$ and δJ is the first variation of J evaluated for the function $\vec{x}$, δJ is the linear approximation to the difference in the functional J caused by $\vec{x}$, $\vec{x}+\delta\vec{x}$.

- The increment of a functional

$$\begin{aligned}\Delta J(\vec{x}, \delta\vec{x}) &= J(\vec{x}+\delta\vec{x}) - J(\vec{x}) \\ &= J(\vec{x}) + \left.\frac{\partial J}{\partial \vec{x}}\right|_{\vec{x}} \delta\vec{x} + \frac{1}{2}\left.\frac{\partial^2 J}{\partial \vec{x}^2}\right|_{\vec{x}} (\delta\vec{x})^2 + \cdots - J(\vec{x}) \\ &= \left.\frac{\partial J}{\partial \vec{x}}\right|_{\vec{x}} \delta\vec{x} + \frac{1}{2}\left.\frac{\partial^2 J}{\partial \vec{x}^2}\right|_{\vec{x}} (\delta\vec{x})^2 + H.O.T\end{aligned}$$

$$\begin{aligned} where \quad \left.\frac{\partial J}{\partial \vec{x}}\right|_{\vec{x}} \delta\vec{x} &\equiv \delta J(\vec{x}, \delta\vec{x});\ first\ variation \\ \frac{1}{2}\left.\frac{\partial^2 J}{\partial \vec{x}^2}\right|_{\vec{x}} (\delta\vec{x})^2 &\equiv \delta^2 J(\vec{x}, \delta\vec{x});\ second\ variation\end{aligned}$$

- The total variation is given as

$$\Delta J(\vec{x}, \delta\vec{x}) = \underbrace{\delta J(\vec{x}, \delta\vec{x})}_{linear\ term\ in\ \delta\vec{x}} + \delta^2 J(\vec{x}, \delta\vec{x}) + g_2(\vec{x}, \delta\vec{x}) \cdot \|\delta\vec{x}\|^2$$

$\delta^2 J$ is a quadratic functional.

If $\lim_{\|\delta\vec{x}\|\to 0} \{g_2(\vec{x}, \delta\vec{x})\} = 0$, then $J(\vec{x})$ is twice differentiable.

It is tacitly assumed that functionals are twice differentialble, and their second variation is uniquely defined.

- Theorem 1: A <u>necessary condition</u> for the functional $J\left(\vec{x}^*(t)\right)$ to have a minimum/maximum is that

  $$\delta^2 J(\vec{x}^*(t)) \geq 0 \ \ / \ \ \delta^2 J(\vec{x}^*(t)) \leq 0 \qquad \text{(i)}$$

  for all admissible $\delta\vec{x}(t)$

  Proof: By definition of the total variation, we have

  $$\Delta J\left(\vec{x}, \delta\vec{x}\right) = \underbrace{\delta J\left(\vec{x}, \delta\vec{x}\right)}_{\text{linear term in } \delta\vec{x}} + \delta^2 J\left(\vec{x}, \delta\vec{x}\right) + g_2\left(\vec{x}, \delta\vec{x}\right) \cdot \left\|\delta\vec{x}\right\|^2$$

  $$= \delta J\left(\vec{x}, \delta\vec{x}\right) + \delta^2 J\left(\vec{x}, \delta\vec{x}\right) + \varepsilon\left\|\delta\vec{x}\right\|^2 \qquad \text{(ii)}$$

  $$\textit{where } \ \varepsilon \equiv g_2\left(\vec{x}, \delta\vec{x}\right), \ \textit{and } \ \varepsilon \to 0 \ \textit{as} \ \left\|\delta\vec{x}\right\| \to 0$$

  According to the fundamental theorem, $\delta J(\vec{x}^*(t), \delta\vec{x}(t)) = 0$ for all admissible $\delta\vec{x}(t)$

  Hence Equation (ii) becomes

  $$\Delta J\left(\vec{x}^*, \delta\vec{x}\right) = \delta^2 J\left(\vec{x}^*, \delta\vec{x}\right) + \varepsilon\left\|\delta\vec{x}\right\|^2 \qquad \text{(iii)}$$

  Thus, for sufficiently small $\left\|\delta\vec{x}\right\|$,

  $$\operatorname{sign} \Delta J\left(\vec{x}^*, \delta\vec{x}\right) = \operatorname{sign} \delta^2 J\left(\vec{x}^*, \delta\vec{x}\right)$$

  Now suppose that $\delta^2 J\left(\vec{x}^*, \delta\vec{x}_0\right) < 0$ for some admissible $\delta\vec{x}_0$.

  Then, however small $\alpha \neq 0$, might be, we have

  $$\delta^2 J\left(\vec{x}^*, \alpha\delta\vec{x}_0\right) = \alpha^2 \delta^2 J\left(\vec{x}^*, \delta\vec{x}_0\right) < 0$$

  Hence, Equation (iii) can be made negative for some arbitrarily small $\left\|\delta\vec{x}\right\|$

  This contradicts the hypothesis that $J\left(\vec{x}^*(t)\right)$ is a minimum, i.e.,

  $$\Delta J\left(\vec{x}, \delta\vec{x}\right) = J\left(\vec{x}^* + \delta\vec{x}\right) - J\left(\vec{x}^*\right) \geq 0$$

  for all sufficiently small $\left\|\delta\vec{x}\right\|$, which completes the proof

Again, $\delta^2 J(x(t)) \geq 0$ is a necessary but not sufficient condition for the functional $J\left(\vec{x}^*(t)\right)$ to be minimum

- Discussing the sufficient condition requires to introduce the concept of 'Strong Positiveness':
  A quadratic functional $\delta^2 J(\delta x(t))$ defined on some normed linear space $\Re$ is strongly positive if there exists a constant k >0 such that

  $$\delta^2 J(\delta x(t)) \geq k\|\delta x\|^2 \text{ for all } \delta\vec{x}(t)$$

- Theorem 2: A <u>sufficient condition</u> for a functional $J\left(\vec{x}^*(t)\right)$ to be a minimum is that
  - the first variation $\delta J(\vec{x}^*(t), \delta\vec{x}(t)) = 0$
  - the second variation $\delta^2 J(\vec{x}^*(t), \delta\vec{x}(t))$ is <u>strongly positive</u>

Proof: We have $\delta J(\vec{x}^*, \delta\vec{x})$ =0 for all admissible $\delta\vec{x}$, and hence

$$\Delta J\left(\vec{x}^*, \delta\vec{x}\right) = \delta^2 J\left(\vec{x}^*, \delta\vec{x}\right) + \varepsilon\|\delta\vec{x}\|^2 \text{ where } \varepsilon \to 0 \text{ } as \text{ } \|\delta\vec{x}\| \to 0.$$

Moreover, from the strong positiveness of $\delta^2 J(\vec{x}^*, \delta\vec{x})$,

$$\delta^2 J(\vec{x}^*, \delta\vec{x}) \geq k\|\delta x\|^2 \text{ where } k = \text{constant} > 0.$$

Thus, for sufficiently small ε_1, $|\varepsilon| < k/2$ if $\|\delta x\| < \varepsilon_1$,

then, it follows that

$$\Delta J\left(\delta\vec{x}\right) = \delta^2 J\left(\delta\vec{x}\right) + \varepsilon\|\delta\vec{x}\|^2 > \frac{1}{2}k\|\delta x\|^2 > 0,$$

Which concludes that $J\left(\vec{x}^*(t)\right)$ is a minimum [Q.E.D.]

- $J = \int_{t_0}^{t_f} g(x, \dot{x}, t)dt$ $\quad$ $x(t_0), x(t_f)$ are given

$$\Delta J = \int_{t_0}^{t_f} g(x,\dot{x},t)\,dt - \int_{t_0}^{t_f} g\left(x^*,\dot{x}^*,t\right)dt$$

$$= \int_{t_0}^{t_f} \left\{ \begin{array}{l} g\left(x^*,\dot{x}^*,t\right) + \frac{\partial g}{\partial x}\Big|_{x^*,\dot{x}^*}\delta x + \frac{\partial g}{\partial \dot{x}}\Big|_{x^*,\dot{x}^*}\delta \dot{x} \\ + \frac{1}{2}\left[\frac{\partial^2 g}{\partial x^2}\Big|_{x^*,\dot{x}^*}(\delta x)^2 + 2\frac{\partial^2 g}{\partial x \partial \dot{x}}\Big|_{x^*,\dot{x}^*}\delta x \delta \dot{x} + \frac{\partial^2 g}{\partial \dot{x}^2}\Big|_{x^*,\dot{x}^*}(\delta \dot{x})^2 \right] + H.O.T \end{array} \right\} dt$$

$$- \int_{t_0}^{t_f} g\left(x^*,\dot{x}^*,t\right)$$

$$\delta^2 J = \int_{t_0}^{t_f} \tfrac{1}{2}\left[g_{xx}(\delta x)^2 + 2g_{x\dot{x}}\delta x \delta \dot{x} + g_{\dot{x}\dot{x}}(\delta \dot{x})^2 \right] dt$$

A (weak) necessary condition for a minimum / maximum is that $\delta^2 J \ge 0$ / $\delta^2 J \le 0$ for all admissible variations of $\delta x(t) \in C^1$, i.e.,

$$\boxed{\begin{array}{ll} \delta J = 0,\quad \delta^2 J \ge 0 \rightarrow & \text{minimum J} \\ \delta J = 0,\quad \delta^2 J \le 0 \rightarrow & \text{maximum J} \end{array}}$$

- Accessory Problem

Note that

$$\delta^2 J = \int_{t_0}^{t_f} w\left(\delta x, \delta \dot{x}, t\right) dt,\quad w = \frac{1}{2}\left[g_{xx}(\delta x)^2 + 2g_{x\dot{x}}\delta x \delta \dot{x} + g_{\dot{x}\dot{x}}(\delta \dot{x})^2 \right] \quad (1)$$

where $\delta x(t_0) = \delta x(t_f) = 0$ and $\delta x(t)$ is arbitrary

The problem of extremizing $\delta^2 J$ with respect to δx is said to be the <u>Accessory Problem</u>

Necessary condition

$$\frac{d}{dt}\left(\frac{\partial w}{\partial(\delta \dot{x})} \right) - \frac{\partial w}{\partial(\delta x)} = 0 \qquad \text{Jacobi differential equation}$$

Jacobi differential equation is the Euler-Lagrange equation for Accessory Problem

Let the solution to the Jacobi differential equation be $\delta v(t)$

Then, with $\delta x \to \delta v$ and $\delta \dot{x} \to \delta \dot{v}$,

$$\frac{d}{dt}\left[g_{\dot{x}\dot{x}}\delta\dot{v} + g_{x\dot{x}}\delta v\right] - \left(g_{x\dot{x}}\delta\dot{v} + g_{xx}\delta v\right) = 0$$

$$\Rightarrow \frac{d}{dt}\left(g_{\dot{x}\dot{x}}\delta\dot{v}\right) + \frac{d}{dt}\left(g_{x\dot{x}}\right)\delta v + g_{x\dot{x}}\delta\dot{v} - \left(g_{x\dot{x}}\delta\dot{v} + g_{xx}\delta v\right) = 0$$

$$\Rightarrow \frac{d}{dt}\left(g_{\dot{x}\dot{x}}\delta\dot{v}\right) + \left[\frac{d}{dt}\left(g_{x\dot{x}}\right) - g_{xx}\right]\delta v = 0 \qquad (2)$$

From (1),

$$2\delta^2 J = \int_{t_0}^{t_f}\left[g_{\dot{x}\dot{x}}\left(\delta\dot{x}\right)^2 + g_{x\dot{x}}\frac{d}{dt}\left(\delta x\right)^2 + g_{xx}\left(\delta x\right)^2\right]dt$$

$$= g_{\dot{x}\dot{x}}\delta\dot{x}\delta x\Big|_{t_0}^{t_f} + g_{x\dot{x}}\left(\delta x\right)^2\Big|_{t_0}^{t_f} - \int_{t_0}^{t_f}\left[\frac{d}{dt}\left(g_{\dot{x}\dot{x}}\delta\dot{x}\right)\delta x - \left(g_{xx} - \frac{d}{dt}g_{x\dot{x}}\right)\left(\delta x\right)^2\right]dt$$

$$\Rightarrow \delta^2 J = -\frac{1}{2}\int_{t_0}^{t_f}\delta x\left[\frac{d}{dt}\left(g_{\dot{x}\dot{x}}\delta\dot{x}\right) - \left(g_{xx} - \frac{d}{dt}g_{x\dot{x}}\right)\delta x\right]dt$$

$$\Rightarrow \delta^2 J = \frac{1}{2}\int_{t_0}^{t_f}\delta x\left[\left(g_{xx} - \frac{d}{dt}g_{x\dot{x}}\right)\delta x - \frac{d}{dt}\left(g_{\dot{x}\dot{x}}\delta\dot{x}\right)\right]dt \qquad (3)$$

If $\delta x \to \delta v$, then $\delta^2 J = 0$ since (2) is satisfied.

(3) can be multiplied and divided by $\delta v\left(t\right)$:

$$\delta^2 J = \frac{1}{2}\int_{t_0}^{t_f}\frac{\delta x}{\delta v}\left[\left(g_{xx} - \frac{d}{dt}g_{x\dot{x}}\right)\delta x\delta v - \frac{d}{dt}\left(g_{\dot{x}\dot{x}}\delta\dot{x}\right)\delta v\right]dt$$

$$\text{where}\left(g_{xx} - \frac{d}{dt}g_{x\dot{x}}\right)\delta x\delta v = \left[\left(g_{xx} - \frac{d}{dt}g_{x\dot{x}}\right)\delta v\right]\delta x = \left[\frac{d}{dt}\left(g_{\dot{x}\dot{x}}\delta\dot{v}\right)\right]\delta x \qquad \text{from } (2)$$

$$= \frac{1}{2}\int_{t_0}^{t_f}\frac{\delta x}{\delta v}\left[\frac{d}{dt}\left(g_{\dot{x}\dot{x}}\delta\dot{v}\right)\delta x - \frac{d}{dt}\left(g_{\dot{x}\dot{x}}\delta\dot{x}\right)\delta v\right]dt$$

$$= \frac{1}{2}\int_{t_0}^{t_f}\frac{\delta x}{\delta v}\left[\frac{d}{dt}\left(g_{\dot{x}\dot{x}}\delta\dot{v}\right)\delta x + \left(g_{\dot{x}\dot{x}}\delta\dot{v}\right)\frac{d}{dt}\delta x - \frac{d}{dt}\left(g_{\dot{x}\dot{x}}\delta\dot{x}\right)\delta v - \left(g_{\dot{x}\dot{x}}\delta\dot{x}\right)\frac{d}{dt}\delta v\right]dt$$

$$= \frac{1}{2}\int_{t_0}^{t_f} \frac{\delta x}{\delta v}\left[\frac{d}{dt}\left(g_{\dot{x}\dot{x}}\delta\dot{v}\delta x - g_{\dot{x}\dot{x}}\delta\dot{x}\delta v\right)\right]dt$$

$$= \frac{1}{2}\int_{t_0}^{t_f} \frac{\delta x}{\delta v}\frac{d}{dt}\left[g_{\dot{x}\dot{x}}\left(\delta\dot{v}\delta x - \delta\dot{x}\delta v\right)\right]dt$$

$$= \frac{1}{2}\frac{\delta x}{\delta v}g_{\dot{x}\dot{x}}\left(\delta\dot{v}\delta x - \delta\dot{x}\delta v\right)\Big|_{t_0}^{t_f} - \frac{1}{2}\int_{t_0}^{t_f} g_{\dot{x}\dot{x}}\left(\delta\dot{v}\delta x - \delta\dot{x}\delta v\right)\frac{d}{dt}\left(\frac{\delta x}{\delta v}\right)dt$$

$$\text{where } \left(\delta\dot{v}\delta x - \delta\dot{x}\delta v\right)\frac{d}{dt}\left(\frac{\delta x}{\delta v}\right) = \left(\delta\dot{v}\delta x - \delta\dot{x}\delta v\right)\frac{\delta\dot{x}\delta v - \delta x\delta\dot{v}}{\left(\delta v\right)^2} = -\left(\frac{\delta x\delta\dot{v} - \delta\dot{x}\delta v}{\delta v}\right)^2 = -\left(\delta\dot{x} - \frac{\delta x\delta\dot{v}}{\delta v}\right)^2$$

$$= \frac{1}{2}\int_{t_0}^{t_f} g_{\dot{x}\dot{x}}\left(\delta\dot{x} - \frac{\delta x\delta\dot{v}}{\delta v}\right)^2 dt$$

Recall that the necessary conditions for minimum/maximum requires that

$\delta^2 J \geq 0$ / $\delta^2 J \leq 0$ for admissible $\delta x(t) \neq 0$ and $\delta x(t) \neq \delta v(t)$.

$\Rightarrow g_{\dot{x}\dot{x}} \geq 0$ / $g_{\dot{x}\dot{x}} \leq 0$ along every point on the extremal

Legenre derived for the scalar case, Clebsch extended into the vector case, which is now known as the Legenre – Clebsch condition

There might be finite number of times when $\left[\delta\dot{x} - \dfrac{\delta x\delta\dot{v}}{\delta v}\right]$ might not be finite.

- <u>Sufficient Conditions for Optimality</u>

I. if an extremal $\vec{x}^*(t)$ satisfies the Euler – Lagrange equation

II. if $g(x,\dot{x},t)$ satisfies Legendre-Clebsch condition

$g_{\dot{x}\dot{x}} \geq 0$ / $g_{\dot{x}\dot{x}} \leq 0$

III. if there does not exist any conjugate point,

then, the extremal $\vec{x}^*(t)$ minimizes/maximizes the given performance index

4 The Variational Approach to Optimal Control Problems

- **Preview** from Sussman, H. J. and Willems, J. C., 300 Yeasrs of Optimal Control: from the Brachstochrone to the Maximum Principle, *IEEE Control Systems*, June 1997, pp. 12-44
  - The Optimal Control Theory was born in 1950~1960's in the former Soviet Union, with the work on the Pontryagin Maximum Principle by L. S. Pontryagin and his group
  - Some believe that this new theory was no more than a minor adding to the classical Calculus of Variations, essentially involving the incorporation of inequality constraints
  - The Optimal Control is significantly richer/broader than the Calculus of Variations, and differs in some fundamental ways

  - The Calculus of Variations deals mainly with the following standard form

    Minimize $J = \int_a^b L\left(q(t), \dot{q}(t), t\right) dt$

    subject to $q(a) = q_a, \quad q(b) = q_b$ (1)

    Or, equivalently

    Minimize $J = \int_a^b L\left(q(t), u(t), t\right) dt$

    subject to $\begin{array}{l} \dot{q}(t) = u(t),\ t \in [a,b] \\ q(a) = q_a, \quad q(b) = q_b \end{array}$ (2)

    - The distinctive feature is that the minimization takes place in the space of **all** curves
    - Nothing interesting happens on the level of the set of curves under consideration
    - All the nontrivial features of the problem arise because of the the Lagrangian *L*

- Optimal Control Problems, by contrast, involve a minimization over **a set of curves** which is determined by some dynamical constraints and usually represented by a differential equation with some choice of the **control function** $u(t)$

$$\dot{q}(t) = f(q(t), u(t), t) \tag{3}$$

That is, in the Optimal Control Problem, there are at least two objects of interests:

  - **the dynamics *f*** to be satisfied
  - **the cost functional *J*** to be minimized

  - In particular, the Optimal Control theory contains, at the opposite extreme from the calculus of variations, problems where the Lagragian *L* is *1*, i.e. completely trivial, and therefore all the interesting action occurs because of the dynamics *f*
(Minimum Time Problem, i.e., the integral *J* of (2) with *L=1*)

- The above contrast does not imply that the Calculus of Variations do not cannot deal with the Minimum Time Problem:
  - Bernoulli's Brachstochrone Problem is a true minimum time problem of the kind that is studied today in optimal control theory
  - Brachstochrone problem is the first one ever to deal with a dynamical behavior and explicitly ask for the optimal selection of a path
  - A large part of the subsequent history of the calculus of variations can be intrepretated as the search for **simplest** and **most general** statement of the **necessary conditions for optimality**
  - This statement is provided by **the Maximum Principle of Optimal Control Theory**

4.1 Necessary Conditions for Optimal Control

4.1.1 $\left(\vec{x}\left(t_f\right), t_f\right)$ Specified

- Minimize the Performance Measure

$$J(\vec{u}) = h(\vec{x}(t_f), t_f) + \int_{t_0}^{t_f} g(\vec{x}(t), \vec{u}(t), t)dt$$

subject to the dynamical constraints

$$\dot{\vec{x}}(t) = \vec{f}(\vec{x}, \vec{u}, t)$$

while satisfying the boundary conditions

$$\vec{x}(t_0) = \vec{x}_0, \quad \vec{x}(t_f) = \vec{x}_f, \quad t_f \text{ fixed}$$

where $\vec{x} \in R^n$, $\vec{u} \in R^m$, $m \le n$, $\vec{u} \in R^m$
$\vec{x}$ and $\vec{u}$ are unconstrained

  - Alternate Statement: Find an admissible control $\vec{u}^*(t)$ that causes the system

$$\dot{\vec{x}}(t) = \vec{f}(\vec{x}, \vec{u}, t)$$

to follow an admissible trajectory $\vec{x}^*(t)$ that extrimizes the performance measure

$$J(\vec{u}) = h(\vec{x}(t_f), t_f) + \int_{t_0}^{t_f} g(\vec{x}(t), \vec{u}(t), t)dt$$

$$\vec{x}(t_0) = \vec{x}_0, \quad \vec{x}(t_f) = \vec{x}_f, \quad t_f \text{ fixed}$$

$\vec{x} \in R^n$, $\vec{u} \in R^m$, $m \le n$
$\vec{x}$ and $\vec{u}$ are unconstrained

Derivations of Necessary Conditions begins by defining the Augmented Functional:

$$J_a = h(\vec{x}(t_f), t_f) + \int_{t_0}^{t_f} \left[g + \vec{\lambda}^T (\vec{f} - \dot{\vec{x}}) \right] dt, \quad \vec{\lambda} \in R^n \text{ Lagrange Multiplier}$$

Take the first variation yields

$$\delta J_a = \frac{\partial h}{\partial \vec{x}(t_f)} \delta \vec{x}(t_f) + \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial \vec{x}} \delta \vec{x} + \frac{\partial g}{\partial \vec{u}} \delta \vec{u} + \vec{\lambda}^T \left\{ \frac{\partial \vec{f}}{\partial \vec{x}} \delta \vec{x} + \frac{\partial \vec{f}}{\partial \vec{u}} \delta \vec{u} - \delta \dot{\vec{x}} \right\} + \delta \vec{\lambda}^T \left(\vec{f} - \dot{\vec{x}} \right) \right] dt$$

$$= -\vec{\lambda}^T \delta \vec{x} \Big|_{t_0}^{t_f} + \int_{t_0}^{t_f} \left[\left(\frac{\partial g}{\partial \vec{x}} + \vec{\lambda}^T \frac{\partial \vec{f}}{\partial \vec{x}} + \dot{\vec{\lambda}}^T \right) \delta \vec{x} + \left(\frac{\partial g}{\partial \vec{u}} + \vec{\lambda}^T \frac{\partial \vec{f}}{\partial \vec{u}} \right) \delta \vec{u} + \left(\vec{f} - \dot{\vec{x}} \right)^T \delta \vec{\lambda} \right] dt$$

$$= 0$$

from which the first order necessary conditions are obtained as

$$\dot{\vec{x}}(t) = \vec{f}\left(\vec{x}, \vec{u}, t\right), \quad \vec{x}\left(t_0\right) = \vec{x}_0, \quad \vec{x}\left(t_f\right) = \vec{x}_f$$

$$\dot{\vec{\lambda}}(t) = -\left[\frac{\partial g}{\partial \vec{x}}\left(\vec{x}, \vec{u}, t\right) + \vec{\lambda}^T(t) \frac{\partial \vec{f}}{\partial \vec{x}}\left(\vec{x}, \vec{u}, t\right) \right]$$

$$\vec{0} = \frac{\partial g}{\partial \vec{u}}\left(\vec{x}, \vec{u}, t\right) + \vec{\lambda}^T(t) \frac{\partial \vec{f}}{\partial \vec{u}}\left(\vec{x}, \vec{u}, t\right)$$

$\Rightarrow 2n$ differential equations for $\left(\vec{x}, \vec{\lambda}\right)$ with $2n$ boundary conditions $\left(\vec{x}\left(t_0\right),\ \vec{x}\left(t_f\right)\right)$

$\Rightarrow$Two Point Boundary Value Problem (TPBVP)

$\Rightarrow 2n+m$ equations for $2n+m$ unknowns

If we define Hamitonian as

$$H\left(\vec{x}(t), \vec{\lambda}(t), \vec{u}(t), t\right) \equiv g\left(\vec{x}(t), \vec{u}(t), t\right) + \vec{\lambda}^T(t) \vec{f}\left(\vec{x}(t), \vec{u}(t), t\right)$$

The first order necessary conditions are compactly expressed as

$$\dot{\vec{x}}(t) = \frac{\partial H}{\partial \vec{\lambda}} = \vec{f}, \quad \vec{x}\left(t_0\right) = \vec{x}_0, \quad \vec{x}\left(t_f\right) = \vec{x}_f \qquad \text{State Equation}$$

$$\dot{\vec{\lambda}}(t) = -\frac{\partial H}{\partial \vec{x}} \qquad \text{Costate (Adjoint) Equation}$$

$$\vec{0} = \frac{\partial H}{\partial \vec{u}} \qquad \text{Optimality Condition}$$

- Example: Minimize

$$J = \tfrac{1}{2} \int_0^1 \left(x^2(t) + u^2(t) \right) dt$$

subject to the dynamics and boundary conditions:

$$\dot{x}(t) = u(t), \quad x(0) = 1, \ x(1) = 0$$

Define Hamiltonian as

$$H = g + \vec{\lambda}^T \vec{f} = \tfrac{1}{2}\left(x^2 + u^2\right) + \lambda u$$

Then, the first order necessary conditions are

$$\left.\begin{aligned} \dot{x} &= \frac{\partial H}{\partial \vec{\lambda}} \\ \dot{\lambda} &= -\frac{\partial H}{\partial \vec{x}} \\ 0 &= \frac{\partial H}{\partial \vec{u}} \end{aligned}\right\} \Rightarrow \begin{cases} \dot{x} = u \\ \dot{\lambda} = -x \\ u = -\lambda \end{cases}$$

$$\Rightarrow \ddot{x} = \dot{u} = -\dot{\lambda} = x$$

$$\Rightarrow x = c_1 e^t + c_2 e^{-t}$$

$$\left.\begin{aligned} x(0) &= 1 \\ x(1) &= 0 \end{aligned}\right\} \Rightarrow \begin{cases} c_1 + c_2 = 1 \\ c_1 e + \frac{c_2}{e} = 0 \end{cases} \Rightarrow \begin{cases} c_1 = \frac{1}{1-e^2} \\ c_2 = \frac{-e^2}{1-e^2} \end{cases}$$

$$\Rightarrow \begin{cases} x^*(t) = \frac{1}{1-e^2} e^t - \frac{e^2}{1-e^2} e^{-t} \\ \lambda^*(t) = -\dot{x}^*(t) \\ u^*(t) = \dot{x}^*(t) = \frac{1}{1-e^2} e^t + \frac{e^2}{1-e^2} e^{-t} \end{cases}$$

4.1.2 $\vec{x}(t_f)$ Free and t_f Specified

- Minimize the Performance Measure

$$J(\vec{u}) = h\left(\vec{x}(t_f), t_f\right) + \int_{t_0}^{t_f} g\left(\vec{x}(t), \vec{u}(t), t\right) dt$$

subject to the dynamical constraints

$$\dot{\vec{x}}(t) = \vec{f}\left(\vec{x}, \vec{u}, t\right)$$

while satisfying the boundary conditions

$$\vec{x}(t_0) = \vec{x}_0, \quad \vec{x}(t_f) = free, \quad t_f \text{ specified}$$

where $\vec{x} \in R^n$, $\vec{u} \in R^m$, $m \le n$
$\vec{x}$ and $\vec{u}$ are unconstrained

Define Hamiltonian as

$$H \equiv g + \vec{\lambda}^T \vec{f}$$

$$J_a = h\left(\vec{x}(t_f), t_f \right) + \int_{t_0}^{t_f} \left[H - \vec{\lambda}^T \dot{\vec{x}} \right] dt$$

$$\begin{aligned}
\delta J_a &= \frac{\partial h}{\partial \vec{x}(t_f)} \delta\vec{x}(t_f) + \int_{t_0}^{t_f} \left[\frac{\partial H}{\partial \vec{u}} \delta\vec{u} + \frac{\partial H}{\partial \vec{x}} \delta\vec{x} + \frac{\partial H}{\partial \vec{\lambda}} \delta\vec{\lambda} - \delta\vec{\lambda}^T \dot{\vec{x}} - \vec{\lambda}^T \delta\dot{\vec{x}} \right] dt \\
&= \left[\frac{\partial h}{\partial \vec{x}(t_f)} - \vec{\lambda}^T(t_f) \right] \delta\vec{x}(t_f) + \int_{t_0}^{t_f} \left[\frac{\partial H}{\partial \vec{u}} \delta\vec{u} + \left(\frac{\partial H}{\partial \vec{x}} + \dot{\vec{\lambda}}^T \right) \delta\vec{x} + \delta\vec{\lambda}^T \left(\frac{\partial H}{\partial \vec{\lambda}}^T - \dot{\vec{x}} \right) \right] dt \\
&= 0
\end{aligned}$$

$$\dot{\vec{x}}(t) = \frac{\partial H}{\partial \vec{\lambda}} = \vec{f}, \quad \vec{x}(t_0) = \vec{x}_0$$

$$\dot{\vec{\lambda}}(t) = -\frac{\partial H}{\partial \vec{x}}, \qquad \vec{\lambda}(t_f) = \left. \frac{\partial h}{\partial \vec{x}} \right|_{t=t_f} \quad \text{Transversality Condition (TVC)}$$

$$\vec{0} = \frac{\partial H}{\partial \vec{u}}$$

$\Rightarrow 2n$ differential equations for $(\vec{x}, \vec{\lambda})$ with $2n$ boundary conditions $\left(\vec{x}(t_0),\ \vec{\lambda}(t_f) \right)$

$\Rightarrow$Two Point Boundary Value Problem (TPBVP)

$\Rightarrow$ $2n+m$ equations for $2n+m$ unknowns

In general, if a part of terminal boundary conditions are specified such that

$$\vec{x}_i(t_f) = \vec{x}_f(given), \qquad i = 1, 2, \cdots, k$$

$$\vec{x}_j(t_f) = free, \qquad j = k+1, k+2, \cdots, n$$

Then the transversality Condition is given as

$$\vec{\lambda}_j(t_f) = \left.\frac{\partial h}{\partial \vec{x}_j}\right|_{t=t_f}$$

Again, ultimately yields the TPBVP for *2n* differential equations with *2n* split boundary conditions

4.1.3 Optimal Control Problem for General Boundary Conditions

- Minimize the Performance Measure

$$J(\vec{u}) = h(\vec{x}(t_f), t_f) + \int_{t_0}^{t_f} g(\vec{x}(t), \vec{u}(t), t)dt$$

subject to the dynamical constraints

$$\dot{\vec{x}}(t) = \vec{f}(\vec{x}, \vec{u}, t)$$

while satisfying the boundary conditions

$$\vec{x}(t_0) = \vec{x}_0, \quad \vec{m}(\vec{x}(t_f), t_f) = 0, \quad t_f \text{ free}$$

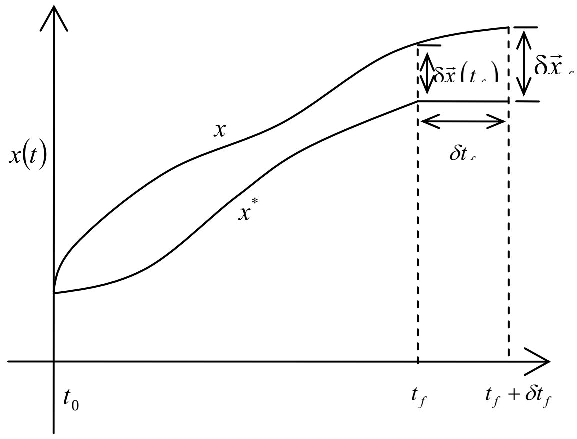
where $\vec{x} \in R^n$, $\vec{u} \in R^m$, $\vec{m} \in R^k$, $m \leq n$, $k \leq n$
$\vec{x}$ and $\vec{u}$ are unconstrained

Define Hamiltonian as

$$H \equiv g + \vec{\lambda}^T \vec{f}$$

$$J_a = h(\vec{x}(t_f), t_f) + \vec{v}^T(t)\vec{m}(\vec{x}(t_f), t_f) + \int_{t_0}^{t_f} \left(H(\vec{x}(t), \vec{\lambda}(t), \vec{u}(t), t) - \vec{\lambda}(t)^T \vec{f}(\vec{x}(t), \vec{u}(t), t)\right) dt$$

$$\begin{aligned}\delta J_a = & \frac{\partial h}{\partial \vec{x}(t_f)}\delta\vec{x}_f + \frac{\partial h}{\partial t_f}\delta t_f + \vec{v}^T \frac{\partial \vec{m}}{\partial \vec{x}(t_f)}\delta\vec{x}_f + \vec{v}^T \frac{\partial \vec{m}}{\partial t_f}\delta t_f + \delta\vec{v}^T \vec{m}(\vec{x}(t_f), t_f) \\ & + \int_{t_0}^{t_f}\left[\frac{\partial H}{\partial \vec{u}}\delta\vec{u} + \frac{\partial H}{\partial \vec{x}}\delta\vec{x} + \frac{\partial H}{\partial \vec{\lambda}}\delta\vec{\lambda} - \vec{\lambda}^T\delta\dot{\vec{x}} - \delta\vec{\lambda}^T\dot{\vec{x}}\right]dt \\ & + \int_{t_f}^{t_f+\delta t_f}\left(H - \vec{\lambda}\dot{\vec{x}}\right)dt \qquad \left(\Leftarrow \int_{t_f}^{t_f+\delta t_f}\left(H - \vec{\lambda}\dot{\vec{x}}\right)dt \simeq \left.\left(H - \vec{\lambda}\dot{\vec{x}}\right)\right|_{t_f^*}\delta t_f\right)\end{aligned}$$



$$\delta\vec{x}\left(t_f\right)=\delta\vec{x}_f-\dot{\vec{x}}^*(t_f^*)\delta t_f$$

$$\begin{aligned}
\delta J_a = & \frac{\partial h}{\partial\vec{x}\left(t_f\right)}\delta\vec{x}_f+\frac{\partial h}{\partial t_f}\delta t_f+\vec{v}^T\frac{\partial\vec{m}}{\partial\vec{x}\left(t_f\right)}\delta\vec{x}_f+\vec{v}^T\frac{\partial\vec{m}}{\partial t_f}\delta t_f-\vec{\lambda}^T\delta\vec{x}\left(t_f\right) \\
& +\int_{t_0}^{t_f}\left[\frac{\partial H}{\partial\vec{u}}\delta\vec{u}+\frac{\partial H}{\partial\vec{x}}\delta\vec{x}+\dot{\vec{\lambda}}^T\delta\vec{x}+\delta\vec{\lambda}^T\left(\frac{\partial H}{\partial\vec{\lambda}}^{\mathrm{T}}-\dot{\vec{x}}\right)\right]dt \\
& +\left[H\left(t_f^*\right)-\vec{\lambda}^T\left(t_f^*\right)\dot{\vec{x}}\left(t_f^*\right)\right]\delta t_f \\
= & \left[\frac{\partial \mathrm{h}}{\partial\vec{\mathrm{x}}\left(\mathrm{t_f}\right)}+\nu^{\mathrm{T}}\frac{\partial\vec{\mathrm{m}}}{\partial\vec{\mathrm{x}}\left(\mathrm{t_f}\right)}-\vec{\lambda}^{\mathrm{T}}\left(\mathrm{t_f}\right)\right]\delta\vec{\mathrm{x}}_{\mathrm{f}} \\
& +\left[\frac{\partial\mathrm{h}}{\partial\mathrm{t_f}}+\vec{\mathrm{v}}^{\mathrm{T}}\frac{\partial\vec{\mathrm{m}}}{\partial\mathrm{t_f}}+\vec{\lambda}^{\mathrm{T}}\left(\mathrm{t_f}\right)\dot{\vec{\mathrm{x}}}\left(\mathrm{t_f}\right)+\mathrm{H}\left(\mathrm{t_f}\right)-\vec{\lambda}^{\mathrm{T}}\left(\mathrm{t_f}\right)\dot{\vec{\mathrm{x}}}\left(\mathrm{t_f}\right)\right]\delta\mathrm{t_f} \\
& +\int_{\mathrm{t_0}}^{\mathrm{t_f}}\left[\frac{\partial\mathrm{H}}{\partial\vec{\mathrm{u}}}\delta\vec{\mathrm{u}}+\left(\frac{\partial\mathrm{H}}{\partial\vec{\mathrm{x}}}+\dot{\vec{\lambda}}^T\right)\delta\vec{\mathrm{x}}+\delta\vec{\lambda}^{\mathrm{T}}\left(\frac{\partial\mathrm{H}}{\partial\vec{\lambda}}^{\mathrm{T}}-\dot{\vec{\mathrm{x}}}\right)\right]\mathrm{dt} \\
= & \, 0
\end{aligned}$$

1st Order Necessary Conditions

$$\dot{\vec{x}}=\frac{\partial H}{\partial\vec{\lambda}}^T=\vec{f},\ \vec{x}\left(t_0\right)=\vec{x}_0$$ State Equations

$$\dot{\vec{\lambda}}=-\frac{\partial H}{\partial\vec{x}}^T,\qquad \vec{\lambda}\left(t_f\right)=\left[\frac{\partial h}{\partial\vec{x}\left(t_f\right)}+\vec{v}^T\frac{\partial\vec{m}}{\partial\vec{x}\left(t_f\right)}\right]_{t=t_f}$$

Costate (Adjoint) Equations, TVC

$$\vec{0}=\frac{\partial H}{\partial\vec{u}}$$ Optimality Conditions

$$0=H\left(t_f\right)+\left[\frac{\partial h}{\partial t_f}+\vec{v}^T\frac{\partial \vec{m}}{\partial t_f}\right]_{t=t_f} \qquad \text{Transversality Condition for } t_f$$

$\Rightarrow 2n$ differential equations for $\left(\vec{x},\vec{\lambda}\right)$ with $2n$ boundary conditions $\left(\vec{x}\left(t_0\right),\ \vec{\lambda}\left(t_f\right)\right)$

$\Rightarrow$Two Point Boundary Value Problem (TPBVP)

$\Rightarrow 2n+m+k+1$ equations for $2n+m+k+1$ unknowns

Table 5-1: Summary of Boundary Conditions for Optimal Control Problem

Problem	Description	Substitution	Boundary –condition equations	Remark
t_f fixed	1. $\mathbf{x}(t_f)=\mathbf{x}_f$ specified final state	$\delta\mathbf{x}_f=\delta\mathbf{x}(t_f)=\mathbf{0}$ $\delta t_f=0$	$\mathbf{x}^*(t_0)=\mathbf{x}_0$ $\mathbf{x}^*(t_f)=\mathbf{x}_f$	$2n$ equations to determine $2n$ constraints of integration
	2. $\mathbf{x}(t_f)$ free	$\delta\mathbf{x}_f=\delta\mathbf{x}(t_f)$ $\delta t_f=0$	$\mathbf{x}^*(t_0)=\mathbf{x}_0$ $\frac{\partial h}{\partial \mathbf{x}}(\mathbf{x}^*(t_f))-\Lambda^*(t_f)=\mathbf{0}$	$2n$ equations to determine $2n$ constraints of integration
	3. $\mathbf{x}(t_f)$ on the surface $\boldsymbol{m}\left(\boldsymbol{x}(t)\right)=0$	$\delta\mathbf{x}_f=\delta\mathbf{x}(t_f)$ $\delta t_f=0$	$\mathbf{x}^*(t_0)=\mathbf{x}_0$ $\frac{\partial h}{\partial \mathbf{x}}(\mathbf{x}^*(t_f))-\Lambda^*(t_f)=\sum_{i=1}^{k}\nu_i\left[\frac{\partial m_i}{\partial \mathbf{x}}(\mathbf{x}^*(t_f))\right]$ $\mathbf{m}(\mathbf{x}^*(t_f))=\mathbf{0}$	$(2n+k)$ equations to determine $2n$ constraints of integration and the variables $\nu_1,\ldots,\nu_k$
t_f free	4. $\mathbf{x}(t_f)=\mathbf{x}_f$ specified final state	$\delta\mathbf{x}_f=\mathbf{0}$	$\mathbf{x}^*(t_0)=\mathbf{x}_0$ $\mathbf{x}^*(t_f)=\mathbf{x}_f$ $\mathrm{H}\left(\mathrm{x}^*(t_f),\mathrm{u}(t_f),\Lambda(t_f),t_f\right)+\frac{\partial h}{\partial t}\left(\mathrm{x}^*(t_f),t_f\right)=0$	$(2n+1)$ equations to determine $2n$ constraints of integration t_f
	5. $\mathbf{x}(t_f)$ free		$\mathbf{x}^*(t_0)=\mathbf{x}_0$ $\frac{\partial h}{\partial \mathbf{x}}(\mathbf{x}^*(t_f))-\Lambda^*(t_f)=\mathbf{0}$ $\mathrm{H}\left(\mathrm{x}^*(t_f),\mathrm{u}^*(t_f),\Lambda^*(t_f),t_f\right)+\frac{\partial h}{\partial t}\left(\mathrm{x}^*(t_f),t_f\right)=0$	$(2n+1)$ equations to determine $2n$ constraints of integration t_f
	6. $\mathbf{x}(t_f)$ on the moving $\boldsymbol{\theta}(t)$	$\delta\mathbf{x}_f=\frac{d\theta}{dt}(t_f)\delta t_f$	$\mathbf{x}^*(t_0)=\mathbf{x}_0$ $\mathbf{x}^*(t_f)=\boldsymbol{\theta}(t_f)$	$(2n+1)$ equations to determine $2n$ constraints of integration t_f

			$H\left(x^*\left(t_f\right),u^*\left(t_f\right),\Lambda^*\left(t_f\right),t_f\right)+\frac{\partial h}{\partial t}\left(x^*\left(t_f\right),t_f\right)$ $+\left[\frac{\partial h}{\partial \mathbf{x}}\left(x^*\left(t_f\right),t_f\right)-\Lambda^*\left(t_f\right)\right]^T\frac{d\boldsymbol{\theta}}{dt}\left(t_f\right)=0$	
	7. $\mathbf{x}(t_f)$ on the surface $\mathbf{m}(\mathbf{x}(t))=\mathbf{0}$		$\mathbf{x}^*(t_0)=\mathbf{x}_0$ $\frac{\partial h}{\partial \mathbf{x}}(\mathbf{x}^*(t_f))-\Lambda^*(t_f)=\sum_{i=1}^{k}\nu_i\left[\frac{\partial m_i}{\partial \mathbf{x}}(\mathbf{x}^*(t_f))\right]$ $\mathbf{m}(\mathbf{x}(t_f))=\mathbf{0}$ $H\left(x^*\left(t_f\right),u^*\left(t_f\right),\Lambda^*\left(t_f\right),t_f\right)+\frac{\partial h}{\partial t}\left(x^*\left(t_f\right),t_f\right)=0$	$(2n+k+1)$ equations to determine $2n$ constraints of integration, the variables $\nu_1,\ldots,\nu_k$, and t_f
	8. $\mathbf{x}(t_f)$ on the surface $\mathbf{m}(\mathbf{x}(t),t)=\mathbf{0}$		$\mathbf{x}^*(t_0)=\mathbf{x}_0$ $\frac{\partial h}{\partial \mathbf{x}}(\mathbf{x}^*(t_f))-\Lambda^*(t_f)=\sum_{i=1}^{k}\nu_i\left[\frac{\partial m_i}{\partial \mathbf{x}}(\mathbf{x}^*(t_f),t_f)\right]$ $\mathbf{m}(\mathbf{x}(t_f),t_f)=\mathbf{0}$ $H\left(x^*\left(t_f\right),u^*\left(t_f\right),\Lambda^*\left(t_f\right),t_f\right)+\frac{\partial h}{\partial t}\left(x^*\left(t_f\right),t_f\right)$ $=\sum_{i=1}^{k}\nu_i\left[\frac{\partial m_i}{\partial t}(x^*\left(t_f\right),t_f\right]$	$(2n+k+1)$ equations to determine $2n$ constraints of integration, the variables $\nu_1,\ldots,\nu_k$, and t_f

4.1.4 Stationarity of Hamiltonian

- If $\vec{u}(t)$ is extremum (candidate optimal), then the time derivative of the Hamiltonian is

$$\begin{aligned}\frac{dH}{dt}&=\frac{\partial H}{\partial t}+\frac{\partial H}{\partial \vec{x}}\frac{d\vec{x}}{dt}+\frac{\partial H}{\partial \vec{\lambda}}\frac{d\vec{\lambda}}{dt}+\frac{\partial H}{\partial \vec{u}}\frac{d\vec{u}}{dt}\\&=\frac{\partial H}{\partial t}+\frac{\partial H}{\partial \vec{x}}\frac{\partial H}{\partial \vec{\lambda}}-\frac{\partial H}{\partial \vec{\lambda}}\frac{\partial H}{\partial \vec{x}}+0\\&=\frac{\partial H}{\partial t}\end{aligned}$$

  Now if f and g are not explicit functions of time t, then neither is H, which leads to

$$\frac{dH}{dt}=\frac{\partial H}{\partial t}\equiv 0$$

- Further, if t_f is free, $h=0$ and $\vec{m}=\vec{0}$, then the transversality condition gives

$$\begin{cases} H\left(t_f\right) = -\left[\dfrac{\partial h}{\partial t_f} + \vec{v}^T \dfrac{\partial \vec{m}}{\partial t_f}\right]_{t=t_f} = 0 \\ \dfrac{\mathrm{dH}}{\mathrm{dt}} \equiv 0 \Rightarrow H(t) \equiv constant \end{cases}$$

$$\Rightarrow \mathrm{H(t)} \equiv 0$$

4.1.5 Implication of Initial Costate

- Consider minimization of the Performance Measure

  $$J\left(x\right) = \int_{t_0}^{t_f} g\left(\vec{x}, \vec{u}, t\right) dt$$

  subject to

  $$\dot{\vec{x}}(t) = \vec{f}(\vec{x}, \vec{u}, t)$$

  $\vec{x}(t_0)$; unspecified, t_0 is fixed

  $\vec{x}(t_f)$; specified, t_f is fixed

  Define Hamiltonian as

  $$H \equiv g + \vec{\lambda}^T \vec{f}$$

  $$J_a = \int_{t_0}^{t_f} \left(H - \vec{\lambda}^T \dot{\vec{x}}\right) dt$$

  $$\begin{aligned} \delta J_a &= \int_{t_0}^{t_f} \left[\frac{\partial H}{\partial \vec{u}} \delta \vec{u} + \frac{\partial H}{\partial \vec{x}} \delta \vec{x} + \frac{\partial H}{\partial \vec{\lambda}} \delta \vec{\lambda} - \dot{\vec{\lambda}} \delta \dot{\vec{x}} - \dot{\vec{x}} \delta \vec{\lambda}\right] dt \\ &= -\vec{\lambda} \delta \vec{x} \Big|_{t_0}^{t_f} + \int_{t_0}^{t_f} \left[\frac{\partial H}{\partial \vec{u}} \delta \vec{u} + \left(\frac{\partial H}{\partial \vec{x}} + \dot{\vec{\lambda}}\right) \delta \vec{x} + \delta \vec{\lambda} \left(\frac{\partial H}{\partial \vec{\lambda}} - \dot{\vec{x}}\right)\right] dt \\ &= \vec{\lambda}\left(t_0\right) \delta \vec{x}\left(t_0\right) + \int_{t_0}^{t_f} \left[\frac{\partial H}{\partial \vec{u}} \delta \vec{u} + \left(\frac{\partial H}{\partial \vec{x}} + \dot{\vec{\lambda}}\right) \delta \vec{x} + \delta \vec{\lambda} \left(\frac{\partial H}{\partial \vec{\lambda}} - \dot{\vec{x}}\right)\right] dt \\ &= 0 \end{aligned}$$

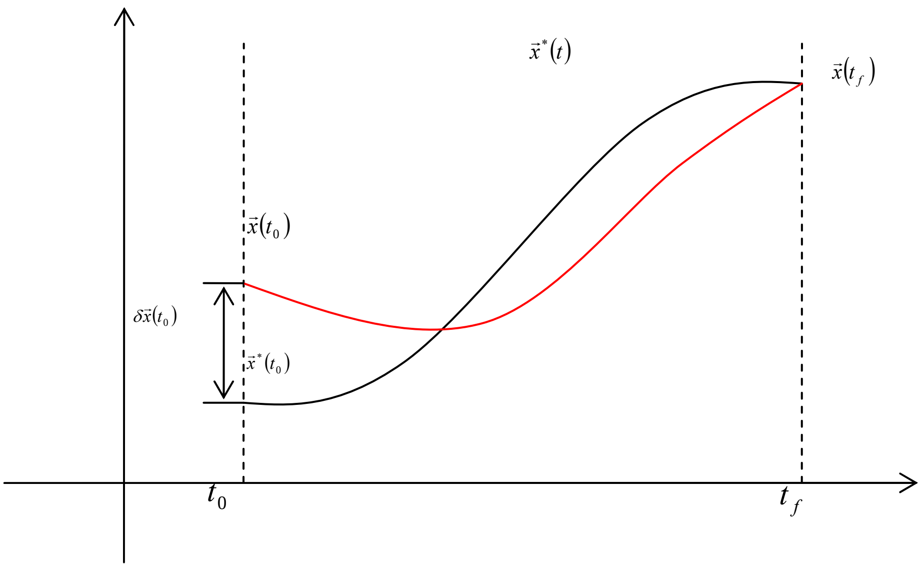
$$\Rightarrow \begin{cases} \dot{\vec{x}} = \dfrac{\partial H}{\partial \vec{\lambda}} \\ \dot{\vec{\lambda}} = -\dfrac{\partial H}{\partial \vec{x}} \\ \vec{0} = \dfrac{\partial H}{\partial \vec{u}} \end{cases}$$

$\Rightarrow \delta J_a = \vec{\lambda}(t_0)\delta\vec{x}(t_0)$ Transversality Condition

$\delta J = \delta J_a$ since all equations are satisfied.

$$\Rightarrow \vec{\lambda}(t_0) = \frac{\partial J}{\partial \vec{x}(t_0)}$$

$\vec{\lambda}(t_0)$ implies the sensitivity of the cost with respect to initial conditions.



$$\begin{aligned} &\vec{x}^*(t_0) &&\Rightarrow \quad J^* \\ &\vec{x}^*(t_0) + \delta\vec{x}(t_0) &&\Rightarrow \quad J^* + \delta J \end{aligned}$$

4.2 Linear Quadratic Problem

- For an optimal control problem posed for Linear State Equations and Quadratic Performance Indices, the optimal controller can be obtained in the form of Linear Full-State Feedback
  - Many dynamical systems of interest can be approximated as linear
  - Many meaningful performance index can be formed as quadratic

4.2.1 Finite Horizon Terminal Controller: $\vec{x}(t_f)$ Free, t_f Specified

- Minimize the performance index

$$J = \frac{1}{2}\vec{x}^T(t_f)L\vec{x}(t_f) + \frac{1}{2}\int_{t_0}^{t_f}\left[\vec{x}^T(t)Q(t)\vec{x}(t) + \vec{u}^T(t)R(t)\vec{u}(t)\right]dt$$

  subject to the dynamical system

$$\dot{\vec{x}}(t) = A(t)\vec{x}(t) + B(t)\vec{u}(t), \quad \vec{x}(t_0) = \vec{x}_0, \quad \vec{x}(t_f) = \mathit{free}, \quad t_f \text{ is fixed}$$

  where

  $\vec{x} \in R^n, \vec{u} \in R^m, A \in R^{n\times n}, B \in R^{m\times m}$

  $L \in R^{n\times n}$ and $Q \in R^{n\times n}$ are symmetric positive semi-definite[3]

  $R \in R^{m\times m}$ is symmetric positive definite

  - The quadratic performance index implies that it is desired to maintain the states close to the origin without an excessive expenditure of control effort
  - Assumption: the state and control are unbounded

  Define Hamiltonian as

[3] A symmetric matrix $A \in \mathbf{R}^{n\times n}$ is positive definite and positive semi-definite if $\vec{x}^T A\vec{x} > 0$ and $\vec{x}^T A\vec{x} \ge 0$, respectively, for all non-zero $x \in \mathbf{R}^n$.

$$H = \frac{1}{2}\vec{x}^T Q\,\vec{x} + \frac{1}{2}\vec{u}^T R\,\vec{u} + \vec{\lambda}^T\left(A\vec{x} + B\vec{u}\right)$$

$$\dot{\vec{x}} = \frac{\partial H}{\partial \vec{\lambda}} = A\vec{x} + B\vec{u}, \qquad \vec{x}\left(t_0\right) = \vec{x}_0$$

$$\dot{\vec{\lambda}} = -\frac{\partial H}{\partial \vec{x}} = -Q\vec{x} - A^T\vec{\lambda}, \quad \vec{\lambda}\left(t_f\right) = L\vec{x}\left(t_f\right) \qquad \left(\Leftarrow \vec{\lambda}\left(t_f\right) = \left.\frac{\partial h}{\partial \vec{x}}\right|_{t=t_f}\right)$$

$$\vec{0} = \frac{\partial H}{\partial \vec{u}} \quad \Rightarrow R\,\vec{u} + B^T\vec{\lambda} = 0$$

$$\Rightarrow \vec{u} = -R^{-1}B^T\vec{\lambda}, R > 0$$

Substituting the Optimality Condition into Hamitonian, State/Costate equations yields

$$H = \frac{1}{2}\begin{bmatrix}\vec{x}\\ \vec{\lambda}\end{bmatrix}^T \begin{bmatrix} Q & A^T \\ A & -BR^{-1}B^T\end{bmatrix}\begin{bmatrix}\vec{x}\\ \vec{\lambda}\end{bmatrix}$$

$$\begin{bmatrix}\dot{\vec{x}}\\ \dot{\vec{\lambda}}\end{bmatrix} = \begin{bmatrix} A & -BR^{-1}B^T \\ -Q & -A^T\end{bmatrix}\begin{bmatrix}\vec{x}\\ \vec{\lambda}\end{bmatrix}, \begin{bmatrix}\vec{x}\left(t_0\right) = \vec{x}_0 \\ \vec{\lambda}\left(t_f\right) = L\vec{x}\left(t_f\right)\end{bmatrix}$$

- Solution by Riccati Transformation

Define $\vec{\lambda}\left(t\right)$ as a linear function of states $\vec{x}\left(t\right)$:

$$\vec{\lambda}\left(t\right) = K\left(t\right)\vec{x}\left(t\right)$$

$$\Rightarrow \dot{\vec{\lambda}} = \dot{K}\vec{x} + K\dot{\vec{x}}$$

$$\left(\Uparrow \begin{cases}\dot{\vec{\lambda}} = -Q\vec{x} - A^T\vec{\lambda} = -Q\vec{x} - A^TK\vec{x} \\ \dot{\vec{x}} = A\vec{x} + B\vec{u} = A\vec{x} - BR^{-1}B^T\vec{\lambda}\end{cases}\right)$$

$$= \dot{K}\vec{x} + K\left(A\vec{x} - BR^{-1}B^T\vec{\Lambda}\right)$$

$$\Rightarrow \vec{0} = \left[\dot{K} + KA - KBR^{-1}B^TK + A^TK + Q\right]\vec{x}$$

Since this equation must be satisfied at all times, we obtain the following differential Riccati Equation:

$$\dot{K}(t) = -A^T(t)K(t) - K(t)A(t) - Q(t) + K(t)B(t)R^{-1}(t)B^T(t)K(t) \qquad (1)$$

Boundary Condition is obtained from the Transversality Condition:

$$\begin{cases} \vec{\lambda}(t_f) = L\vec{x}(t_f) \\ \vec{\lambda}(t_f) = K(t_f)\vec{x}(t_f) \end{cases} \Rightarrow K(t_f) = L \tag{2}$$

Solving (1) with (2) requires backward integration from t_f to t_0

- As the set of matrix (Q,R,L) are all symmetric, K is an $n \times n$ symmetric matrix, and thus the Riccati equations is not n^2 , but $n(n+1)/2$ first-order differential equations.
- Can be integrated numerically and be stored

Then, the linear, time-varying optimal control law can be obtained as

$$\vec{u}(t) = -R^{-1}B^T\vec{\lambda}(t) = -R^{-1}B^T K(t)\vec{x}(t)$$

Optimal Trajectory is obtained from

$$\dot{\vec{x}} = A\vec{x} + B\vec{u} = \left(A - BR^{-1}B^T K\right)\vec{x}, \quad \vec{x}(t_0) = \vec{x}_0$$

- Solution by State Transition Matrix

Consider the state and costate equations:

$$\begin{bmatrix} \dot{\vec{x}} \\ \dot{\vec{\lambda}} \end{bmatrix} = \begin{bmatrix} A & -BR^{-1}B^T \\ -Q & -A^T \end{bmatrix} \begin{bmatrix} \vec{x} \\ \vec{\lambda} \end{bmatrix} \equiv \bar{A} \begin{bmatrix} \vec{x} \\ \vec{\lambda} \end{bmatrix}, \quad \begin{bmatrix} \vec{x}(t_0) = \vec{x}_0 \\ \vec{\lambda}(t_f) = L\vec{x}(t_f) \end{bmatrix}$$

Since the state transition matrix $\Phi(t,t_0)$ is given by defintion

$$\Rightarrow \begin{bmatrix} \vec{x}(t) \\ \vec{\lambda}(t) \end{bmatrix} = \Phi(t,t_0) \begin{bmatrix} \vec{x}(t_0) \\ \vec{\lambda}(t_0) \end{bmatrix} = \begin{bmatrix} \phi_{11}(t,t_0) & \phi_{12}(t,t_0) \\ \phi_{21}(t,t_0) & \phi_{22}(t,t_0) \end{bmatrix} \begin{bmatrix} \vec{x}_0 \\ \vec{\lambda}(t_0) \end{bmatrix}$$

$$\Rightarrow \begin{cases} \vec{x}(t_f) = \phi_{11}(t_f,t_0)\vec{x}_0 + \phi_{12}(t_f,t_0)\vec{\lambda}(t_0) & (1) \\ \vec{\lambda}(t_f) = \phi_{21}(t_f,t_0)\vec{x}_0 + \phi_{22}(t_f,t_0)\vec{\lambda}(t_0) & (2) \\ \vec{\lambda}(t_f) = L\vec{x}(t_f) & (3) \end{cases}$$

Introducing (1) into (3), and equating (2) and (3) leads to

$$\phi_{21}\left(t_f,t_0\right)\vec{x}_0+\phi_{22}\left(t_f,t_0\right)\vec{\lambda}\left(t_0\right)=L\left[\phi_{11}\left(t_f,t_0\right)\vec{x}_0+\phi_{12}\left(t_f,t_0\right)\vec{\lambda}\left(t_0\right)\right]$$
$$\Rightarrow\left[\phi_{22}\left(t_f,t_0\right)-L\phi_{12}\left(t_f,t_0\right)\right]\vec{\lambda}\left(t_0\right)=-\left[\phi_{21}\left(t_f,t_0\right)-L\phi_{11}\left(t_f,t_0\right)\right]\vec{x}_0$$
$$\Rightarrow\vec{\lambda}\left(t_0\right)=-\left[\phi_{22}\left(t_f,t_0\right)-L\phi_{12}\left(t_f,t_0\right)\right]^{-1}\left[\phi_{21}\left(t_f,t_0\right)-L\phi_{11}\left(t_f,t_0\right)\right]\vec{x}_0$$

Since any time $t\le t_f$ can be initial time, the linear, time-varying optimal control law can be obtained as

$$\vec{u}\left(t\right)=-R^{-1}B^T\vec{\lambda}\left(t\right)=R^{-1}B^T\left[\phi_{22}\left(t_f,t\right)-L\phi_{12}\left(t_f,t\right)\right]^{-1}\left[\phi_{21}\left(t_f,t\right)-L\phi_{11}\left(t_f,t\right)\right]\vec{x}(t)$$

Optimal Trajectory is obtained from

$$\dot{\vec{x}}=A\vec{x}+B\vec{u}=\left(A+BR^{-1}B^T\left[\phi_{22}\left(t_f,t\right)-L\phi_{12}\left(t_f,t\right)\right]^{-1}\left[\phi_{21}\left(t_f,t\right)-L\phi_{11}\left(t_f,t\right)\right]\right)\vec{x}$$
$$\vec{x}\left(t_0\right)=\vec{x}_0$$

In case $\overline{A}$ is time-invariant, i.e., $\left(A,B,Q,R\right)$ are all constant, the state transition matrix is given as

$$\Phi(t,t_0)=exp\left(\overline{A}(t-t_0)\right)$$

4.2.2 Finite Horizon Terminal Controller: $\left(\vec{x}\left(t_f\right),t_f\right)$ Free

- Consider a special case where $\left(A,B,Q,R\right)$ are all constant and $L\equiv 0_{n\times n}$

Minimize the performance index

$$J=\frac{1}{2}\int_{t_0}^{t_f}\left[\vec{x}^TQ\,\vec{x}+\vec{u}^TR\,\vec{u}\right]dt$$

subject to the dynamical system

$$\dot{\vec{x}}=A\vec{x}+Bu,\quad \vec{x}\left(t_0\right)=\vec{x}_0,\quad \vec{x}\left(t_f\right)=free,\quad t_f=free \text{ is fixed}$$

where

$$\vec{x}\in R^n,\vec{u}\in R^m,A\in R^{n\times n},B\in R^{m\times m}$$

$Q \in R^{n\times n}$ are symmetric positive semi-definite
$R \in R^{m\times m}$ is symmetric positive definite

Define Hamiltonian as

$$H = \frac{1}{2}\vec{x}^T Q\,\vec{x} + \frac{1}{2}\vec{u}^T R\,\vec{u} + \vec{\lambda}^T\left(A\vec{x} + B\vec{u}\right)$$

$$\dot{\vec{x}} = \frac{\partial H}{\partial \vec{\lambda}} = A\vec{x} + B\vec{u}, \qquad \vec{x}\left(t_0\right) = \vec{x}_0$$

$$\dot{\vec{\lambda}} = -\frac{\partial H}{\partial \vec{x}} = -Q\vec{x} - A^T\vec{\lambda}, \quad \vec{\lambda}\left(t_f\right) = \vec{0} \qquad \left(\Leftarrow \vec{\lambda}\left(t_f\right) = \left.\frac{\partial h}{\partial \vec{x}}\right|_{t=t_f}\right)$$

$$\vec{0} = \frac{\partial H}{\partial \vec{u}} \quad \Rightarrow R\,\vec{u} + B^T\vec{\lambda} = 0$$

$$\Rightarrow \vec{u} = -R^{-1}B^T\vec{\lambda}, R > 0$$

$$0 = H(t_f) \qquad \left(\Leftarrow 0 = H(t_f) + \frac{\partial h(x(t_f),t_f)}{\partial t_f}\right)$$

Substituting the Optimality Condition into Hamitonian, State/Costate equations yields

$$H = \frac{1}{2}\begin{bmatrix}\vec{x}\\ \vec{\lambda}\end{bmatrix}^T \begin{bmatrix}Q & A^T\\ A & -BR^{-1}B^T\end{bmatrix}\begin{bmatrix}\vec{x}\\ \vec{\lambda}\end{bmatrix}, \ H(t_f) = 0$$

$$\begin{bmatrix}\dot{\vec{x}}\\ \dot{\vec{\lambda}}\end{bmatrix} = \begin{bmatrix}A & -BR^{-1}B^T\\ -Q & -A^T\end{bmatrix}\begin{bmatrix}\vec{x}\\ \vec{\lambda}\end{bmatrix}, \ \begin{bmatrix}\vec{x}\left(t_0\right) = \vec{x}_0\\ \vec{\lambda}\left(t_f\right) = \vec{0}\end{bmatrix}$$

Similarly, employing the Ricatti Transformation, we obtain the following differential Riccati Equation with its boundary condition

$$\dot{K} = -A^TK - KA - Q + KBR^{-1}B^TK, \ K\left(t_f\right) = 0$$

Now, recall the stationarity of Hamiltonian with the transversality condition

$$H(t) \equiv constant, \quad H(t_f) = 0$$
$$\Rightarrow H(t) \equiv 0$$
$$\Rightarrow H(t) = \frac{1}{2}\left(\vec{x}^T Q \vec{x} + \vec{\lambda}^T A \vec{x} + \vec{x}^T A^T \vec{\lambda} - \vec{\lambda}^T B R^{-1} B^T \vec{\lambda}\right) \equiv 0$$
$$\left(\Uparrow \ \vec{\lambda} = K\vec{x}\right)$$
$$\Rightarrow H(t) = \frac{1}{2}\vec{x}^T \left[A^T K + K^T A + Q - KBR^{-1}B^T K\right]\vec{x} \equiv 0, \forall \vec{x}$$
$$\Rightarrow 0_{n\times n} \ = A^T K + K^T A + Q - KBR^{-1}B^T K \quad \text{(Algebraic Riccati Equation)}$$

Note that the Riccati Gain K is constant, not a function of time

- Specail Case: Infinite Horizon Regulator Problem

Intuitively, in order for an optimal controller to stabilize the system

$$\lim_{t\to\infty} \vec{x}(t) = \vec{0}$$

As $\vec{u}(t) = -R^{-1}B^T K(t)\vec{x}(t)$,

$$\lim_{t\to\infty} \vec{u}(t) = \vec{0}$$

It can be formally proved that for an infinite horizon regulator problem, the Constant Gain K satisfies the following Algebraic Ricatti Equation:

$$0 = A^T K + KA + Q - KBR^{-1}B^T K$$

4.2.3 Linear Tracking Problem

- Can be considered as a Generalization of Linear Quadratic regulator: Minimize the performance index

$$J = \frac{1}{2}\left[\vec{x}(t_f) - \vec{r}(t_f)\right]^T L\left[\vec{x}(t_f) - \vec{r}(t_f)\right]$$
$$+\frac{1}{2}\int_{t_0}^{t_f}\left[\left[\vec{x}(t) - \vec{r}(t)\right]^T Q(t)\left[\vec{x}(t) - \vec{r}(t)\right] + \vec{u}^T(t) R(t)\vec{u}(t)\right]dt$$

subject to the dynamical system

$$\dot{\vec{x}}(t) = A(t)\vec{x}(t) + B(t)\vec{u}(t), \ \vec{x}(t_0) = \vec{x}_0, \ t_f \text{ is fixed}$$

where

$\vec{r}(t_f)$ is a desired target point, which is known a priori

$\vec{r}(t)$ is a desired reference trajectory to be tracked

Define Hamiltonian as

$$H=\frac{1}{2}\left[\left(\vec{x}-\vec{r}\right)^T Q\left(\vec{x}-\vec{r}\right)+\vec{u}^T R\,\vec{u}\right]+\vec{\lambda}^T\left(A\vec{x}+B\vec{u}\right)$$

$$\dot{\vec{x}}=\frac{\partial H}{\partial \vec{\lambda}}=A\vec{x}+B\vec{u}, \qquad \vec{x}\left(t_0\right)=\vec{x}_0$$

$$\dot{\vec{\lambda}}=-\frac{\partial H}{\partial \vec{x}}=-Q\vec{x}-A^T\vec{\lambda}+Q\vec{r},\ \ \vec{\lambda}\left(t_f\right)=\left.\frac{\partial h}{\partial \vec{x}}\right|_{t_f}=L\left[\vec{x}\left(t_f\right)-\vec{r}\left(t_f\right)\right]$$

$$\vec{0}=\frac{\partial H}{\partial \vec{u}}\Rightarrow \vec{u}=-R^{-1}B^T\vec{\lambda}$$

Define Riccati Transformation

$$\vec{\lambda}(t)=K(t)\vec{x}(t)+S(t)$$

$$\Rightarrow\begin{cases}\dot{\vec{\lambda}}=\dot{K}\vec{x}+K\dot{\vec{x}}+\dot{S}=\dot{K}\vec{x}+K\left(A\vec{x}+B\vec{u}\right)+\dot{S}\\ \dot{\vec{\lambda}}=-Q\vec{x}-A^T\vec{\lambda}+Q\vec{r}\end{cases}$$

$$\Rightarrow \dot{K}\vec{x}+K\left[A\vec{x}-BR^{-1}B^T\left(K\vec{x}+S\right)\right]+\dot{S}=-Q\vec{x}-A^T\left[K\vec{x}+S\right]+Q\vec{r}$$

$$\Rightarrow\left[\dot{K}+KA-KBR^{-1}B^TK+A^TK+Q\right]\vec{x}+\left[-KBR^{-1}B^TS+\dot{S}+A^TS-Q\vec{r}\right]=0$$

- Since this equation must hold for all $\vec{x}$, we obtain

$$\dot{K}=-KA+KBR^{-1}B^TK-A^TK-Q$$
$$\dot{S}=-\left(A^T-KBR^{-1}B^T\right)S+Q\vec{r}$$

The associated boundary conditions are obtained from

$$\begin{cases}\vec{\lambda}\left(t_f\right)=L\vec{x}\left(t_f\right)-L\vec{r}\left(t_f\right)\\ \vec{\lambda}\left(t_f\right)=K\left(t_f\right)\vec{x}\left(t_f\right)+S\left(t_f\right)\end{cases}$$

$$\Rightarrow\begin{cases}K\left(t_f\right)=L\\ S\left(t_f\right)=-L\vec{r}\left(t_f\right)\end{cases}$$

Note that $\vec{r}(t)$ is completely known a priori

Solve for K first, and then for S

$$\vec{u}(t) = -R^{-1}B^{T}\vec{\lambda} = -R^{-1}B^{T}\left[K\vec{x} + S\right]$$

If $\vec{r}(t) = 0$, then $S(t) = 0$, which reduces the tracking problem to the linear quadratic terminal controller problem

4.2.4 Stability, Controllability and Observability

- Stability in the sense of Lyapunov
  - A zero solution or an equilibrium point $\vec{x}(t) \equiv \vec{0}$ of a dynamical system

    $$\dot{\vec{x}}(t) = \vec{f}(\vec{x}(t), t)$$

    is said to be stable: for $\forall \varepsilon > 0$, $\exists \delta(\varepsilon) > 0$ such that if $\|\vec{x}(t_0)\| < \delta$, then $\|\vec{x}(t)\| < \varepsilon, \forall t \geq t_0$
  - A zero solution or an equilibrium point $\vec{x}(t) \equiv \vec{0}$ of a dynamical system

    $$\dot{\vec{x}}(t) = \vec{f}(\vec{x}(t), \vec{u}(t), t)$$

    is said to be asymptotically stable: if it is Lyapunov-stable and

    $$\lim_{t\to\infty} \vec{x}(t) = \vec{0}$$
  - A zero solution of a linear system

    $$\dot{\vec{x}}(t) = A\vec{x}(t)$$

    is asymptotically stable iff all of the eigenvalues of A have negative real parts

- Controllability
  - A system is said to be controllable at time t_0 if it is possible by means of an unconstrained control vector to transfer the system from any initial state $\vec{x}(t_0)$ to any other state in a finite interval of time
  - A system

$$\dot{\vec{x}}(t) = A\vec{x}(t) + B\vec{u}(t)$$

is completely cotrollable, iff the $nm \times n$ controllability matrix

$$\begin{bmatrix} B & AB & \cdots & (A)^{n-1}B \end{bmatrix}$$

is of rank n

- Observability
  - A system is said to be observable at time t_0 if, with the system in state $\vec{x}(t_0)$, it is possible to determine this state from the observation of the output over a finite time interval.
  - Important because, in practice, the difficulty encountered with state feedback control is that some of the state variables are not accessible for direct measurememt
  - In that case, with the measurement, it becomes necessary to estimate the unmeasurable state variables in order to construct the control signals.
  - A system

    $$\dot{\vec{x}}(t) = A\vec{x}(t)$$

    $$\vec{y}(t) = C\vec{x}(t)$$

    is completely observable iff the $n \times nm$ observability matrix

    $$\begin{bmatrix} C^T & A^T C^T & \cdots & \left(A^T\right)^{n-1} C^T \end{bmatrix}$$

    is of rank n

4.2.5 State-Dependent Riccati Equation (SDRE) Controller

- First proposed by J. R. Cloutier, "State-Dependent Riccati Equation Techniques: An Overview", Proceedings of the American Control Conference, Albuquerque, NM, 1997, pp. 932-936

- General process of SDRE technique is the same as that of LQR method
  - However, the SDRE technique updates the dynamic model matrices **A** and **B** at each time step, because these matrices in the SDRE technique are dependent on the states as shown in Figure 1
  - It affects the value of gain matrix **K**, so that the controller can choose proper control values for the dynamic systems at any time

- Using the SDRE techniques requires us to write the state-space equation in the form of

$$\dot{\vec{x}}(t) = \vec{f}(\vec{x}(t)) + \vec{g}(\vec{x}(t))\vec{u}(t)$$
$$\Rightarrow \dot{\vec{x}}(t) = A(\vec{x}(t))\vec{x} + B(\vec{x}(t))\vec{u}(t)$$

  where $\vec{x}(t)$, $\vec{u}(t)$, $A(\vec{x}(t))$ and $B(\vec{x}(t))$ denote states vector, control vector, dynamics matrix and control matrix, respectively
- Unlike the LQR method, $A(\vec{x}(t))$ and $B(\vec{x}(t))$ in the SDRE technique are not constant any more, but are dependent on the states

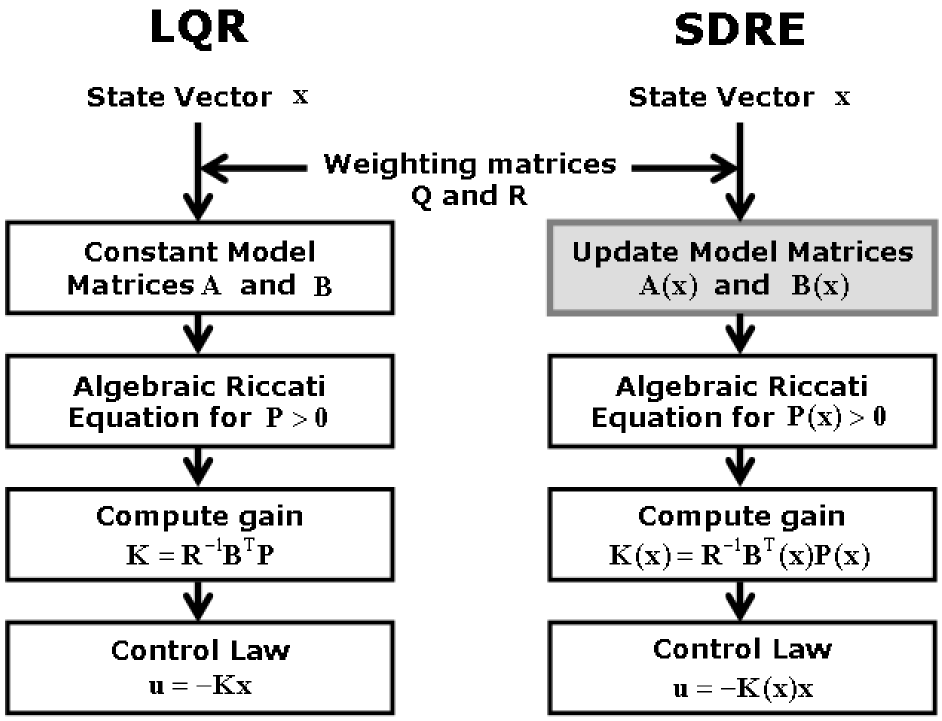


Figure 1 Description of the SDRE technique

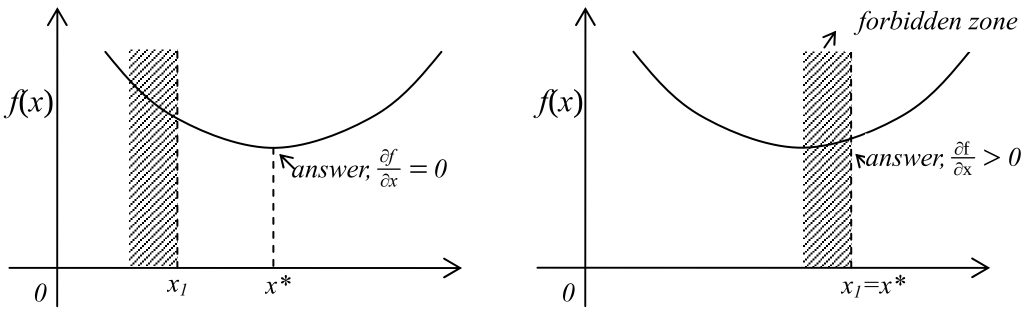
4.3 Pontryagin's Minimum Principle and Constraints

4.3.1 Pontryagin's Minimum Principle

- It commonly occurs in reaslistic situations that admissible states and controls are bounded
  - Physically realizable controls generally have magnitude limitations

- State constraints often arise because of safety or structural restrictions
- Then, questions arise:
  - When $\delta\vec{u}(t)$ is arbitrary, $\delta J\left(\vec{u}^*,\delta\vec{u}\right)=0 \Rightarrow \vec{u}^*$ is optimal
  - When $\delta\vec{u}(t)$ is not arbitrary, $\delta J\left(\vec{u}^*,\delta\vec{u}\right)=?$

- Function Minimization with inequality constraints
  Min $f(x)$, subject to $x \geq x_1$



The first order necessary condition for minimizing $f(x)$ is, as shown in the above figure,

$$\frac{\partial f}{\partial x} \geq 0$$

- Optimal Control Problem with Control Constraints
  Minimize

$$J = h\left(\vec{x}\left(t_f\right),t_f\right) + \int_{t_0}^{t_f} g\left(\vec{x}\left(t\right),\vec{u}\left(t\right),t\right)dt$$

subject to a dynamical system with boundary conditions

$$\dot{\vec{x}} = \vec{f}\left(\vec{x}\left(t\right),\vec{u}\left(t\right),t\right),\ \ \vec{x}\left(t_0\right) = \vec{x}_0,\ \ \vec{x}\left(t_f\right) = free$$

and control bounds

$$\left|u_i\right| \leq u_{i0}$$

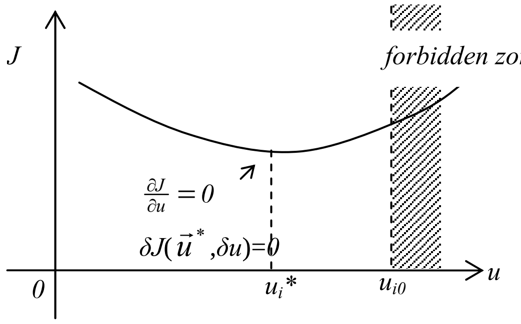
Assume that boundary conditions are satisfied

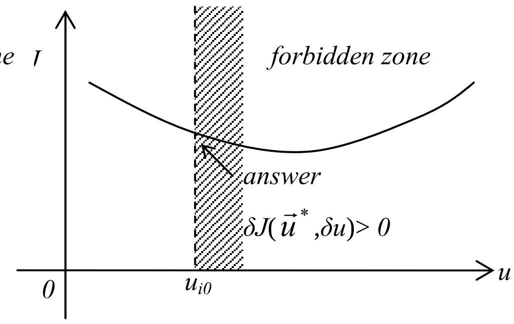
$$\delta J=\left[\frac{\partial h}{\partial\vec{x}}+\vec{v}^T\frac{\partial\vec{m}}{\partial\vec{x}}-\vec{\lambda}^T\right]_{t=t_f}\delta\vec{x}_f+\left[\frac{\partial h}{\partial t}+\vec{v}^T\frac{\partial\vec{m}}{\partial t}+H\right]_{t=t_f}\delta t_f$$

$$+\int_{t_0}^{t_f}\left[\frac{\partial H}{\partial\vec{u}}\delta\vec{u}+\left(\frac{\partial H}{\partial\vec{x}}+\dot{\vec{\lambda}}^T\right)\delta\vec{x}+\delta\vec{\lambda}^T\left(\frac{\partial H}{\partial\vec{\lambda}}-\dot{\vec{x}}\right)\right]dt$$

If the state and costate differential equations are satisfied, then

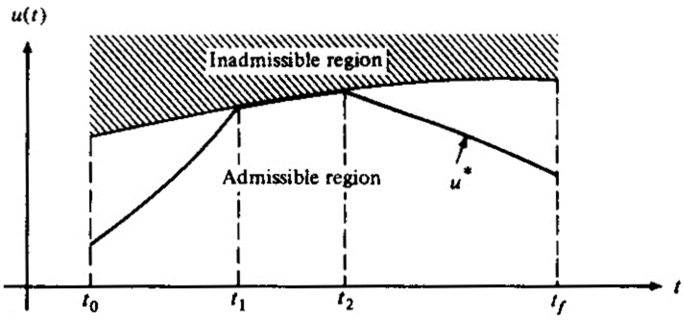
$$\delta J=\int_{t_0}^{t_f}\frac{\partial H}{\partial\vec{u}}\delta\vec{u}dt$$



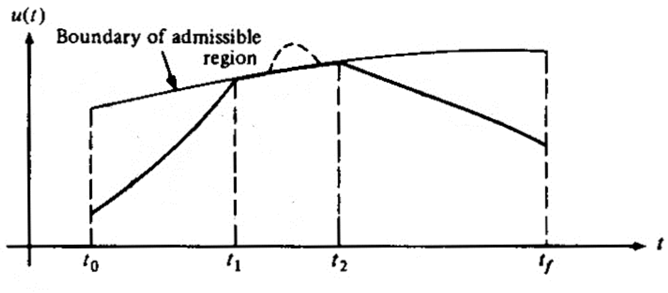


The above diagram shows that $\delta J\geq 0$ for minimum J

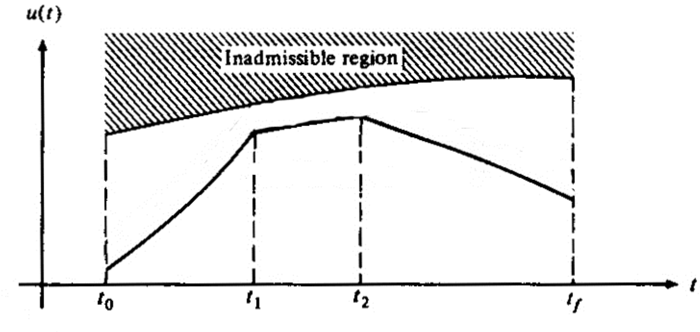
- Case (A) $\delta J=0$: extremal control that lies on a boundary



- Case (B) $\delta J>0$: extremal control that is constrained by a boundary



■ Case (C) $\delta J = 0$: extremal control lies within the boundary during any portion of time interval $\lfloor t_0, t_f)$



Thus, the cases (A)~(C) yield $\delta J(u^*, \delta u) \geq 0$ for the problem with control constraints

$$
\begin{aligned}
\Delta H &= H\left(\vec{x}^*, \vec{u}^* + \delta\vec{u}, \vec{\lambda}^*, t\right) - H\left(\vec{x}^*, \vec{u}^*, \vec{\lambda}^*, t\right) \\
&= H\left(\vec{u}^*\right) + \frac{\partial H}{\partial \vec{u}}\Big|_{u^*} \delta\vec{u} + \left(H.O.T.\right) - H\left(\vec{u}^*\right) \\
&\simeq \frac{\partial H}{\partial \vec{u}}\Big|_{u^*} \delta\vec{u}
\end{aligned}
$$

$$
\begin{aligned}
\delta J &= \int_{t_0}^{t_f} \frac{\partial H}{\partial \vec{u}} \delta\vec{u}\, dt \\
&\simeq \int_{t_0}^{t_f} \Delta H\, dt \\
&= \int_{t_0}^{t_f} \left[H\left(\vec{x}^*, \vec{u}^* + \delta\vec{u}, \vec{\lambda}^*, t\right) - H\left(\vec{x}^*, \vec{\lambda}^*, \vec{u}^*, t\right) \right] dt \\
&\geq 0
\end{aligned}
$$

For all admissible $\delta\vec{u}$, this can happen, if

$$H\left(\vec{x}^*, \vec{u}^* + \delta\vec{u}, \vec{\lambda}^*, t\right) \geq H\left(\vec{x}^*, \vec{u}^*, \vec{\lambda}^*, t\right)$$

for all $t \in \left[t_0, t_f \right]$

● Pontryagin's Minimum Principle

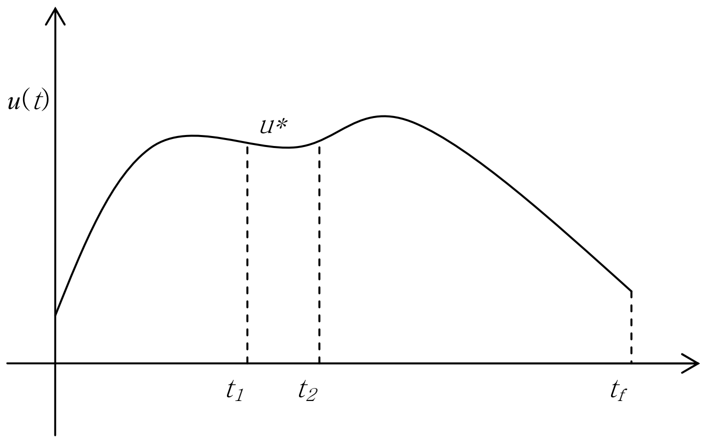
If $\vec{x}^*(t)$, $\vec{\lambda}^*(t)$ and $\vec{u}^*(t)$ are extremal states, costates and controls, respectively, then the optimal control (whether it is constrained or unconstrained) minimizes the Hamiltonian, i.e.,

$$\vec{u}^*(t) = arg \min_{\vec{u} \in \Omega} H\left(\vec{x}^*(t), \vec{\lambda}^*(t), \vec{u}(t), t\right)$$

Proof by Contradiction: Suppose

$$\mathrm{H}\left(\vec{x}^*, \vec{u}^* + \delta\vec{u}, \vec{\lambda}^*, t\right) < \mathrm{H}\left(\vec{x}^*, \vec{u}^*, \vec{\lambda}^*, t\right) \text{ in } [t_1, t_2]$$

where $[t_1, t_2]$ is an arbitrary small, but nonzero interval



$$\vec{U} = \vec{U}^* \qquad 0 \le t < t_1$$

$$\vec{U} = \vec{U}^* \qquad t_2 < t < t_f$$

$$\vec{U} = \vec{U}^* + \delta\vec{U} \qquad t_1 \le t \le t_2$$

Then, the following inequality holds:

$$
\begin{aligned}
\delta J &= \int_{t_0}^{t_f}\left[H\left(\vec{x}^*,\vec{\lambda}^*,\vec{u},t\right)-H\left(\vec{x}^*,\vec{\lambda}^*,\vec{u}^*,t\right)\right]dt \\
&= \int_{t_0}^{t_1}\left[H\left(\vec{x}^*,\vec{\lambda}^*,\vec{u}^*,t\right)-H\left(\vec{x}^*,\vec{\lambda}^*,\vec{u}^*,t\right)\right]dt \\
&\quad + \int_{t_1}^{t_2}\left[H\left(\vec{x}^*,\vec{\lambda}^*,\vec{u},t\right)-H\left(\vec{x}^*,\vec{\lambda}^*,\vec{u}^*,t\right)\right]dt \\
&\quad + \int_{t_2}^{t_f}\left[H\left(\vec{x}^*,\vec{\lambda}^*,\vec{u}^*,t\right)-H\left(\vec{x}^*,\vec{\lambda}^*,\vec{u}^*,t\right)\right]dt \\
&= \int_{t_1}^{t_2}\left[H\left(\vec{x}^*,\vec{\lambda}^*,\vec{u},t\right)-H\left(\vec{x}^*,\vec{\lambda}^*,\vec{u}^*,t\right)\right]dt \\
&< 0,
\end{aligned}
$$

which violates the optimality condition $\delta J \ge 0$, since $[t_1, t_2]$ can be anywhere in $[0, t_f)$. Thus, if $H\left(\vec{x}^*,\vec{\lambda}^*,\vec{u}^*+\delta\vec{u},t\right) < H\left(\vec{x}^*,\vec{\lambda}^*,\vec{u}^*,t\right)$ for any $t \in [0, t_f)$, then it is always possible to construct an admissible control which makes $\Delta J < 0 \left(\simeq \delta J < 0\right)$, which contradicts the optimality of the control $u^*(t)$, i.e., $\delta J \ge 0$

The above arguments conclude that a necessary condition for $\vec{u}^*$ to minimize the functional J is

$$H\left(\vec{x}^*,\vec{\lambda}^*,\vec{u}^*+\delta\vec{u},t\right) \ge H\left(\vec{x}^*,\vec{\lambda}^*,\vec{u}^*,t\right)$$

for all $t \in [0, t_f)$ and for all admissible control.

- (Fist Order) Necessary Conditions for Optimality
  If Hamiltonian is defined as

  $$H \equiv g\left(\vec{x},\vec{u},t\right) + \vec{\lambda}^T \vec{f}\left(\vec{x},\vec{u},t\right)$$

  then, the Necessary Conditions for $\vec{u}^*(t)$ to be an optimal control are

$$\dot{\vec{x}} = \frac{\partial H}{\partial \vec{\lambda}}$$

$$\dot{\vec{\lambda}} = -\frac{\partial H}{\partial x}$$

$$\vec{u}^*(t) = \arg\min_{\vec{u}\in\Omega} H\left(\vec{x}^*(t), \vec{\lambda}^*(t), \vec{u}(t), t\right)$$

$$\left(\Updownarrow H\left(\vec{x}^*(t), \vec{\lambda}^*(t), \vec{u}(t), t\right) \geq H\left(\vec{x}^*(t), \vec{\lambda}^*(t), \vec{u}^*(t), t\right)\right)$$

$$\left[\frac{\partial h}{\partial \vec{x}} + \nu^T \frac{\partial m}{\partial \vec{x}} - \vec{\lambda}^T\right]_{t=t_f} \delta x_f + \left[H + \frac{\partial h}{\partial t} + \nu^T \frac{\partial m}{\partial t}\right]_{t=t_f} \delta t_f = 0$$

Note that the Minimum Principle holds when admissible controls are not bounded, in which case

$$\vec{u}^*(t) = arg\,\min_{\vec{u}\in\Omega} H\left(\vec{x}^*(t), \vec{\lambda}^*(t), \vec{u}(t), t\right) \quad \Rightarrow \quad \frac{\partial H}{\partial \vec{u}^*} = 0$$

- Example 1: Minimum-Time Problem
  Minimize

$$J = \int_0^{t_f} 1dt = t_f$$

subject to the dynamics, boundary conditions, and control bounds:

$$\dot{x}_1 = x_2 \quad x_1(0) = x_{10}, \quad x_1(t_f) = 0$$

$$\dot{x}_2 = u \quad x_2(0) = x_{20}, \quad x_2(t_f) = 0$$

$$|u| \leq 1$$

The control objective is to transfer the system from an initial state $\vec{x}_0$ to the origin in minimum time

Define Hamiltonian as

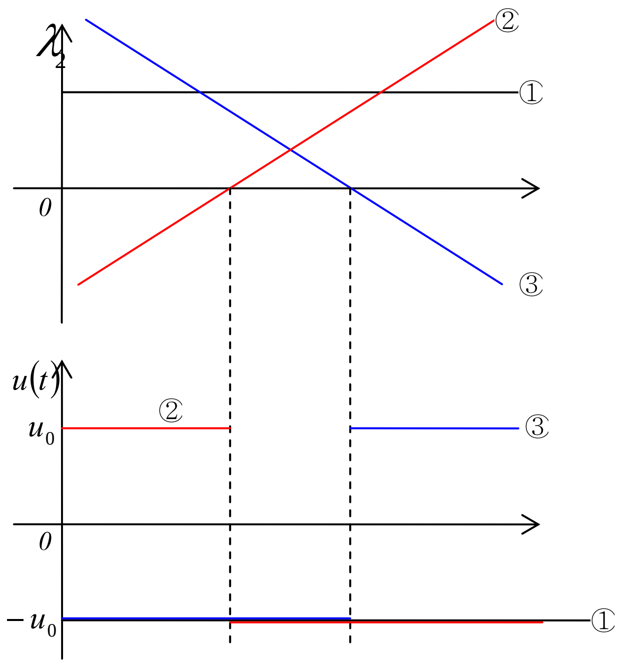
$$H = 1 + \lambda_1 x_2 + \lambda_2 u$$

$$\left.\begin{array}{l} \dot{H} = 0 \Rightarrow H(t) = \text{constant} \\ H(t_f) + \dfrac{\partial h(x(t_f), t_f)}{\partial t_f} = 0 \end{array}\right\} \Rightarrow H(t) \equiv 0$$

$$\dot{x}_1 = x_2 \qquad x_1(0) = x_{10}, \quad x_1(t_f) = 0$$
$$\dot{x}_2 = u \qquad x_2(0) = x_{20}, \quad x_2(t_f) = 0$$

$$\dot{\lambda}_1 = 0 \quad \Rightarrow \quad \lambda_1 = \text{constant}$$
$$\dot{\lambda}_2 = -\lambda_1 \Rightarrow \quad \lambda_2 = -\lambda_1 t + \lambda_{20}$$
$$\vec{u}^*(t) = \arg\min_{\vec{u}\in\Omega} H\left(\vec{x}^*(t), \vec{\lambda}^*(t), \vec{u}(t), t\right)$$
$$\Rightarrow u^* = \begin{cases} -1 & if \quad \lambda_2 > 0 \\ +1 & if \quad \lambda_2 < 0 \end{cases}$$
$$\Rightarrow u^* = -sign(\lambda_2)$$

cf) What if $\lambda_2 \equiv 0$ for a finite time span?



$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = \pm 1 \end{cases}$$

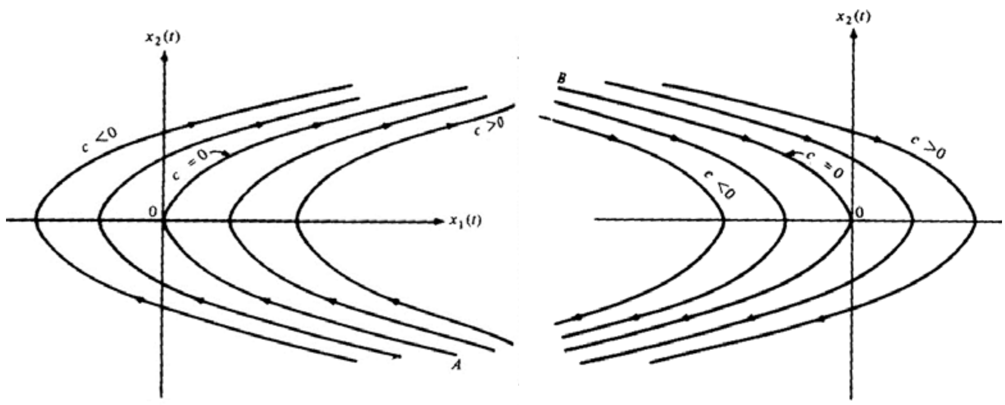
$$\Rightarrow \begin{cases} dx_1 = x_2 dt \\ dx_2 = \pm dt \end{cases}$$

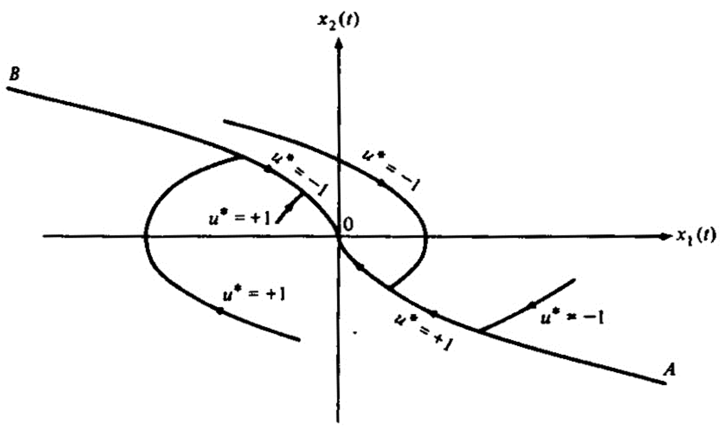
$$\Rightarrow \quad \frac{dx_1}{dx_2} = \frac{x_2}{\pm 1}$$

$$\Rightarrow \pm \int dx_1 = \int x_2 dx_2$$

$$\Rightarrow \pm x_1 = \frac{1}{2} x_2^2 + c \quad for \quad u = \pm 1$$

cf) Same results can be obtained if the dynamic equations are integrated with respect to time backward from the terminal time t_f





switching point(to get origin)

switching curve(line)

$c = 0$ yields the equation of the switching curve (line):

$$x_1(t) = -\frac{1}{2}x_2(t)|x_2(t)|$$

$$\Rightarrow s \equiv x_1 + \frac{1}{2}x_2|x_2|$$

Note that we have obtained the optimal control law, that is, the optimal control at every time *t* is known as a function of the state variables (x_1, x_2)

S(***x***(*t*)) > 0 implies ***x***(*t*) lies above the switching curve (*A*-*O*-*B*)

S(***x***(*t*)) < 0 implies ***x***(*t*) lies below the switching curve (*A*-*O*-*B*)

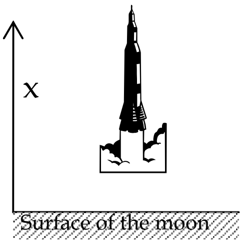
S(***x***(*t*)) = 0 implies ***x***(*t*) lies on the switching curve (*A*-*O*-*B*)

In terms of this switching function, the optimal control law is

$$u^*(t) = \begin{cases} -1, & S(x(t)) > 0 \\ +1, & S(x(t)) < 0 \\ -1, & S(x(t)) = 0, x_2(t) > 0 \\ +1, & S(x(t)) = 0, x_2(t) < 0 \\ 0, & x_1(t) = x_2(t) = 0 \end{cases} \quad \Leftrightarrow u^*(t) = -sign(\lambda_2)$$

- Example 2: Minimum-Time Problem

  A lunar roket makes a soft landing on the surface of the Moon



m	: spacecraft mass
$-M \le \dot{m} \le 0$	: spacecraft mass rate, control variable
g	: gravity constant on the Moon's surface
T	: thrust ($T = -k\dot{m}$)
k	: constant relative exhaust velocity of gas

Equations of motion

$$m(t)\ddot{x}(t) = -gm(t) + T(t)$$

$$\Rightarrow m(t)\ddot{x}(t) = -gm(t) - k\dot{m}(t)$$

Let $x_1 = x$, $x_2 = \dot{x}$, $x_3 = m$, $u = \dot{m}$

$$\dot{x}_1(t) = x_2(t)$$
$$\dot{x}_2(t) = -g - \frac{k}{x_3(t)} u(t) \qquad (1)$$
$$\dot{x}_3(t) = u(t)$$

Problem Formulation: Minimize

$$J = \int_{t_0}^{t_f} 1 dt$$

subject to the equations of motin (2) and control bounds

$$-M \le u \le 0$$

Define Hamiltonian as

$$\begin{aligned} H &= 1 + \lambda_1 x_2 + \lambda_2 \left(-g - \frac{k}{x_3} u \right) + \lambda_3 u \\ &= 1 + \lambda_1 x_2 - \lambda_2 g + \left(-\lambda_2 \frac{k}{x_3} + \lambda_3 \right) u \end{aligned}$$

Optimal control

$$\vec{u}^*(t) = \arg \min_{\vec{u} \in \Omega} H\left(\vec{x}^*(t), \vec{\lambda}^*(t), \vec{u}(t), t \right)$$

$$\Rightarrow u^*(t) = \begin{cases} 0, & -\lambda_2 \frac{k}{x_3} + \lambda_3 < 0 \\ -M, & -\lambda_2 \frac{k}{x_3} + \lambda_3 > 0 \\ \text{undetermined}, & -\lambda_2 \frac{k}{x_3} + \lambda_3 \equiv 0 \ \textit{for a finite time span} \end{cases}$$

Obtaining an explicit solution for $u^*(t)$ requires us to solve a nonlinear two point boundary value problem using the numerical technique described in the previous sections.

The optimal control for Minimum Time problem often yields Bang-Bang control, that is, the optimal control switches between its maximum and minimum admissible value

- Example 3: Minimum-Fuel Problem

  Minimize

$$J = \int_{t_0}^{t_f} |u(t)| dt$$

subject to the dynamics

$$\dot{x}(t) = u(t), \; x(t_0) = x_0, \; x(t_f) = 0, \; t_f \text{ is free}$$

and control bounds

$$|u(t)| \le 1.0$$

The control objective is for the system to transferr from an arbitrary initial state x_0 to the origin, while minimizing fuel consumption

Define Hamiltonian as

$$H = |u| + \lambda u$$

$$\dot{x}(t) = u(t), \; x(t_0) = x_0, \; x(t_f) = 0, \; t_f \text{ is free}$$

$$\dot{\lambda}(t) = 0 \;\Rightarrow \lambda(t) = \text{constant}$$

$$u^*(t) = \arg\min_{\bar{u}\in\Omega} H\left(x^*(t), \lambda^*(t), u(t), t\right)$$

$$\Rightarrow u^*(t) = \begin{cases} 1 & , \quad \lambda < -1 \\ 0 & , \quad -1 < \lambda < 1 \\ -1 & , \quad \lambda > 1 \\ \text{undetermined} & , \quad \lambda = \pm 1 \; \textit{for a finite time} \end{cases}$$

- Example 4: Weighted Combination of Time and Fuel

  Minimize

$$J = \int_{t_0}^{t_f} \left[w + u^2(t) \right] dt, \; w > 0 \text{ is chosen to weigh the relative importance}$$

subject to the dynamics

$$\dot{x}(t) = -ax(t) + u(t), \; x(t_0) = x_0, \; x(t_f) = 0, \; a > 0, \; t_f \text{ is free}$$

and control bounds

$$|u(t)| \le 1.0$$

The control objective is for the system to transfer from an arbitrary initial state x_0 to the origin, while minimizing the performance index representing the weighted combination of time and fuel

Define Hamiltonian as

$$H = w + u^2 - \lambda a x + \lambda u$$

$$\dot{x}(t) = -ax(t) + u(t),\ x(t_0) = x_0,\ x(t_f) = 0,\ a > 0,\ t_f$$

$$\dot{\lambda}(t) = -\frac{\partial H}{\partial x} = a\lambda(t) \Rightarrow \lambda(t) = C_1 e^{at}$$

$$u^*(t) = \arg\min_{\vec{u}\in\Omega} H\left(x^*(t), \lambda^*(t), u(t), t\right)$$

If the control bound is not active, i.e., $\left|\mathrm{u}(\mathrm{t})\right| < 1.0$, then

$$\frac{\partial \mathrm{H}}{\partial \mathrm{u}} = 2\mathrm{u} + \lambda = 0 \Leftrightarrow \mathrm{u} = -\frac{1}{2}\lambda \Leftrightarrow |\lambda| < 2 \quad \text{when} \quad \left|\mathrm{u}(\mathrm{t})\right| < 1.0$$

If $|\lambda| \geq 2$, then the control that minimizes H is

$$u^*(t) = \begin{cases} +1, & \lambda \leq -2 \\ -1, & \lambda \leq -2 \end{cases}$$

In summary, the optimal control logic is

$$u^*(t) = \begin{cases} +1, & \lambda \leq -2 \\ -\frac{1}{2}\lambda, & -2 \leq \lambda \leq 2 \\ -1, & \lambda \leq -2 \end{cases}$$

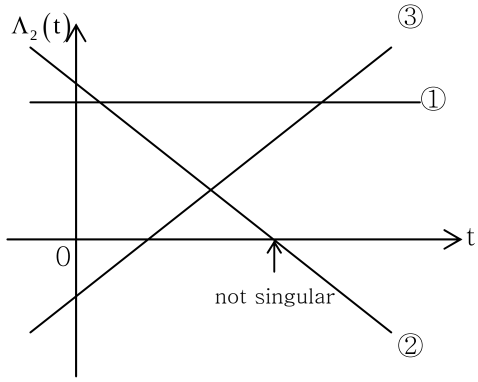
- Often, but NOT always,
  - Minimum-Time problems lead to Bang-Bang control
  - Minimum-Fuel problems lead to Bang-Off-Bang control
  - Minimum-Energy problems lead to a smooth continuous control

4.3.2 Singular Optimal Control Problems

- If there exists a time interval $[t_1, t_2]$ of finite duration such that the necessary condition for optimality provides no information on the relationship between $x^*(t)$, $\lambda^*(t)$ and $u^*(t)$, then the optimal control problem is said to be singular; the associated states, control and interval $[t_{1,}t_2]$ are said to be singular arc, singular control and singular interval, respectively

- Example 1: Minimize

$$J=\int_{t_0}^{t_f}1dt$$



subject to the dynamics

$$\dot{x}_1(t)=x_2(t),\quad x_1(t_0)=x_{10},\quad x_1(t_f)=x_{1f}$$
$$\dot{x}_2(t)=u(t),\quad x_2(t_0)=x_{20},\quad x_2(t_f)=x_{2f}$$

control bounds

$$|u(t)|\le 1.0$$

t_f is free

Define Hamiltonian as

$$H=1+\lambda_1 x_2+\lambda_2 u$$

$$\dot{x}_1(t)=x_2(t),\quad x_1(t_0)=x_{10},\quad x_1(t_f)=x_{1f}$$
$$\dot{x}_2(t)=u(t),\quad x_2(t_0)=x_{20},\quad x_2(t_f)=x_{2f}$$

$$\dot{\lambda}_1=0 \qquad \Rightarrow \lambda_1=\text{constant}$$
$$\dot{\lambda}_2=-\lambda_1 \;\Rightarrow\; \lambda_2(t)=-\lambda_1 t+\lambda_{20}$$

$$\mathrm{u}=-\mathrm{sign}(\lambda_2)$$

$$\begin{cases} H(t_f)+\dfrac{\partial h}{\partial t}+\nu^T\dfrac{\partial m}{\partial t}=0 \;\Rightarrow\; H(t_f)=0 \\ H(t)=\text{constant} \end{cases} \quad\Rightarrow\quad H(t)\equiv 0$$

It turns out that the singular control cannot occur for this problem, since it violates the transversality condition

Suppose $\lambda_2\equiv 0$ for a finite time span such that singular control region exists

Since

$$\lambda_2(t) = -\lambda_1 t + \lambda_{20} \;\Rightarrow\; \lambda_1(t) \equiv 0,$$

the Hamiltonian becomes

$$H = 1 + \lambda_1 x_2 + \lambda_2 u \equiv 1,$$

which violates the transversality condition

Hence, the singular interval does not exist

- Example 2: Minimize

$$J = \tfrac{1}{2}\int_{t_0}^{t_f} x_1^2 dt$$

subject to the dynamics

$$\begin{aligned}
\dot{x}_1 &= x_2 + u, \quad x_1(t_0) = x_{10}, \quad x_1(t_f) = x_{1f} \\
\dot{x}_2 &= -u, \qquad x_2(t_0) = x_{20}, \quad x_2(t_f) = x_{2f}
\end{aligned}$$

t_f is fixed and u is unbounded

Define Hamiltonian as

$$H = \frac{1}{2}x_1^2 + \lambda_1\left(x_2 + u\right) - \lambda_2 u$$

$$\begin{aligned}
\dot{x}_1 &= x_2 + u, \quad x_1(t_0) = x_{10}, \quad x_1(t_f) = x_{1f} \\
\dot{x}_2 &= -u, \qquad x_2(t_0) = x_{20}, \quad x_2(t_f) = x_{2f}
\end{aligned}$$

$$\begin{aligned}
\dot{\lambda}_1 &= -x_1 \\
\dot{\lambda}_2 &= -\lambda_1
\end{aligned}$$

$$\frac{\partial H}{\partial u} = 0 \quad\Rightarrow\quad \lambda_1 - \lambda_2 = 0$$

Note that we cannot determine the control logic directly from the first order necessary condition $\partial H / \partial u = 0$

Suppose that $\lambda_1 - \lambda_2 = 0$ for a finite time span

$$\dot{\lambda}_1 - \dot{\lambda}_2 = 0 \;\Rightarrow -x_1 + \lambda_1 = 0 \;\Rightarrow\; \lambda_1 = x_1$$

$$\Rightarrow \ddot{\lambda}_1 - \ddot{\lambda}_2 = 0 \;\Rightarrow -\dot{x}_1 + \dot{\lambda}_1 = -(x_2 + u) - x_1 = 0 \;\Rightarrow\; u = -(x_1 + x_2)$$

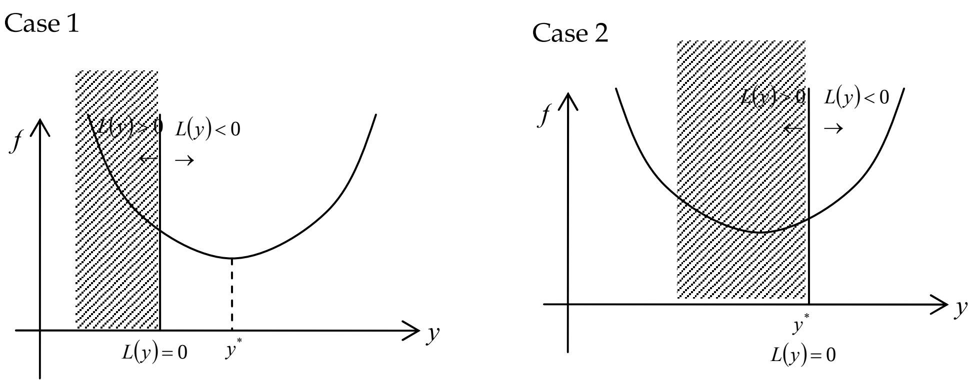
The singular control is obtained by taking consecutive time derivaties *2n, n=1,2,…* times, the associated state trajectory is said to be the singular arc of the *n*-th order

4.3.3 Inequality Constraints on Control Variables

- Function Minimization with Inequality Constraints:
  Min $f(y)$, subject to $L(y) \le 0$

  f and *L* are vectors, possibly in different dimension

  Case 1: Constraint is not active and can be ignored
  Case 2: Constraint is active



$$y^* : L(y^*) < 0, \frac{\partial f}{\partial y} = 0 \qquad\qquad y^* : L(y^*) = 0, \frac{\partial f}{\partial y} > 0 \Rightarrow \left.\frac{\partial L}{\partial y}\right|_{y^*} = 0$$

$$df = \left.\frac{\partial f}{\partial y}\right|_{y^*} dy \ge 0 \qquad (1)\;\; dy : \text{small perturbation about } y^*$$

$$dL = \left.\frac{\partial L}{\partial y}\right|_{y^*} dy \le 0 \qquad (2)$$

In order for (1) and (2) to be consistent with each other, it is necessary that either

$$\mathrm{sign}\left(\frac{\partial f}{\partial y}\right) = -\mathrm{sign}\left(\frac{\partial L}{\partial y}\right) \quad \text{or} \quad \frac{\partial f}{\partial y} = 0,$$

which can be expressed as

$$\frac{\partial f}{\partial y} + \lambda \frac{\partial L}{\partial y} = 0, \quad \lambda \geq 0 \quad \text{where} \quad \begin{cases} \lambda = 0, & \text{if} \quad \dfrac{\partial f}{\partial y} = 0 \\ \lambda > 0, & \text{if} \quad \dfrac{\partial f}{\partial y} \neq 0 \end{cases}$$

These two cases can be treated analytically by adjoining $f(y)$ and $L(y)$

Define

$$K(y, \lambda) = f(y) + \lambda L(y)$$

where λ is a Lagrange multiplier

Then, the necessary conditions for minimum with respect to y becomes

$$\frac{\partial K}{\partial y} = 0 \Leftrightarrow \frac{\partial f}{\partial y} + \lambda \frac{\partial L}{\partial y} = 0 \quad \text{and} \quad L(y) \leq 0$$

If the constraint is not active (Case 1), then

$$\begin{cases} L(y) < 0 \\ \lambda = 0, \ \dfrac{\partial f}{\partial y} = 0 \end{cases}$$

If the constraint is active (Case 2), then

$$\begin{cases} L(y) = 0 \\ \left(\lambda > 0, \ \dfrac{\partial f}{\partial y} > 0, \ \dfrac{\partial L}{\partial y} < 0\right) \Leftrightarrow sign\left(\dfrac{\partial f}{\partial y}\right) = -sign\left(\dfrac{\partial L}{\partial y}\right) \end{cases}$$

cf) For a maximum, $sign(\lambda)$ must be switched

cf) This concept can be used to numerically find optimal solutions

cf) This concept does not violate the Pontryagin's Minimum Principle

- Optimal Control Problem with Control Constraints

Minimize

$$J = h\left(\vec{x}\left(t_f\right), t_f\right) + \int_{t_0}^{t_f} g\left(\vec{x}\left(t\right), \vec{u}\left(t\right), t\right) dt$$

subject to a dynamical system with boundary conditions

$$\dot{\vec{x}} = \vec{f}\left(\vec{x}\left(t\right), \vec{u}\left(t\right), t\right), \quad \vec{x}\left(t_0\right) = \vec{x}_0, \quad \vec{x}\left(t_f\right) = free$$

and inequality constraints

$$C(u(t), t) \leq 0 \tag{1}$$

Define Hamiltonian as

$$H^0 = g + \lambda f \tag{2}$$

$$\delta J = \int_{t_0}^{t_f} H_u^0 \delta u \, dt$$

$$= \int_{t_0}^{t_f} \delta H^0 \, dt$$

when all state and costate necessary conditions are satisfied

Since we must have $\delta J \geq 0$ for all admissible δu

$$\delta H^0 = \frac{\partial H^0}{\partial u} \delta u \geq 0$$

$$\delta C = \frac{\partial C}{\partial u} \delta u \leq 0 \quad (\textit{analogous to the case of function minimization})$$

Define Hamiltonian as

$$H = g + \lambda^T f + \mu^T C \tag{3}$$

Then, the necessary condition on H is

$$\dot{x}(t) = f(x(t), u(t), t)$$

$$\dot{\lambda}(t) = -\frac{\partial g}{\partial x} - \lambda^T \frac{\partial f}{\partial x}$$

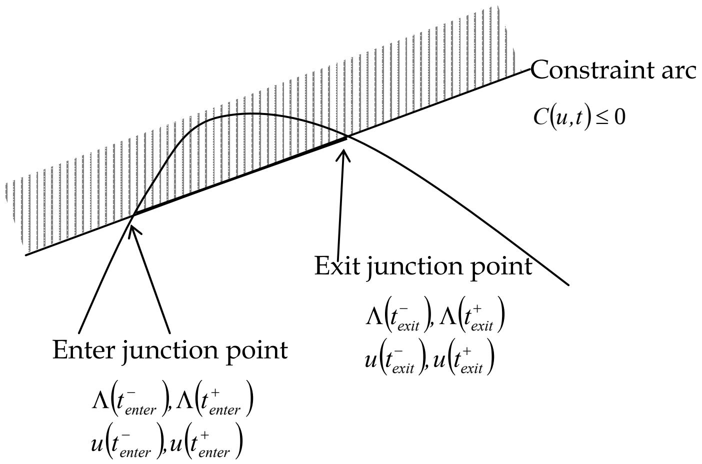
$$0 = \frac{\partial g}{\partial u} + \lambda^T \frac{\partial f}{\partial u} + \mu^T \frac{\partial C}{\partial u}, \quad \mu \geq 0 \tag{4}$$

$\mu > 0 \;\Rightarrow\; C = 0 \;\Rightarrow\;$ (1) determines $u(t)$ on contrained arc

$\mu = 0 \;\Rightarrow\; C < 0 \;\Rightarrow\;$ (4) determines $u(t)$ in uncontrained region

- In solving a particular problem, constrained and unconstrained arcs must be pieced together to satisfy all the necessary conditions.

- At the junction points of constrained and unconstrained arcs, the control may or may not be continuous; if it is discontinuous, the point is called a corner.
- These seems no way to determine the existence of corners a priori
- If $u(t)$ is continuous at a junction point, it follows from the continuity of λ, H_u and H that $\mu(t)$ is continuous across the junction point



4.3.4 Inequality Constraints on State/Control Variables

- State/Control Inequality Constraints

$$C(x,u,t) \le 0 \tag{1}$$

Define Hamiltonian as

$$H = g + \lambda^T f + \mu^T C, \quad \begin{cases} \mu \ge 0, & C = 0 \\ \mu = 0, & C < 0 \end{cases} \tag{2}$$

$$\dot{x} = f$$

$$\dot{\lambda} = -\frac{\partial H}{\partial x} = \begin{cases} -\dfrac{\partial g}{\partial x} - \lambda^T \dfrac{\partial f}{\partial x}, & C < 0 \\ -\dfrac{\partial g}{\partial x} - \lambda^T \dfrac{\partial f}{\partial x} - \mu^T \dfrac{\partial C}{\partial x}, & C = 0 \end{cases} \tag{3}$$

$$0 = \frac{\partial g}{\partial u} + \lambda^T \frac{\partial f}{\partial u} + \mu^T \frac{\partial C}{\partial u}, \quad \begin{cases} \mu = 0, & C < 0 \\ \mu \ge 0, & C = 0 \end{cases} \tag{4}$$

4.3.5 Inequality Constraints on State Variables

- State Inequality Constraints

$$S(x,t) \le 0 \tag{1}$$

Take successive time derivatives of (1) and substitute $f(x,u,t)$ for $\dot{x}$, until u appears explicitly

If u appears first in the q^{th} time derivative, (1) is said to be the q^{th}-order state variable inequality constraint

Now $S^{(q)}(x,u,t)$, the q^{th} time derivative of S, plays the same role as $C(x,u,t)$ in the previous section on state/control inequality constraints:

Define Hamiltonian as

$$H = g + \lambda^T f + \mu^T S^{(q)}(x,u,t), \quad \begin{cases} S^{(q)} = 0, & S = 0 \\ \mu = 0, & S < 0 \end{cases} \tag{2}$$

Tangency constraints apply if the path is on the boundary

$$N(x,t) \equiv \begin{Bmatrix} S(x,t) \\ S^{(1)}(x,t) \\ \vdots \\ S^{(q-1)}(x,t) \end{Bmatrix} = 0, \tag{3}$$

which form a set of interior boundary conditions

Consequently $\lambda(t)$ are, in general, discontinuous at junction points between constrained and unconstrained arcs

Necessary condtions

$$\dot{x} = f$$

$$\dot{\lambda} = -\frac{\partial H}{\partial x} = -\left(\frac{\partial g}{\partial x} + \lambda^T \frac{\partial f}{\partial x} + \mu^T \frac{\partial S^{(q)}}{\partial x} \right)$$

$$0=\frac{\partial g}{\partial u}+\lambda^T\frac{\partial f}{\partial u}+\mu^T\frac{\partial S^{(q)}}{\partial u},\quad \begin{cases}\mu=0, & S^{(q)}<0\\ \mu\geq 0, & S^{(q)}=0\end{cases} \tag{4}$$

4.3.6 Equality Constraints on Control Variables

- Control Equality Constraints

$$C(u,t)=0 \tag{1}$$

where $u(t):R\to R^m$ is the m-dimensional control vector and $C(u(t),t):R^m\times R\to R^p$ is p-dimensional vector

Define Hamiltonian as

$$H=g+\lambda^T f+\mu^T C$$

Necessary Conditions

$$\dot{x}=f \tag{2}$$

$$\dot{\lambda}=-\frac{\partial g}{\partial \vec{x}}-\lambda^T\frac{\partial f}{\partial \vec{x}} \tag{3}$$

$$0=\frac{\partial g}{\partial u}+\lambda^T\frac{\partial f}{\partial u}+\mu^T\frac{\partial C}{\partial u} \tag{4}$$

We have $2n+m+p$ equations of (1)~(4) for $2n+m+p$ unknowns of (x,λ,u,μ)

4.3.7 Equality Constraints on State/Control Variables

- Mixed State/Control Equality Constraints

$$C(x,u,t)=0 \tag{1}$$

where $u(t):R\to R^m$ is the m-dimensional control vector, $\partial C/\partial u\neq 0, \forall u\in R^m$ and $C(u(t),t):R^m\times R\to R^p$ is p-dimensional vector

Define Hamiltonian as

$$H=g+\lambda^T f+\mu^T C$$

Necessary Conditions

$$\dot{x} = f \tag{2}$$

$$\dot{\lambda} = -\frac{\partial g}{\partial \vec{x}} - \lambda^T \frac{\partial f}{\partial \vec{x}} - \mu^T \frac{\partial C}{\partial x} \tag{3}$$

$$0 = \frac{\partial g}{\partial u} + \lambda^T \frac{\partial f}{\partial u} + \mu^T \frac{\partial C}{\partial u} \tag{4}$$

We have *2n+m+p* equations of (1)~(4) for *2n+m+p* unknowns of (x, λ, u, μ)

4.3.8 Equality Constraints on State Variables

- Equality constraint

$$S(x,t) = 0 \tag{1}$$

  - (1) can be added to a boundary condition at $t=0$ or $t=t_f$, or one component of x can be replaced by the remaining $n-1$ components using equation (1)
  - If $\dot{S}$ does not explicitly involve control variables, we may take another time derivative and substitute $\dot{x}=f$, until explicit control variables does occur

$$\begin{bmatrix} S(x,t) \\ S^1(x,t) \\ \vdots \\ S^{(q-1)}(x,t) \end{bmatrix} = 0 \tag{2}$$

$$S^{(q)}(x,u,t) = \frac{d^q S}{dt^q}(x,u,t) = 0, \tag{3}$$

(2) can be added to a set of boundary conditions at $t=0$ or $t=t_f$, or q components of x can be replaced by the remaining $n-q$ components using equation(2)

(3) plays the role of mixed state/control variable equality constraints

Define Hamiltonian as

$$H = g + \lambda^T f + \mu^T S^{(q)}(x,u,t)$$

$$0 = \frac{\partial g}{\partial u} + \lambda^T \frac{\partial f}{\partial u} + \mu^T \frac{\partial S^{(q)}}{\partial u}$$

$$\dot{\lambda} = -\frac{\partial g}{\partial x} - \lambda^T \frac{\partial f}{\partial x} - \mu^T \frac{\partial S^{(q)}}{\partial x}$$

- Example: Minimum-Energy Problem

  Minimize

$$J = \tfrac{1}{2}\int_{t_0}^{t_f} \left(u_1^2 + u_2^2\right) dt$$

subject to the equations of motion

$\dot{x}_1 = x_2$
$\dot{x}_2 = u_1$
$\dot{x}_3 = x_4$
$\dot{x}_4 = u_2$

and a state equality (path) constraint

$$S(x) = x_1^2 + x_3^2 - 1 = 0$$

$x(0)$ is given, $x(t_f)$ is free and t_f is fixed

Path constraint

$$\dot{S}(x) = 2x_1\dot{x}_1 + 2x_3\dot{x}_3 = 2x_1x_2 + 2x_3x_4 = 0$$

$$\ddot{S}(x) = 2\left(\dot{x}_1x_2 + x_1\dot{x}_2 + \dot{x}_3x_4 + x_3\dot{x}_4\right) = 2\left(x_2^2 + x_1u_1 + x_4^2 + x_3u_2\right) = 0 \qquad (1)$$

Define Hamiltonian as

$$H = \frac{1}{2}\left(u_1^2 + u_2^2\right) + \lambda_1 x_2 + \lambda_2 u_1 + \lambda_3 x_4 + \lambda_4 u_2 + \mu\left(x_2^2 + x_1u_1 + x_4^2 + x_3u_2\right)$$

State Equations

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= u_1 \\ \dot{x}_3 &= x_4 \\ \dot{x}_4 &= u_2 \end{aligned} \qquad (2)$$

Costate Equations

$$\dot{\lambda} = -\frac{\partial H}{\partial x} = -\left(\frac{\partial g}{\partial x} + \lambda^T \frac{\partial f}{\partial x} + \mu^T \frac{\partial S^{(2)}}{\partial x}\right) \tag{3}$$

Optimality Conditions

$$\frac{\partial H}{\partial u_1} = u_1 + \lambda_2 + \mu x_1 = 0 \Rightarrow u_1 = -\left(\lambda_2 + \mu x_1\right) \tag{4}$$

$$\frac{\partial H}{\partial u_2} = u_2 + \lambda_4 + \mu x_3 = 0 \Rightarrow u_2 = -\left(\lambda_4 + \mu x_3\right) \tag{5}$$

Substituting (4) and (5) into (1) leads to

$$\begin{aligned}
& x_2^2 + x_1 u_1 + x_4^2 + x_3 u_2 = 0 \\
\Rightarrow\ & x_2^2 - x_1\left(\lambda_2 + \mu x_1\right) + x_4^2 - x_3\left(\lambda_4 + \mu x_3\right) = 0 \\
\Rightarrow\ & x_2^2 + x_4^2 - x_1\lambda_2 - x_3\lambda_4 - \mu\left(x_1^2 + x_3^2\right) = 0 \\
\Rightarrow\ & \left(x_1^2 + x_3^2\right)\mu = x_2^2 + x_4^2 - x_1\lambda_2 - x_3\lambda_4 \\
& \left(\Uparrow\ S(x) = x_1^2 + x_3^2 - 1 = 0\right)
\end{aligned}$$

$$\Rightarrow \mu = x_2^2 + x_4^2 - x_1\lambda_2 - x_3\lambda_4 \tag{6}$$

Boundary Conditions

$x(0)$ is given $\Rightarrow$ 4 boundary conditions

$x(t_f)$ must be on $S(x(t_f)) = 0 \quad \Rightarrow \quad x_1^2(t_f) + x_3^2(t_f) - 1 = 0$

$x(t_f)$ is free $\lambda\left(t_f\right) = \nu^T \dfrac{\partial S(x(t_f))}{\partial \vec{x}} \quad \left(\Leftarrow \lambda(t_f) = \dfrac{\partial h(x(t_f))}{\partial x} + \nu^T \dfrac{\partial m(x(t_f))}{\partial x}\right)$

We have *12* equations of (1)~(5) for *12* unknowns of $(x, \lambda, u, \mu, \nu)$

4.3.6 Discontinuities in the State Vairables at Interior Points

- Optimal Control Problem with Interior Boundary Conditions:
  Minimize

$$J = h\left(\vec{x}\left(t_f\right), t_f\right) + \int_{t_0}^{t_f} g\left(\vec{x}\left(t\right), \vec{u}\left(t\right), t\right) dt$$

subject to a dynamical system

$$\dot{\vec{x}} = \vec{f}\left(\vec{x}\left(t\right), \vec{u}\left(t\right), t\right), \quad \vec{x}\left(t_0\right) = \vec{x}_0$$

with a set of interior boundary conditions

$$N\left(\vec{x}(t_1), t_1\right) = 0 \tag{1}$$

Here $t_1 \in \left[t_0, t_f\right]$ is an intermediate time and N is a q-component vector function

This becomes a three-point boundary-value problem instead of two-point boundary-value problem
(1) represents a set of terminal constraints for the part of the path from t_0 to t_1

- Example:
  Minimize

  $$J = \int_0^1 g\left(x, u\right) dt$$

  subject to the dyanmics

  $$\dot{x} = f(x, u, t), \quad x(0) = x_0, \quad x(1) = 0$$

  with an interior point constraint representing a jump at t_1:

  $$x\left(t_1^+\right) = x\left(t_1^-\right) + 1, \quad t_1 \in \left[0, 1\right]$$

  Derive Necessary Conditions for Optimality and determine whether the Hamiltonian is continuous/constant or not

  Start by bisecting the time interval of integration

  $$J = \int_0^{t_1^-} g\left(x, u\right) dt + \int_{t_1^+}^1 g\left(x, u\right) dt$$

  Adjoin the dynamics to the performance index by using the Lagrange multiplier functions $\lambda(t)$ and ν

$$J_a = \nu\left(x\left(t_1^+\right) - x\left(t_1^-\right) - 1\right)$$
$$+\int_0^{t_1^-}\left[g\left(x,u\right) + \lambda\left(f - \dot{x}\right)\right]dt + \int_{t_1^+}^{1}\left[g\left(x,u\right) + \lambda\left(f - \dot{x}\right)\right]dt$$

Define

$$H \equiv g\left(x,u\right) + \lambda f$$

$$\Phi \equiv \nu\left(x\left(t_1^+\right) - x\left(t_1^-\right) - 1\right)$$

$$\begin{aligned}
\Delta J_a \simeq \delta J_a &= J\left(x+\delta x, u+\delta u, \lambda+\delta\lambda\right) - J\left(x,u,\lambda\right) \\
&= \Phi\left[x\left(t_1^-\right)+\delta x\left(t_1^-\right), x\left(t_1^+\right)+\delta x\left(t_1^+\right), \delta t_1\right] - \Phi\left[x\left(t_1^-\right), x\left(t_1^+\right), t_1\right] \\
&+\int_0^{t_1^-}\left[H\left(x+\delta x, u+\delta u, \lambda+\delta\lambda\right) - H\left(x,u,\lambda\right) - \left\{\lambda\left(\dot{x}+\delta\dot{x}\right) + \dot{x}\delta\lambda - \lambda\dot{x}\right\}\right]dt \\
&+\left[H\left(t_1^-\right) - \lambda\dot{x}\left(t_1^-\right)\right]\delta t_1 \\
&+\int_{t_1^+}^{1}\left[H\left(x+\delta x, u+\delta u, \lambda+\delta\lambda\right) - H\left(x,u,\lambda\right) - \left\{\lambda\left(\dot{x}+\delta\dot{x}\right) + \dot{x}\delta\lambda - \lambda\dot{x}\right\}\right]dt \\
&-\left[H\left(t_1^+\right) - \lambda\dot{x}\left(t_1^+\right)\right]\delta t_1 \\
&= \frac{\partial\Phi}{\partial x\left(t_1^-\right)}\delta x_{t_1^-} + \frac{\partial\Phi}{\partial x\left(t_1^+\right)}\delta x_{t_1^+} + \frac{\partial\Phi}{\partial t_1}\delta t_1 \\
&+\int_0^{t_1^-}\left[\frac{\partial H}{\partial x}\delta x + \frac{\partial H}{\partial u}\delta u + \frac{\partial H}{\partial\lambda}\delta\lambda - \dot{x}\delta\lambda - \lambda\delta\dot{x}\right]dt + \left(H - \lambda\dot{x}\right)_{t_1^-}\delta t_1 \\
&+\int_{t_1^+}^{1}\left[\frac{\partial H}{\partial x}\delta x + \frac{\partial H}{\partial u}\delta u + \frac{\partial H}{\partial\lambda}\delta\lambda - \dot{x}\delta\lambda - \lambda\delta\dot{x}\right]dt - \left(H - \lambda\dot{x}\right)_{t_1^+}\delta t_1
\end{aligned}$$

Integration by parts leads to

$$\begin{aligned}
\delta J_a &= \frac{\partial\Phi}{\partial x\left(t_1^-\right)}\delta x_{t_1^-} + \frac{\partial\Phi}{\partial x\left(t_1^+\right)}\delta x_{t_1^+} + \frac{\partial\Phi}{\partial t_1}\delta t_1 \\
&-\lambda\delta x\Big|_0^{t_1^-} + \int_0^{t_1^-}\left[\left(\frac{\partial H}{\partial x} + \dot{\lambda}\right)\delta x + \frac{\partial H}{\partial u}\delta u + \left(\frac{\partial H}{\partial\lambda} - \dot{x}\right)\delta\lambda\right]dt + \left(H - \lambda\dot{x}\right)_{t_1^-}\delta t_1 \\
&-\lambda\delta x\Big|_{t_1^+}^{1} + \int_{t_1^+}^{1}\left[\left(\frac{\partial H}{\partial x} + \dot{\lambda}\right)\delta x + \frac{\partial H}{\partial u}\delta u + \left(\frac{\partial H}{\partial\lambda} - \dot{x}\right)\delta\lambda\right]dt - \left(H - \lambda\dot{x}\right)_{t_1^+}\delta t_1
\end{aligned}$$

Recall that

$$\delta x_{t_1^\pm} = \delta x\left(t_1^\pm\right) + \dot{x}\left(t_1^\pm\right)\delta t_1 \;\Rightarrow\; \delta x\left(t_1^\pm\right) = \delta x_{t_1^\pm} - \dot{x}\left(t_1^\pm\right)\delta t_1,$$

which leads to

$$-\lambda \delta x\Big|_0^{t_1^-} \;\Rightarrow\; -[\lambda\delta x(t_1^-) - \lambda\delta x(0)] = -\lambda[\delta x_{t_1^-} - \dot{x}\left(t_1^-\right)\delta t_1]$$

$$-\lambda \delta x\Big|_{t_1^+}^{1} \;\Rightarrow\; -\left[\lambda\delta x(1) - \lambda\delta x(t_1^+)\right] = \lambda\left[\delta x_{t_1^+} - \dot{x}\left(t_1^+\right)\delta t_1\right]$$

and ultimately

$$\begin{aligned}
\delta J_a = &\left[\frac{\partial \Phi}{\partial t_1} + \lambda\dot{x}\left(t_1^-\right) - \lambda\dot{x}\left(t_1^+\right) + H\left(t_1^-\right) - \lambda\dot{x}\left(t_1^-\right) - H\left(t_1^+\right) + \lambda\dot{x}\left(t_1^+\right)\right]\delta t_1 \\
&+\left[\frac{\partial \Phi}{\partial x\left(t_1^-\right)} - \lambda\left(t_1^-\right)\right]\delta x_{t_1^-} + \left[\frac{\partial \Phi}{\partial x\left(t_1^+\right)} + \lambda\left(t_1^+\right)\right]\delta x_{t_1^+} \\
&+\int_0^{t_1^-}\left[\left(\frac{\partial H}{\partial x} + \dot{\lambda}\right)\delta x + \frac{\partial H}{\partial u}\delta u + \left(\frac{\partial H}{\partial \lambda} - \dot{x}\right)\delta\lambda\right]dt \\
&+\int_{t_1^+}^{1}\left[\left(\frac{\partial H}{\partial x} + \dot{\lambda}\right)\delta x + \frac{\partial H}{\partial u}\delta u + \left(\frac{\partial H}{\partial \lambda} - \dot{x}\right)\delta\lambda\right]dt
\end{aligned}$$

The fundamental theorem of the Calculus of Variations, $\delta J = 0$, gives the following necessary conditions for optimality

$$\dot{x} = f(x,u,t),\; x(t_0) = x_0$$

$$\dot{\lambda} = -\frac{\partial H}{\partial x},\quad \lambda\left(t_1^-\right) = \frac{\partial \Phi}{\partial x\left(t_1^-\right)},\quad \lambda\left(t_1^+\right) = -\frac{\partial \Phi}{\partial x\left(t_1^+\right)}$$

$$0 = \frac{\partial H}{\partial u}$$

$$H\left(t_1^+\right) = H\left(t_1^-\right) + \frac{\partial \Phi}{\partial t_1} \quad\Rightarrow\quad H\left(t_1^-\right) = H\left(t_1^+\right) \;\; if \;\; \frac{\partial \Phi}{\partial t_1} = 0$$

- Generalization
  - There can be discontinuities in the state variables as well as discontinuities in the system equations at interior points
  - Performance Index and Constraints may be functions of the state and/or time at several discrete points

Minimize

$$J = \phi\left[x\left(t_0^-\right), x\left(t_0^+\right), \cdots, x\left(t_N^-\right), x\left(t_N^+\right); t_0, \cdots, t_N\right] + \sum_{i=1}^{N} \int_{t_{i-1}^+}^{t_i^-} L^{(i)}\left(x(t), u(t), t\right) dt \tag{1}$$

subject to the dynamics

$$\dot{x} = f^{(i)}\left(x(t), u(t), t\right), \quad t_{i-1} < t < t_i, \quad i = 1, 2, \cdots, N \tag{2}$$

and constraints

$$\psi^{(j)}\left[x\left(t_0^-\right), x\left(t_0^+\right), \cdots, x\left(t_N^-\right), x\left(t_N^+\right); t_0, \cdots, t_N\right] = 0, \quad j = 1, 2, \cdots, N \tag{3}$$

Here $x(t_i^-)$ and $x(t_i^+)$ signify the state vector just before and after $t=t_i$, respectively

Start by adjoining the dynamic equation (2) and the interior point constraints (3) to the performance index (1) by using the Lagrange multiplier $\lambda(t)$ and $\nu^{(i)}$, respectively

$$J_a = \phi + \sum_{j=0}^{N} \left[\nu^{(j)}\right]^T \psi^{(j)} + \sum_{i=1}^{N} \int_{t_{i-1}^+}^{t_i^-} \left[L^{(i)} + \lambda^T f^{(i)} - \lambda^T \dot{x}\right] dt \tag{4}$$

Define

$$H^{(i)} \equiv L^{(i)} + \lambda^T f^{(i)}, \quad i = 1, 2, \cdots, N \tag{5}$$

$$\Phi = \phi + \sum_{j=0}^{N} \left[\nu^{(j)}\right]^T \psi^{(j)} \tag{6}$$

Take the first variation of (4) and perform intergration by parts:

$$\begin{aligned}
\delta J_a = & \sum_{i=0}^{N} \left[\frac{\partial \Phi}{\partial t_i} \delta t_i + \frac{\partial \Phi}{\partial x\left(t_i^-\right)} \delta x_{t_i}^- + \frac{\partial \Phi}{\partial x\left(t_i^+\right)} \delta x_{t_i}^+\right] \\
& + \sum_{i=1}^{N} \left\{\left[H^{(i)} - \lambda^T \dot{x}\right]_{t=t_i^-} \delta t_i - \left[H^{(i+1)} - \lambda^T \dot{x}\right]_{t=t_i^+} \delta t_i\right\} \\
& + \sum_{i=1}^{N} \left\{\left[-\lambda \delta x\right]_{t_{i-1}^+}^{t_i^-} + \int_{t_{i-1}^+}^{t_i^-} \left[\left(\dot{\lambda}^T + \frac{\partial H^{(i)}}{\partial x}\right) \delta x + \frac{\partial H^{(i)}}{\partial u} \delta u + \left(\frac{\partial H^{(i)}}{\partial \lambda} - \dot{x}\right) \delta \lambda\right] dt\right\} \\
= & \ 0
\end{aligned} \tag{7}$$

Eliminate $\delta x(t_i^{\pm})$ from equation (7) by using the following relations

$$\delta x_{t_i}^{\pm} = \delta x\left(t_i^{\pm}\right) + \dot{x}\left(t_i^{\pm}\right)\delta t_i \;\Rightarrow\; \delta x\left(t_i^{\pm}\right) = \delta x_{t_i}^{\pm} - \dot{x}\left(t_i^{\pm}\right)\delta t_i \tag{8}$$

and rearrange as follows:

$$\begin{aligned}\partial J_a = & \sum_{i=0}^{N}\left[\frac{\partial \Phi}{\partial t_i} + H^{(i)}\left(t_i^-\right) - H^{(i+1)}\left(t_i^+\right) + \lambda\dot{x}(t_i^-) - \lambda\dot{x}(t_i^+) - \lambda\dot{x}(t_i^-) + \lambda\dot{x}(t_i^+)\right]\delta t_i \\ & + \sum_{i=1}^{N}\left[\left(\frac{\partial \Phi}{\partial x\left(t_i^-\right)} - \lambda^T\left(t_i^-\right)\right)\delta x_{t_i}^-\right] + \sum_{i=0}^{N-1}\left[\left(\frac{\partial \Phi}{\partial x\left(t_i^+\right)} + \lambda^T\left(t_i^+\right)\right)\delta x_{t_i}^+\right] \\ & + \sum_{i=1}^{N}\int_{t_{i-1}^+}^{t_i^-}\left[\left(\dot{\lambda}^T + \frac{\partial H^{(i)}}{\partial x}\right)\delta x + \frac{\partial H^{(i)}}{\partial u}\delta u + \left(\frac{\partial H^{(i)}}{\partial \lambda} - \dot{x}\right)\delta\lambda\right]dt \\ = & \; 0\end{aligned} \tag{9}$$

where note that

$$H^{(0)} = H^{(N+1)} \equiv 0$$

Finally, the fundamental theorem of the Calculus of Variations, $\delta J = 0$, gives the following necessary conditions for optimality

$$\dot{x} = f^{(i)}\left(x(t), u(t), t\right), \;\; t_{i-1} < t < t_i, \;\; i = 1, 2, \cdots, N$$

$$\dot{\lambda} = -\frac{\partial H^{(i)}}{\partial x}, \;\; t_{i-1}^+ < t < t_i^-, \;\; i = 1, \cdots, N \tag{10}$$

$$\lambda\left(t_i^-\right) = \frac{\partial \Phi}{\partial x\left(t_i^-\right)}, \;\; i = 1, \cdots, N \tag{11}$$

$$\lambda\left(t_i^+\right) = -\frac{\partial \Phi}{\partial x\left(t_i^+\right)}, \;\; i = 0, \cdots, N-1 \tag{12}$$

$$0 = \frac{\partial H^{(i)}}{\partial u} \tag{13}$$

$$H^{(i+1)}\left(t_i^+\right) = H^{(i)}\left(t_i^-\right) + \frac{\partial \Phi}{\partial t_i}, \;\; i = 0, \cdots, N \tag{14}$$

$$\delta\psi^{(j)} = \sum_{i=0}^{N}\left[\frac{\partial\psi^{(j)}}{\partial t_i}\delta t_i + \frac{\partial\psi^{(j)}}{\partial x\left(t_i^-\right)}\delta x_i^- + \frac{\partial\psi^{(j)}}{\partial x\left(t_i^+\right)}\delta x_i^+\right] = 0,\quad j = 0,\cdots,N \qquad (15)$$

If $x\left(t_0^+\right)$ is specified, $\delta x_0^+ = 0$ in (9) $\Rightarrow$ (12) for $i = 0$ is not necessary

If t_i is specified, $\delta t_i = 0$ in (9) $\Rightarrow$ (14) is not necessary

We must also choose $\nu^{(j)}$ to satisfy the conditions $\psi^{(j)} = 0$ in (14)

5 Numerical Determination of Optimal Control

5.1 Introduction: Direct vs. Indirect Methods

- Optimal Control Laws can be found for
  - Problems with linear plant dynamics and quadratic performance index by numerically integrating the Riccati differential equation
  - Some simple nonlinear problems by applying the Pontryagin principle

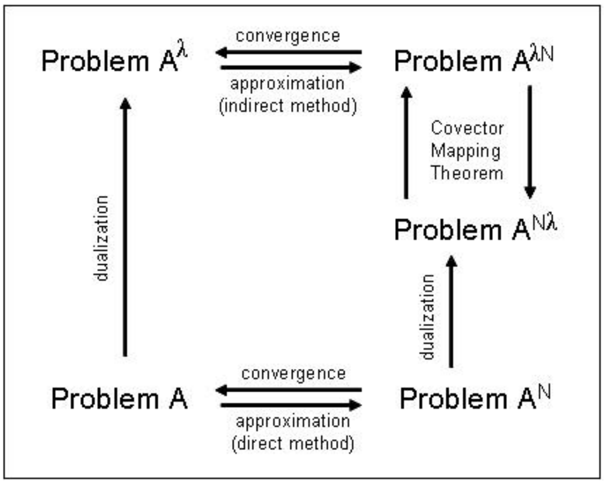
- In general, the variational approach
  - Leads to a nonlinear two-point boundary-value problem (TPBVP)
  - Usually cannot be solved analytically to obtain the optimal control law
  - Sometimes not even easy to obtain an open-loop control
  - Motivates to go for numerical techniques

- **Direct** and **Indirect** Methods:
  - Modern computational (=numerical) techniques for solving optimal control problems
    - Direct Method first discretizes the infinite dimensional optimal control problem to form a finite dimensional parameter optimization problem, and then solves it as a nonlinear programming (NLP) problem
    - Indirect Method first derives the necessary/sufficient conditions for optimality and then solves the resultant ordinary/partial differential equations with the associated boundary conditions
  - Historically they have developed together, while sometimes competing and sometimes cooperating with each other
    - Indirect Method first led a significant growth in both theory and applications with the birth of modern optimal control theory by Bellman and Pontryagin in the 1960s
    - More recently, the direct method has gained the most attention with the advancement of modern computing facilities

- Direct Method

- First discretize the optimal control problem to form a parameter optimization problem
  - Euler
  - Runge-Kutta
  - Hermite-Simpson
  - Pseudospectral methods-Legendre, Gauss, Chevychef, etc.
- Then solve it as a nonlinear programming (NLP) problem
  - Compatible with modern digital computers
  - Have been extensively applied to aerospace control problems, such as lunar guidance, orbit transfers, tether liberation control, magnetic control, and ascent guidance

- Advantages
  - Formulation is straightforward
  - Can be easily applied to even complex engineering problems
  - As the radius of convergence of the associated NLP is relatively large, it can be solved with rather rough initial estimates on decision variables
- Drawbacks
  - Simplicity belies some fundamental theoretical issues lying at the intersection of approximation and control theory
  - Standard convergence theorems employed in the analysis of differential equations are not applicable to discretizations of optimal control problems, and vice versa, in general
  - Not all the direct methods provide discretized solutions which converge to the continuous solution of the original optimal control problem



- Indirect Method
  - First derive the necessary conditions for optimality (or the Hamilton-Jacobi-Bellman equation) from the Minimum Principle (or the Principle of Optimality)
    - There exist many varieties derived from these two fundamental approaches
    - Asserts that the indirect method is, in itself, an active research field
  - Then solve the resultant multi-point-boundary-value-problem (or the HJB PDE) numerically
    - Simple Shooting
    - Multiple Shooting
    - *Generating Function Method

  - Drawbacks
    - Inherent difficulty in deriving the necessary conditions for optimality for complex engineering problems
    - The resultant boundary value problem is highly sensitive to initial estimates
    - Predict the auxiliary costate variables rather accurately without any physical interpretations, in general
  - Advantages
    - (Once technical drawbacks are resolved)
    - Usually shows faster convergence
    - Develops more accurate solutions than the direct optimization method

- Note: Classfication of Direct vs. Indirect method is for convenience of anlaysis
  - Some computational optimal control techniques do not obviously fall into direct or indirect methods: (ex) time-domain finite element method
  - Historical perspectives and characteristic contrasts between Direct and Indirect methods suggest that it is still premature to conclude that one method is superior to the other

- Simple Example: Direct vs. Indirect
  Minimize

$$J = \int_0^1 u^2(t)dt$$

subject to the dyanmics

$$\dot{x} = u, \quad x(t_0) = x_0, \quad x(t_f) = x_f$$

with an interior point constraint at t_1:

$$N(x(t_1), t_1) = 0$$

Show the process of direct and indirect method

5.2 Optimization (of Functions): Gradient Method

- Consider an unconstrained minimization of a function:
  Minimize

$$J = f(x)$$

  - First Order Gradient Method (Steepest Descent Method)

$$x^{k+1} = x^k - \alpha \frac{\partial f}{\partial x}, \quad \frac{\partial f}{\partial x}: \text{ gradient (Jacobian)}$$

where $\alpha > 0$ is the step size and is determined from one-dimensional line search

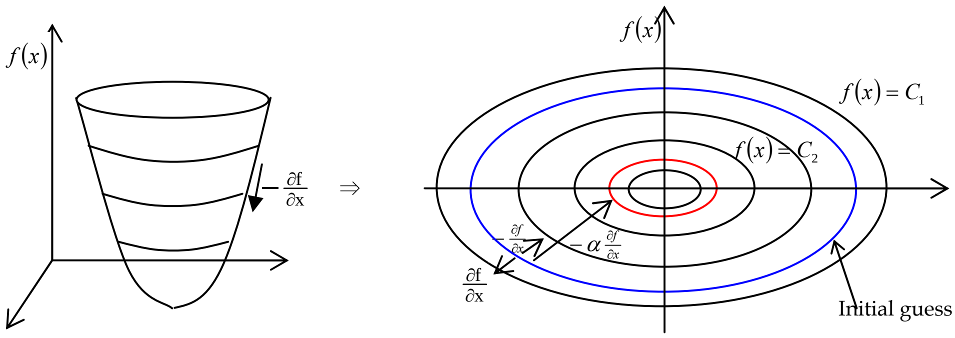
f has a minimum, if the increment in f becomes zero, i.e.,

$$df(x) = \frac{\partial f}{\partial x}\Delta x = 0$$

$$\left.\frac{\partial f}{\partial x}\right|_{x=x^*} = 0 \quad \text{at a relative minimum}$$

Convegence is guaranteed if $\alpha > 0$ is sufficiently small, possibly after many iterations

If f is highly nonlinear, then oscillation occurs in the process



■ Second Order Gradient Method (Gauss-Newton Method)

$$f\left(x^k\right) = f\left(x^* + \Delta x\right), \quad x^k \textit{ is the current guess}$$

$$\simeq f\left(x^*\right) + \left.\frac{\partial f}{\partial x}\right|_{x=x^*}\Delta x + \frac{1}{2}\Delta x^T \left.\frac{\partial^2 f}{\partial x^2}\right|_{x=x^*}\Delta x$$

$$\simeq f\left(x^*\right) + \underbrace{\left.\frac{\partial f}{\partial x}\right|_{x=x^k}}_{G^k;Jacobian}\Delta x + \frac{1}{2}\Delta x^T \underbrace{\left.\frac{\partial^2 f}{\partial x^2}\right|_{x=x^k}}_{H^k;Hessian}\Delta x$$

Miniminze $f(x)$ with respect to Δx

$$\Rightarrow \frac{\partial f}{\partial \Delta x} = 0$$

$$\Rightarrow \frac{\partial\left[f\left(x^*\right) + G^k\Delta x + \frac{1}{2}\Delta x^T H^k \Delta x\right]}{\partial \Delta x}$$

$$\Rightarrow G^k + H^k\Delta x = 0$$

$$\Rightarrow \Delta x = -\left[H^k\right]^{-1} G^k$$

$$\Rightarrow x^{k+1} = x^k - \alpha \left[H^k \right]^{-1} G^K,\ \alpha > 0$$ is the step size

The Hessian must be positive definite

$$H^k > 0$$

$$\Rightarrow \Delta f \simeq \frac{\partial f}{\partial x} \Delta x = -G^k \left[H^k \right]^{-1} G^k < 0$$

$$\Leftrightarrow \Delta f < 0$$

$\Rightarrow f(x)$ decreases in each step

Problems

- *H* may not be positive define for many noninear problems
- *H* is expensive to evaluate $\Rightarrow$ Use Quasi-Newton method to approximate the inverse of H directly

5.3 Simple Optimal Control Problem

- Minimize

$$J = h\left(x\left(t_f\right), t_f\right) + \int_0^{t_f} g\left(x, u, t\right) dt$$

subject to the dyanmics and boundary conditions

$$\dot{x} = f(x, u, t),\ x(t_0) = x_0,\ x(t_f) = free,\ t_f \text{ is fixed}$$

- Numerical Procedure

I. Guess $u(t)$

II. Integrate the state equations forward in time:

$$\dot{x} = f\left(x, u, t\right),\ x(t_0) = x_0$$

Store the state history $x(t)$

III. Integrate the costate equations backward in time:

$$\dot{\lambda} = -\frac{\partial H(x,\lambda,u,t)}{\partial x}, \quad \lambda\left(t_f\right) = \frac{\partial h\left(x\left(t_f\right),t_f\right)}{\partial x\left(t_f\right)}$$

Store the costate history $\lambda\left(t\right)$

IV. Evauate the history of optimality condition:

$$\left\|\frac{\partial H(x,\lambda,u,t)}{\partial u}\right\|$$

V. If $\left\|\partial H/\partial u\right\| < \varepsilon$ for a pre-selected tolerance $\varepsilon > 0$, terminate the iteration procedure

If not, choose

$$\delta u^k(t) = -\alpha \frac{\partial H}{\partial u}(t), \quad \alpha > 0$$

which minimizes the Hamiltonian *H* with respect to u and decreases *J* since

$$\begin{aligned}
\delta J_a &= \left[\frac{\partial h}{\partial x\left(t_f\right)} - \lambda\left(t_f\right)\right]\delta x\left(t_f\right) \\
&+ \int_0^{t_f}\left[\left(\dot{\lambda} + \frac{\partial H}{\partial x}\right)\delta x + \left(\frac{\partial H}{\partial u}\right)\delta u + \left(\frac{\partial H}{\partial \lambda} - \dot{x}\right)\delta\lambda\right]dt \\
&= \cdots + \int_0^{t_f}\frac{\partial H^k}{\partial u}(t)\delta u^k(t)dt \\
&= \cdots + \int_0^{t_f}\frac{\partial H^k}{\partial u}(t)\left[-\alpha\frac{\partial H^k}{\partial u}(t)\right]dt < 0
\end{aligned}$$

VI. Update

$$u^{k+1}(t) = u^k(t) + \delta u^k(t) = u^k(t) - \alpha\frac{\partial H}{\partial u}(t)$$

and go to Step II

5.4 Optimal Control Problem with Terminal Constraints at Fixed Final Time

- Minimize

$$J = \phi(x(t_f)) + \int_{t_0}^{t_f} L(x,u,t)dt$$

subject to the dynamics and boundary conditions

$$\dot{x} = f(x,u,t),\ \ x(t_0) = x_0,\ \ \psi(x(t_f)) = 0,\ \ t_f \textit{ is fixed}$$

Necessary Conditions for a Stationary Path

Define Hamiltonian as

$$H = L + \lambda^T f$$

$$\dot{x} = f(x,u,t),\ \ x(t_0) = x_0$$

$$\dot{\lambda} = -\frac{\partial H}{\partial x} = -\frac{\partial L}{\partial x} - \frac{\partial f}{\partial x}\lambda,\ \ \lambda(t_f) = \frac{\partial \Phi(x(t_f))}{\partial x(t_f)},\ \ \Phi(x(t_f)) = \phi(x(t_f)) + \nu^T \psi(x(t_f))$$

$$0 = \frac{\partial H}{\partial u} = \frac{\partial L}{\partial u} + \lambda^T \frac{\partial f}{\partial u}$$

This forms a TPBVP for $x(t)$ and $\lambda(t)$ where we must also find ν to satisfy the terminal constraints $\psi(x(t_f)) = 0$

$$x(t) : R \rightarrow R^{n\times 1}$$
$$\lambda(t) : R \rightarrow R^{n\times 1}$$
$$u(t) : R \rightarrow R^{m\times 1}$$
$$\psi \in R^{q\times 1}$$
$$\nu \in R^{q\times 1}$$

The augmented performance index is given as

$$J_a = \phi(x(t_f)) + \nu^T \psi(x(t_f)) + \int_{t_0}^{t_f} \left[L(x,u) + \lambda^T (f - \dot{x}) \right] dt$$

$$
\begin{aligned}
\delta J_a &= \frac{\partial \phi}{\partial x_f}\delta x_f + \nu^T \frac{\partial \psi}{\partial x_f}\delta x_f \\
&\quad + \int_{t_0}^{t_f}\left[\left(\frac{\partial L}{\partial x} + \lambda^T \frac{\partial f}{\partial x}\right)\delta x + \left(\frac{\partial L}{\partial u} + \lambda^T \frac{\partial f}{\partial u}\right)\delta u - \lambda^T \delta \dot{x} + \delta\lambda\left(f - \dot{x}\right)\right]dt \\
&= \left[\frac{\partial \phi}{\partial x_f} + \nu^T \frac{\partial \psi}{\partial x_f} - \lambda\left(t_f\right)\right]\delta x_f \\
&\quad + \int_{t_0}^{t_f}\left[\left(\frac{\partial L}{\partial x} + \lambda^T \frac{\partial f}{\partial x} + \dot{\lambda}^T\right)\delta x + \left(\frac{\partial L}{\partial u} + \lambda^T \frac{\partial f}{\partial u}\right)\delta u + \delta\lambda\left(f - \dot{x}\right)\right]dt
\end{aligned}
$$

Inspired by the form of the terminal condition $\Phi(t_f)$, we define the costate history $\lambda(t)$ as

$$
\begin{aligned}
&\Phi(t_f) = \phi(x(t_f)) + \nu^T \psi(x(t_f)) \\
&\Rightarrow \lambda(t) = \lambda^\phi(t) + \nu^T \lambda^\psi(t)
\end{aligned}
$$

Then, the augmented functional is rephrased as

$$
\begin{aligned}
\delta J_a &= \left[\frac{\partial \phi}{\partial x_f} + \nu^T \frac{\partial \psi}{\partial x_f} - \lambda^\phi\left(t_f\right) - \nu^T \lambda^\psi\left(t_f\right)\right]\delta x_f \\
&\quad + \int_{t_0}^{t_f}\left[\begin{array}{l}\left(\frac{\partial L}{\partial x} + \left(\lambda^\phi + \nu^T \lambda^\psi\right)^T \frac{\partial f}{\partial x} + \left(\dot{\lambda}^\phi + \nu^T \dot{\lambda}^\psi\right)^T\right)\delta x \\ \quad + \left(\frac{\partial L}{\partial u} + \left(\lambda^\phi + \nu^T \lambda^\psi\right)^T \frac{\partial f}{\partial u}\right)\delta u + \delta\lambda\left(f - \dot{x}\right)\end{array}\right]dt
\end{aligned}
$$

Once the following conditions are satisfied

$$\lambda^\phi(t_f) = \frac{\partial \phi}{\partial x_f},\ \lambda^\psi\left(t_f\right) = \frac{\partial \psi}{\partial x_f} \Rightarrow \text{ the coefficient of } \delta x_f \text{ vanishes}$$

$$\dot{\lambda}^\phi(t) = -\frac{\partial L}{\partial x} - \lambda^\phi \frac{\partial f}{\partial x},\ \dot{\lambda}^\psi(t) = -\lambda^\psi \frac{\partial f}{\partial x} \Rightarrow \text{ the coefficient of } \delta x \text{ vanishes}$$

$$\dot{x} = f \Rightarrow \text{ the coefficient of } \delta\lambda \text{ vanishes}$$

the augmented function is reduced to

$$
\begin{aligned}
\delta J_a &= \int_{t_0}^{t_f} \left(\frac{\partial L}{\partial u} + \lambda^{\phi} \frac{\partial f}{\partial u} + v^T \lambda^{\psi} \frac{\partial f}{\partial u} \right) \delta u dt \\
&= \int_{t_0}^{t_f} \left(\frac{\partial L}{\partial u} + \lambda^{\phi} \frac{\partial f}{\partial u} \right) \delta u dt + v^T \int_{t_0}^{t_f} \left(\lambda^{\psi} \frac{\partial f}{\partial u} \right) \delta u dt \\
&= \int_{t_0}^{t_f} H_u^{\phi} \delta u dt + v^T \int_{t_0}^{t_f} H_u^{\psi} \delta u dt,\ H_u^{\phi} \equiv \frac{\partial L}{\partial u} + \lambda^{\phi} \frac{\partial f}{\partial u},\ H_u^{\psi} \equiv \lambda^{\psi} \frac{\partial f}{\partial u} \\
&= \int_{t_0}^{t_f} \left(H_u^{\phi} + v^T H_u^{\psi} \right) \delta u dt \\
&\equiv \delta J_1 + v^T \delta \psi
\end{aligned}
$$

For every iteration, it is required that $\delta J_a < 0$ to decrease J, from which δu is chosen to be

$$
\delta u = -k \left[H_u^{\phi} + v^T H_u^{\psi} \right],\ k > 0
$$

since

$$
\delta J_a = \int_{t_0}^{t_f} \left(H_u^{\phi} + v^T H_u^{\psi} \right) \delta u dt = -k \int_{t_0}^{t_f} \left(H_u^{\phi} + v^T H_u^{\psi} \right)^T \left(H_u^{\phi} + v^T H_u^{\psi} \right) dt < 0
$$

Then,

$$
\begin{aligned}
\delta \psi &\equiv \int_{t_0}^{t_f} \left(\lambda^{\psi} \frac{\partial f}{\partial u} \delta u \right) dt \\
&= -k \int_{t_0}^{t_f} \left(\lambda^{\psi} \frac{\partial f}{\partial u} (H_u^{\phi} + v^T H_u^{\psi}) \right) dt \\
&= -k \int_{t_0}^{t_f} \left((H_u^{\phi} + v^T H_u^{\psi})^T H_u^{\psi} \right) dt \\
&= -k \left[\int_{t_0}^{t_f} \left[H_u^{\phi} \right]^T H_u^{\psi} dt + v^T \int_{t_0}^{t_f} \left[H_u^{\psi} \right]^T H_u^{\psi} dt \right] \\
&\equiv -k(g + v^T Q),\ g \in R^{q \times 1},\ Q \in R^{q \times q}
\end{aligned}
$$

We want $\delta \psi \to 0$ until the procedure ends

$$
\begin{aligned}
&\delta \psi \simeq -\eta \psi \\
&\Rightarrow -\eta \psi = -k \left(g + v Q \right) \\
&\Rightarrow v = \left(-\frac{g}{Q} + \frac{\eta \psi}{k Q} \right)
\end{aligned}
$$

Finally, the increment in control u is given as

$$\begin{aligned}\delta u &= -k\left[H_u^{\phi} + \left(-\frac{g}{Q} + \frac{\eta\psi}{kQ}\right)H_u^{\psi}\right] \\ &= -k\left[H_u^{\phi} - \frac{g}{Q}H_u^{\psi}\right] - k\frac{\eta\psi}{kQ}H_u^{\psi} \\ \Rightarrow \delta u &= -k\left[H_u^{\phi} + \nu' H_u^{\psi}\right] - \eta\left(H_u^{\psi}\right)^T Q^{-1}\psi,\ \nu' = -\frac{g}{Q}\end{aligned}$$

- Gradient Algorithm

I. Guess the control history $u(t)$

II. Integrate the state equations forward in time:

$$\dot{x} = f\left(x,u,t\right),\ x(t_0) = x_0$$

Store the state history $x(t)$

III. Evaluate ϕ, ψ, $\frac{\partial\phi}{\partial x}$, $\frac{\partial\psi}{\partial x}$ at t_f

IV. Solve the following pair of costate equation backward in time

$$\begin{cases}\dot{\lambda}^{\phi}(t) = -\frac{\partial L}{\partial x} - \lambda^{\phi}\frac{\partial f}{\partial x}, & \lambda^{\phi}(t_f) = \frac{\partial\phi(x(t_f),t_f)}{\partial x} \\ \dot{\lambda}^{\psi}(t) = -\lambda^{\psi}\frac{\partial f}{\partial x}, & \lambda^{\psi}(t_f) = \frac{\partial\psi(x(t_f),t_f)}{\partial x}\end{cases}$$

Store the histories of $\lambda^{\phi}(t)$ and $\lambda^{\psi}(t)$

V. Find historeis of $H_u^{\phi}(t)$ and $H_u^{\psi}(t)$

VI. Determine $\nu' = -Q^{-1}g$

VII. Develop

$$\delta u(t) = -k\left[H_u^{\phi}(t) + \nu' H_u^{\psi}(t)\right] - \eta\left[H_u^{\psi}(t)\right]^T Q^{-1}\psi$$

If $|\delta u(t)| < \varepsilon$, then stop

If not, update the control

$$u(t) = u(t) + \delta u(t)$$

VIII. Go to Step II

- Example

  Minimize

$$J=\tfrac{1}{2}\left[x_1^2(1)+x_2^2(1)\right]+\tfrac{1}{2}\int_0^1\left(x_1^2+x_2^2+u^2\right)dt$$

subject to the dynamics and boundary conditions

$$\dot{x}_1=x_2, \qquad x_1(0)=1$$
$$\dot{x}_2=-x_1+x_1^3+u, \quad x_2(0)=1$$

Define Hamiltonian as

$$H=\tfrac{1}{2}\left(x_1^2+x_2^2+u^2\right)+\lambda_1 x_2+\lambda_2\left(-x_1+x_1^3+u\right)$$

Then the necessary conditions are derived as

$$\dot{\lambda}_1=-\frac{\partial H}{\partial x}=-x_1+\lambda_2\left(1-3x_1^2\right), \quad \lambda_1(1)=\frac{\partial h}{\partial x_1(1)}=x_1(1)$$
$$\dot{\lambda}_2=-\frac{\partial H}{\partial x}=-x_2-\lambda_1, \qquad \lambda_2(1)=\frac{\partial h}{\partial x_2(1)}=x_2(1)$$
$$0=u+\lambda_2 \;\Leftarrow\; 0=\frac{\partial H}{\partial u}$$

Apply the above Gradient Algorithm:

I. Guess the control history $u(t)$

II. Integrate the state equation forward in time from $t=0$ to $t=1$
Store the state histories of $x_1(t)$ and $x_2(t)$

III. Integrate the costate equation backward in time from $t=1$ to $t=0$,
where $\lambda_1(1)=x_1(1)$ and $\lambda_2(1)=x_2(1)$ are obtained from Step II

IV. Evaluate and store

$$\frac{\partial H}{\partial u}=u+\lambda_2$$

V. If $\left|\frac{\partial H}{\partial u}\right|<\varepsilon,\ \varepsilon>0$, then stop

If not, update the control

$$u^{new}(t)=u^{old}(t)-\alpha\tfrac{\partial H}{\partial u}(t)$$

VI. Go to Step II

5.5 Optimal Control Problem with Equality Constraints

- Minimize

$$J = f(x)$$

subject to the constaint

$$L(x) = 0$$

$$f(x + \Delta x) \simeq f(x) + \frac{\partial f}{\partial x}\Delta x + \frac{1}{2}\Delta x^T \frac{\partial^2 f}{\partial x^2}\Delta x$$

$$\Rightarrow \Delta f(x) = f(x + \Delta x) - f(x) = f_x \Delta x + \frac{1}{2}\Delta x^T f_{xx} \Delta x$$

Reformulate the problem: minimze

$$\bar{J} = \Delta f(x) = f_x \Delta x + \frac{(\Delta x)^T \Delta x}{2k} \qquad (1)$$

subject to

$$dL = L_x \Delta x = -\eta L \qquad (2)$$

where k and η are small constrant

In order to solve this linear-quadratic problem, we adjoin the constraint (2) to the performance index (1):

$$\bar{J}_a = f_x \Delta x + \frac{(\Delta x)^T \Delta x}{2k} + \alpha^T (\eta L + L_x \Delta x) \qquad (3)$$

In order that $d(\Delta f) = 0$ for arbitrary $d(\Delta x)$, it is clearly necessary that

$$f_x + \frac{\Delta x}{k} + \alpha^T L_x = 0$$

$$\Rightarrow \Delta x = -k\left(f_x + \alpha^T L_x\right) \qquad (4)$$

Substituting (4) into (2) allows us to solve for α :

$$L_x\left[-k\left(f_x+\alpha^T L_x\right)\right]=-\eta L$$

$$\Rightarrow \alpha^T=\frac{\eta L}{L_x k L_x^T}-\frac{f_x}{L_x}$$

$$=-\frac{f_x}{L_x}+\frac{\eta L}{k}\frac{1}{L_x L_x^T},\ \lambda\equiv-\frac{f_x}{L_x},\ Q=L_x L_x^T$$

$$\Rightarrow \alpha^T=\lambda+\frac{\eta}{k}LQ^{-1} \tag{5}$$

Define

$$\begin{cases} L_x^*=\dfrac{1}{L_x} \\ \lambda=-f_x L_x^* \\ Q=L_x L_x^T \\ L_x^*=L_x^T Q^{-1} \end{cases} \tag{6}$$

Substituting (5) and (6) into (4) yields

$$\Delta x^T=-k\left[f_x+\left(\lambda+\frac{\eta}{k}LQ^{-1}\right)L_x\right]$$

$$=-k\left(f_x+\lambda L_x\right)-\eta LQ^{-1}L_x$$

$$\Rightarrow \Delta x^T=-kH_x-\eta L L_x^*, H=f+\lambda^T L \tag{7}$$

When starting with a possibly poor guess, one can put $k=0$ and pick $\eta<1$ to limit the size of Δx, so that the linear approximation is reasonable

Then, gradually increase η to 1 over the next few steps, and a feasible solution ($L\sim 0$) should be obtained, after which η=1 can be used with $k\neq 0$ to minimize f

A straight forward numerical method is to guess an initial x and then take small steps Δx in the direction of -H_x^T=(the direction of steepest descent)

This will lead to a local minimum if one exists

If there are several local minima, this method may not find the global minimum

Choosing the step size requires some experience

If the steps are too large, the minimum may be overshoot, while if the step size are too small, it takes many steps to reach the minimum

5.6 Optimal Control Problem with Terminal Constraints at Free Final Time

- Numerical Solution with Gradient Method

Minimize

$$J = \phi\left(x\left(t_f\right), t_f\right) + \int_0^{t_f} L(x,u,t)dt \tag{1}$$

subject to the dynamics and boundary conditions:

$$\dot{x} = f(x,u), \ \ \mathrm{x(t_0) = x_0}, \ \ \psi(\mathrm{x}(\mathrm{t_f}), \mathrm{t_f}) = 0, \ \ \mathrm{t_f} \ \text{ is free}$$

$$\begin{aligned} J_a &= \phi + \nu^T \psi + \int_0^{t_f} \left[L + \lambda^T \left(f - \dot{x} \right) \right] dt \\ &= \phi + \nu^T \psi + \int_0^{t_f} \left[H - \lambda^T \dot{x} \right] dt, \ \ H = L + \lambda^T f \end{aligned}$$

Recall in the case of fixed final time that

$$\begin{aligned} \delta J_a &= \delta J_1 + \nu^T \delta\psi \\ &= \int_{t_0}^{t_f} H_u^{\phi} \delta u dt + \nu^T \int_{t_0}^{t_f} H_u^{\psi} \delta u dt \end{aligned}$$

Holding the same notation, we have

$$\Delta J_1 = \dot{\phi} \Delta t_f + \int_{t_0}^{t_f} H_u^{\phi} \Delta u dt + \frac{1}{2k} \left[\int_{t_0}^{t_f} \Delta u^T \Delta u dt + \Delta t_f^2 \right]$$

$$\Delta\psi = \dot{\psi} \Delta t_f + \int_{t_0}^{t_f} H_u^{\psi} \Delta u dt$$

This algorithm may be interpreted as finding a small variation in the control function $\Delta u(t)$, and a small changes in Δt_f to minimize a linear approximation of ΔJ subject to a quadratic penalty on $\Delta u(t)$ and on Δt_f and to a linear approximation to the terminal constant $\psi = 0$:

Minimize

$$\Delta J\left(\Delta u(t),\Delta t_f\right)\simeq\dot{\phi}\,\Delta t_f+\int_{t_0}^{t_f}H_u^{\phi}\Delta u dt+\frac{1}{2k}\left[\int_{t_0}^{t_f}\Delta u^T\Delta u dt+\Delta t_f^2\right] \tag{3}$$

subject to

$$\Delta\psi=\dot{\psi}\,\Delta t_f+\int_{t_0}^{t_f}H_u^{\psi}\Delta u dt=-\eta\psi \tag{4}$$

where all the coefficients are evaluated at the current estimate of u and t_f

and $0\le\eta\le 1$

k is a scalar parameter for step-size

In order to solve this LQ problem, we adjoin the constraint (4) to the performance index (3) with a Lagrange multiplier vector $\bar{\nu}$

$$\Delta\bar{J}\equiv\Delta J+\bar{\nu}^T\left(\eta\psi+\dot{\psi}\Delta t_f+\int_{t_0}^{t_f}H_u^{\psi}\Delta u dt\right) \tag{5}$$

Take a differential of $\Delta\bar{J}\equiv\Delta\bar{J}\left(\Delta u,\Delta t_f\right)$

$$\begin{aligned}d\left(\Delta\bar{J}\right)&=\frac{\partial\left(\Delta\bar{J}\right)}{\partial\left(\Delta t_f\right)}d\left(\Delta t_f\right)+\frac{\partial\left(\Delta\bar{J}\right)}{\partial\left(\Delta u\right)}d\left(\Delta u\right)\\&=\left(\dot{\phi}+\bar{\nu}^T\dot{\psi}+\frac{\Delta t_f}{k}\right)d\left(\Delta t_f\right)+\int_{t_0}^{t_f}\left[H_u^{\phi}+\bar{\nu}H_u^{\psi}+\frac{\Delta u^T}{k}\right]d\left(\Delta u\right)dt\\&=0\end{aligned} \tag{6}$$

A stationary solution requires the coefficient of $\mathrm{d}\left(\Delta \mathrm{t_f}\right)$ and $d\left(\Delta u\right)$ to vanish:,

$$\Delta\mathrm{u}=-\mathrm{k}\left(\mathrm{H_u^{\phi}}+\bar{\nu}^{\mathrm{T}}\mathrm{H_u^{\psi}}\right)^{\mathrm{T}} \tag{7}$$

$$\Delta\mathrm{t_f}=-\mathrm{k}\left(\dot{\phi}+\bar{\nu}^{\mathrm{T}}\psi\right) \tag{8}$$

Substitution of (7) and (8) into (4) allows us to solve for $\bar{\nu}$

$$-k\dot{\psi}\left(\dot{\phi}+\bar{\nu}^T\dot{\psi}\right)-\int_{t_0}^{t_f}\left[H_u^{\psi}k\left(H_u^{\phi}+\bar{\nu}^T H_u^{\psi}\right)\right]dt=-\eta\psi$$

$$\Rightarrow k\dot{\psi}\dot{\phi}+k\dot{\psi}\bar{\nu}^T\dot{\psi}+\int_{t_0}^{t_f}\left[kH_u^{\psi}H_u^{\phi}+kH_u^{\psi}\bar{\nu}^T H_u^{\psi}\right]dt=\eta\psi$$

$$\Rightarrow k\dot{\psi}\dot{\phi}+k\dot{\psi}\bar{\nu}^T\dot{\psi}+k\int_{t_0}^{t_f}H_u^{\psi}H_u^{\phi}dt+k\bar{\nu}^T\int_{t_0}^{t_f}H_u^{\psi}H_u^{\psi}dt=\eta\psi$$

$$\Rightarrow k\bar{\nu}^T\underbrace{\left(\dot{\psi}\dot{\psi}+\int_{t_0}^{t_f}H_u^{\psi}H_u^{\psi}dt\right)}_{Q}+k\underbrace{\left(\dot{\psi}\dot{\phi}+\int_{t_0}^{t_f}H_u^{\psi}H_u^{\phi}dt\right)}_{=g}=\eta\psi$$

$$\Rightarrow \bar{\nu}^T=\frac{-kg+\eta\psi}{kQ}=\frac{-g}{Q}+\frac{\eta Q^{-1}\psi}{k}$$

$$\Rightarrow \bar{\nu}^T=-\frac{g}{Q}+\frac{\eta Q^{-1}\psi}{k}=\nu+\frac{\eta Q^{-1}\psi}{k},\ \ \nu\equiv-\frac{g}{Q} \qquad (9)$$

Substitution of (9) into (7) and (8) gives the desired increment:

$$\Delta u=-k\left(H_u^{\phi}+\nu^T H_u^{\psi}\right)^T-\eta\left(H_u^{\psi}\right)^T Q^{-1}\psi$$

$$\Delta t_f=-k\left(\dot{\phi}+\nu^T\dot{\psi}\right)-\eta\dot{\psi}^T Q^{-1}\psi$$

- Summary: Numerical Gradient Algorithm

I. Choice k and η

II. Guess t_f and $u(t), t\in\left[t_0,t_f\right]$

III. Integrate the state equation forward in time

$$\dot{x}=f\left(x,u,t\right),x(t_0)=x_0$$

IV. Store the state history of $x(t)$

V. Evaluate ϕ, ψ, $\dot{\phi}$, $\dot{\psi}$ at t_f

VI. Determine

$$\lambda^{\phi}(t_f)=\phi_x^T(x(t_f),t_f)$$
$$\lambda^{\psi}(t_f)=\psi_x^T(x(t_f),t_f)$$

VII. Integrate the costate equation backward in time

$$\begin{cases} \dot{\lambda}^{\phi}(t) = -\dfrac{\partial L}{\partial x} - \lambda^{\phi}\dfrac{\partial f}{\partial x}, & \lambda^{\phi}(t_f) = \dfrac{\partial \phi}{\partial x} \\ \dot{\lambda}^{\psi}(t) = -\lambda^{\psi}\dfrac{\partial f}{\partial x}, & \lambda^{\psi}(t_f) = \dfrac{\partial \psi}{\partial x} \end{cases}$$

Store the histories of $\lambda^{\phi}(t)$ and $\lambda^{\psi}(t)$

VIII. Find historeis of $H_u^{\phi}(t)$ and $H_u^{\psi}(t)$ for $t \in [t_0, t_f]$

$$H_u^{\phi} = -\left(L_u + f_u \lambda^{\phi}\right)^T$$

$$H_u^{\psi} = -\left(f_u \lambda^{\psi}\right)^T$$

IX. Determine $\nu = -Q^{-1}g$

$$g = \dot{\psi}\dot{\phi} + \int_{t_0}^{t_f} H_u^{\psi}\left(H_u^{\varphi}\right)^T dt$$

$$Q = \dot{\psi}\dot{\psi}^T + \int_{t_0}^{t_f} H_u^{\psi}\left(H_u^{\psi}\right)^T dt$$

X. Compute $\Delta u(t), t \in [t_0, t_f]$

$$\Delta u = -k\left(H_u^{\phi}(t) + \nu^T H_u^{\psi}(t)\right)^T - \eta\left(H_u^{\psi}(t)\right)^T Q^{-1}\psi$$

XI. Compute

$$\Delta t_f = -k\left(\dot{\phi} + \nu^T \dot{\psi}\right) - \eta \dot{\psi} Q^{-1} \psi$$

XII. Compute

$$\delta u_{arg} \equiv \sqrt{\frac{1}{t_f}\int_0^{t_f} \left(\Delta u^T \Delta u\right) dt}$$

XIII. If $\delta u_{\mathrm{arg}} < \varepsilon_1$ and $\Delta t_f < \varepsilon_2$, then stop.

If not, update $u(t)$ and t_f

$$u^{new}(t) = u^{old}(t) + \Delta u(t)$$

$$t_f^{\ new} = t_f^{\ old} + \Delta t_f$$

XIV. Go to Step III

5.7 TPBVP: Shooting Method

- Shooting Method (Quasi-Linearization Method)
  Minimize

$$J = h\left(x(t_f), t_f\right) + \int_{t_0}^{t_f} L(x,u,t)dt$$

subject to the dynamics and boundary conditions:

$$\dot{x} = f(x,u,t),\ x(t_0) = x_0,\ x(t_f) = x_f$$

Derive the first order necessary conditions for optimality:

$$\dot{x} = \frac{\partial H}{\partial \lambda} = f(x,u,t),\ x(t_0) = x_0,\ x(t_f) = x_f$$

$$\dot{\lambda} = -\frac{\partial H}{\partial x} \equiv g(x,\lambda,u,t)$$

$$0 = \frac{\partial H}{\partial u} \Rightarrow u = u(x,\lambda)$$

Substituting $u = u(x,\lambda)$ into $\dot{x} = f(x,\lambda,t)$ and $\dot{\lambda} = g(x,\lambda,u,t)$ leads to

$$\dot{x} = f(x,\lambda,t) \tag{1}$$

$$\dot{\lambda} = g(x,\lambda,t) \tag{2}$$

With the definition of $X = \left[x^T \lambda^T\right]^T$,

$$\dot{X} = \begin{bmatrix} \dot{x} \\ \dot{\lambda} \end{bmatrix} = \begin{bmatrix} f \\ g \end{bmatrix} \equiv F\left(X,t\right) \tag{3}$$

The set of differential equations (3) are linearized about a nominal solution $X^k(t)$ for k^{th} trial solution, i.e., the right-hand side of (3) is expanded in Taylor series and only the first-order terms are retained
The linearized equations are given by

$$\dot{X}^k(t) + \Delta\dot{X}^k(t) = F\left(X^k,t\right) + \left.\frac{\partial F(X,t)}{\partial X}\right|_{X^k} \Delta X^k$$

$$\Rightarrow \Delta\dot{X}^k(t) = \left.\frac{\partial F(X,t)}{\partial X}\right|_{X^k} \Delta X^k + \left[F\left(X^k,t\right) - \dot{X}^k(t)\right]$$

$$\left(\Uparrow\ d(t) \equiv F\left(X^k,t\right) - \dot{X}^k(t)\right)$$

$$\Rightarrow \Delta\dot{X}^k(t) = \left.\frac{\partial F(X,t)}{\partial X}\right|_{X^k} \Delta X^k + d(t) \tag{4}$$

Define $Y \equiv \Delta X^k = \begin{bmatrix} \Delta x^k \\ \Delta \lambda^k \end{bmatrix}$ and $A(t) \equiv \left.\frac{\partial F}{\partial X}\right|_{X^k}$

Then

$$\dot{Y}(t) = A(t)Y(t) + d(t) \tag{5}$$

where *d(t)* will go to zero when $Y \equiv \Delta X^k$ is sufficiently small as the iteration progresses

Approximate

$$\dot{Y}(t) = A(t)Y(t) \tag{6}$$

which is a linear equation whose solution is given by the state transition matrix:

$$Y(t) = \Phi(t,t_0)Y(t_0) \tag{7}$$

Introducing $Y(t_0) = [\Delta x_1(t_0)\ 0\ \cdots\ 0]^T = [1\ 0\ \cdots\ 0]^T$ into (7) leads to the first column of $\Phi(t_f,t_0)$ as follows:

$$Y(t_f) = \Phi(t_f,t_0)Y(t_0) = \begin{bmatrix} \Phi_{11} & \Phi_{12} & \cdots & \Phi_{1,2n} \\ \Phi_{21} & \ddots & & \vdots \\ \vdots & & \ddots & \Phi_{2n-1,2n} \\ \Phi_{2n,1} & \cdots & \Phi_{2n,2n-1} & \Phi_{2n,2n} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} \Phi_{11} \\ \Phi_{21} \\ \vdots \\ \Phi_{2n,1} \end{bmatrix}$$

Similarly, introducing $\mathrm{Y}(\mathrm{t}_0) = [0\ \Delta \mathrm{x}_2(\mathrm{t}_0)\ 0\ \cdots\ 0]^{\mathrm{T}} = [0\ 1\ 0\ \cdots\ 0]^{\mathrm{T}}$ into (7) leads to the second column of $\Phi(t_f,t_0)$ as follows:

$$\mathrm{Y}(\mathrm{t_f}) = \Phi(\mathrm{t_f},\mathrm{t_0})\mathrm{Y}(\mathrm{t_0}) \begin{bmatrix} \Phi_{11} & \Phi_{12} & \cdots & \Phi_{1,2n} \\ \Phi_{21} & \ddots & & \vdots \\ \vdots & & \ddots & \Phi_{2n-1,2n} \\ \Phi_{2n,1} & \cdots & \Phi_{2n,2n-1} & \Phi_{2n,2n} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} \Phi_{12} \\ \Phi_{22} \\ \vdots \\ \Phi_{2n,2} \end{bmatrix}$$

Repeating the above procedure and combining all columns finally yields

$$\Phi(t_f, t_0) = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix}$$

cf) If $A(t) \equiv A$ is time-invariant, $\Phi(t_f, t_0)$ is simply given as

$$\Phi(t_f, t_0) = \exp(A(t_f - t_0))$$

Therefore,

$$\Rightarrow \begin{bmatrix} \Delta x(t_f) \\ \Delta\lambda(t_f) \end{bmatrix}_{2n\times 1} = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix}_{2n\times 2n} \begin{bmatrix} \Delta x(t_0) \\ \Delta\lambda(t_0) \end{bmatrix}_{2n\times 1}$$

$$\Rightarrow \Delta x(t_f) = \phi_{11}\Delta x(t_0) + \phi_{12}\Delta\lambda(t_0)$$

$$\Rightarrow \Delta\lambda(t_0) = \phi_{12}^{-1}\Delta x(t_f)$$

$$\Rightarrow \lambda^{k+1}(t_0) = \lambda^k(t_0) + \Delta\lambda^k(t_0)$$

We integrate the original state and costate equations (3) forward in time, with

$$\mathrm{X}(\mathrm{t}_0) = \begin{bmatrix} \mathrm{x}(\mathrm{t}_0) \\ \lambda(\mathrm{t}_0) \end{bmatrix} = \begin{bmatrix} \mathrm{x}(\mathrm{t}_0) \\ \lambda^{\mathrm{k}+1}(\mathrm{t}_0) \end{bmatrix}$$

This leads to a new solution abut which we linearize with respect to x and λ

Repeat the above procedure until $|d(t)| < \varepsilon$ satisfies

- Shooting Algorithm

  I. Guess the initial costate $\lambda^k(t_0)$

  II. Use the optimality condition $\partial H / \partial u = 0$ to develop the relation
  $u = u(x, \lambda)$
  Introduce the state/costate equations (1)~(2) to eliminate u

  III. Integrate and solve the initial value problem

$$\dot{x}^k = f(x^k, \lambda^k, t), \ \ x^k(t_0) = x(t_0)$$
$$\dot{\lambda}^k = g(x^k, \lambda^k, t), \ \ \lambda^k(t_0) = \text{guess}$$

IV. Compare the given terminal condition $x(t_f)$ with the calculated terminal condition $x^k(t_f)$, i.e, evaluate $\Delta x(t_f) \equiv x^k(t_f) - x(t_f)$

If $\left|\Delta x(t_f)\right| < \varepsilon$, stop

Otherwise,

V. Linearize about $x^k(t)$, $\lambda^k(t)$

$$\Delta\dot{x}^k = f'\left(x^k, \lambda^k, \Delta x^k, \Delta\lambda^k, t\right)$$

$$\Delta\dot{\lambda}^k = g'\left(x^k, \lambda^k, \Delta x^k, \Delta\lambda^k, t\right)$$

VI. Obtain the state transition matrix by perturbing $\begin{bmatrix} \Delta x^k(t_0) \\ \Delta\lambda^k(t_0) \end{bmatrix}$

VII. Develop

$$\Delta\lambda^k(t_0) = \phi_{12}^{-1}\left[\Delta x(t_f) - \phi_{11}\Delta x(t_0)\right]$$
$$\Rightarrow \lambda^{k+1}(t_0) = \lambda^k(t_0) + \Delta\lambda^k(t_0)$$

VIII. Go to Step III

5.7. TPBVP with Constant Parameters: Shooting Method

- Shooting Method

Minimize

$$J = h\left(x(t_f), t_f\right) + \int_{t_0}^{t_f} L(x, u, p, t)dt$$

subject to the dynamics and boundary conditions

$$\dot{x} = f(x, u, p, t), \ \ x(t_0) = x_0, \ \ x(t_f) = x_f$$

where p is a constant parameter vector to be determined

Take the first variation:

$$\delta J = \cdots + \int_{t_0}^{t_f} \frac{\partial H}{\partial p} \delta p \, dt = \cdots \delta p \int_{t_0}^{t_f} \frac{\partial H}{\partial p} dt = 0$$

$$\Rightarrow \int_{t_0}^{t_f} \frac{\partial H}{\partial p} dt = 0$$

Define

$$z(t) \equiv \int_{t}^{t_f} \frac{\partial H}{\partial p} dt$$

$$\Rightarrow \dot{z}(t) = -\frac{\partial H}{\partial p}, \; z(t_0) = 0, \; z(t_f) = 0$$

Use the optimality condition $\partial H / \partial u = 0$ to develop $u = u(x, \lambda)$, and introduce the state/costate equation:

$$\begin{aligned}
&\dot{x} = f(x, u(x,\lambda), p, t), && x(t_0) = x_0, \; x(t_f) = x_f, \; x \in R^n \\
&\dot{z} = -\frac{\partial H}{\partial p}(x, u(x,\lambda), p, t), \; z(t_0) = 0, \; z(t_f) = 0 && z \in R^r \\
&\dot{\lambda} = -\frac{\partial H}{\partial x}(x, u(x,\lambda), p, t) && \lambda \in R^n
\end{aligned} \tag{1}$$

Given the above settings, the goal is to find $\lambda(t_0)$ and p

With the definition of $X = \left[x^T \; z^T \; \lambda^T \right]$

$$\dot{X} = \begin{bmatrix} \dot{x} \\ \dot{z} \\ \dot{\lambda} \end{bmatrix} \equiv F\left(X, p, t\right)$$

- Algorithm
  I. Guess the initial value of $\lambda(t_0)$ and p
  II. Integrate the state/costate equation (1) forward in time with

$$X = \begin{bmatrix} x(t_0) \\ z(t_0) \\ \lambda(t_0) \end{bmatrix} \text{ and } p$$

  III. Linearize the state/costate equation about the nominal solution obtained in Step II:

$$\Delta \dot{X} = \frac{\partial F}{\partial X}\Delta X + \frac{\partial F}{\partial p}\Delta p$$

$$\Rightarrow \Delta X(t_f) = \Phi(t_f, t_0)\Delta X(t_0) + \psi \Delta p$$

where $\Phi \in R^{(2n+r)\times(2n+r)}$ and $\psi \in R^{(2n+r)\times r}$ must be determined by the following formulae:

$$\Phi_{ij} = \frac{\partial \Delta X_i(t_f)}{\partial \Delta X_j(t_0)},\ \psi_{ij} = \frac{\partial \Delta X_i(t_f)}{\partial \Delta p_j}$$

- Method A: Finite Difference Method (FDM)

  Change the first state variable slightly from its nominal value, while hold all the other quantities at their nominal values

  $$x_1(t_0) = x_1^N(t_0) + \varepsilon$$
  $$x_i(t_0) = x_i^N(t_0),\ i \neq 1$$

  Integrate the $\dot{\text{x}}, \dot{\lambda}, \dot{\text{z}}$ equation forward in time to obtain $X(t_f)$

  Develop

  $$\Phi_{ij} = \frac{X(t_f) - X^N(t_f)}{x_1(t_0) - x_1^N(t_0)} = \frac{X(t_f) - X^N(t_f)}{\varepsilon}$$

  Repeat a similar procedure with p to obtain ψ_{ij}

- Method B:

  Define the initial condition:

  $$\Delta X\left(t_0\right) = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix},\ \Delta p = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

  Integrating the associated $\Delta \dot{X}$ equation forward in time leads to

  $$\Delta X(t_f) = \Phi_{i1}$$

  Repeat a similar procedure to obtain Φ_{ij} matrix

  Then, repeat the same procedure by varying Δp to obtain ψ_{ij} matrix

Method B gives more accurate results than FDM, but requires more work (analytical partial derivative).

IV. After finding Φ and ψ, we are ready to solve for $\Delta\lambda(t_0)$ and Δp

$$\begin{bmatrix} \Delta X_1(t_f) \\ \Delta\lambda(t_f) \end{bmatrix} = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{bmatrix} \Delta X_1(t_0) \\ \Delta\lambda(t_0) \end{bmatrix} + \begin{bmatrix} \psi_1 \\ \psi_2 \end{bmatrix} \Delta p, \ \Delta X_1 = \begin{bmatrix} \Delta x \\ \Delta z \end{bmatrix}$$

$$\Rightarrow \Delta X_1(t_f) = \phi_{11}\Delta X_1(t_0) + \phi_{12}\Delta\lambda(t_0) + \psi_1\Delta p$$

$$\Rightarrow \Delta X_1(t_f) = \begin{bmatrix} \phi_{12} & \psi_1 \end{bmatrix} \begin{bmatrix} \Delta\lambda(t_0) \\ \Delta p \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \Delta\lambda(t_0) \\ \Delta p \end{bmatrix} = \begin{bmatrix} \phi_{12} & \psi_1 \end{bmatrix}^{-1} \Delta X_1(t_f)$$

V. Update $\lambda(\mathrm{t}_0)$ and p:

$$\lambda^{k+1}(t_0) = \lambda^k(t_0) + \Delta\lambda^k(t_0)$$
$$p^{k+1} = p^k + \Delta p^k$$

VI. Go to Step II

- Example

Minimize

$$J = \int_0^{t_f} \left(1 + u^2\right) dt$$

subject to

$$\dot{x} = -x + x^3 + u, \ x(0) = 1, \ x(t_f) = 0, \ t_f \text{ is free}$$

We first convert into a fixed final time problem by introducing a free parameter p:

$$t \equiv p^2\tau \ \ (\text{why instead of } t \equiv p\tau\,?)$$
$$\Rightarrow dt = p^2 d\tau$$

where τ is a non-dimensionalized time index $\tau \in [0,1]$

Let $A' \equiv dA/d\tau$ (while $\dot{A} \equiv dA/dt$)

Then, the performance index, dynamic equations are converted into

$$J=\int_0^1\left(1+u^2\right)p^2d\tau \qquad \Leftarrow\ J=\int_0^{t_f}\left(1+u^2\right)dt$$

$$x'=\left(-x+x^3+u\right)p^2,\ x(0)=1,\ x(1)=0 \qquad \Leftarrow\ \dot{x}=-x+x^3+u$$

$$p'=0 \qquad \Leftarrow\ p \text{ is a free constant parameter}$$

Define Hamiltonian as

$$H=\left(1+u^2\right)p^2+\lambda\left(-x+x^3+u\right)p^2+z\cdot 0$$

Then, the first order necessary conditions are

$$x'=(-x+x^3+u)p^2,\ x(0)=1,\ x(1)=0$$

$$p'=0$$

$$\lambda'=\lambda_1 p^2(1-3x^2) \qquad \Leftarrow \lambda'=-\frac{\partial H}{\partial x}$$

$$z'=-2p(1+u^2)-2p\lambda(-x+x^3+u),\ z(0)=0,\ z(1)=0 \quad \Leftarrow z'=-\frac{\partial H}{\partial p}$$

$$0=p^2(2u+\lambda),\ p\neq 0\ \Rightarrow\ u=-\frac{1}{2}\lambda \qquad \Leftarrow 0=\frac{\partial H}{\partial u}$$

Introducing the optimality condition into the state/costate equations leads to

$$x'=\left(-x+x^3-\frac{1}{2}\lambda\right)p^2,\ x\left(0\right)=1,\quad x\left(1\right)=0 \tag{1-1}$$

$$z'=-2p\left(1+\frac{1}{4}\lambda^2\right)-2p\lambda\left(-x+x^3-\frac{1}{2}\lambda\right),\ z\left(0\right)=0,\quad z\left(1\right)=0 \tag{1-2}$$

$$\lambda'=\lambda p^2\left(1-3x^2\right) \tag{1-3}$$

Recall that we look for $\lambda(0)$ and p:

I. Guess the values of $\lambda(0)$ and p

II. Integrate the state/costate equations (1-1)~(1-3)

$$X'=F\left(X,p\right),\ X\equiv\left[x\ \ z\ \ \lambda\right]^T$$

III. Linearize about the nominal solution generated from the forward integration of the nonlinear equations:

$$\Delta X'=\frac{\partial F}{\partial X}\Delta X+\frac{\partial F}{\partial p}\Delta p$$

$$
\Rightarrow \begin{bmatrix} \Delta x' \\ \Delta z' \\ \Delta \lambda' \end{bmatrix} = \begin{bmatrix} p^2\left(-1+3x^2\right) & 0 & -\frac{1}{2}p^2 \\ 2p\lambda\left(1-3x^2\right) & 0 & -2px+2px^3+p\lambda \\ -6\lambda p^2 x & 0 & p^2\left(1-3x^2\right) \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta z \\ \Delta \lambda \end{bmatrix} + \begin{bmatrix} 2p\left(-x+x^3-\frac{1}{2}\lambda\right) \\ -2+\frac{1}{2}\lambda^2+2\lambda x-2\lambda x^3 \\ 2p\lambda\left(1-3x^2\right) \end{bmatrix} \Delta p \tag{2}
$$

$$
\Rightarrow \Delta X(1) = \Phi \Delta X(0) + \psi \Delta p \tag{3}
$$

Let $\Delta X_i(0) = \varepsilon (=1)$ while all the other $\Delta X_j(0) = 0, i \neq j$ and $\Delta p = 0$

Integrate (2) and introduce into (3) yields Φ

$$
\Delta X(1) = \Phi(*, i)\,\varepsilon
$$

$$
\Rightarrow \Phi = \left[\Phi(\cdot,1), \Phi(\cdot,2), \Phi(\cdot,3)\right]
$$

Similarly procedure with Δp yiedls ψ

$$
\Delta X(1) = \psi \Delta p
$$

$$
\Rightarrow \psi = \Delta X(1) / \Delta p
$$

- After finding Φ and ψ, we are ready to solve for $\Delta\lambda(t_0)$ and Δp

$$
\begin{bmatrix} \Delta x(1) \\ \Delta z(1) \\ \Delta\lambda(1) \end{bmatrix} = \begin{bmatrix} \phi_{11} & \phi_{12} & \phi_{13} \\ \phi_{21} & \phi_{22} & \phi_{23} \\ \phi_{31} & \phi_{32} & \phi_{33} \end{bmatrix} \begin{bmatrix} \Delta x(0) \\ \Delta z(0) \\ \Delta\lambda(0) \end{bmatrix} + \begin{bmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \end{bmatrix} \Delta p
$$

$$
\Rightarrow \begin{bmatrix} \Delta x(1) \\ \Delta z(1) \end{bmatrix} = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \cancel{\begin{bmatrix} \Delta x(0) \\ \Delta z(0) \end{bmatrix}} + \begin{bmatrix} \phi_{13} \\ \phi_{23} \end{bmatrix} \Delta\lambda(0) + \begin{bmatrix} \psi_1 \\ \psi_2 \end{bmatrix} \Delta p
$$

$$
\Rightarrow \begin{bmatrix} \Delta\lambda(0) \\ \Delta p \end{bmatrix} = \begin{bmatrix} \phi_{13} & \psi_1 \\ \phi_{23} & \psi_2 \end{bmatrix}^{-1} \begin{bmatrix} \Delta x(1) \\ \Delta z(1) \end{bmatrix}
$$

- Update $\lambda(0)$ and p by

$$
\lambda^{k+1}(0) = \lambda^k(0) + \Delta\lambda^k(0)
$$

$$
p^{k+1} = p^k + \Delta p
$$

- Repeat the above procedure until the following tolerance criterion is satisfied:

$$\left|\Delta x(1)\right| = \left|x(1) - x^{k+1}(1)\right| < \varepsilon$$
$$\left|\Delta z(1)\right| = \left|z(1) - z^{k+1}(1)\right| < \varepsilon$$

- The answer is $p = 0.9822 \;\Rightarrow\; t_f = p^2 = 0.9648$

5.8. Optimal Control Problem by Control Parameterization

- Minimize

  $$J = \int_0^{t_f} g(x,u,t)dt$$

  subject to the dynamic equation

  $$\dot{x} = f(x,u,t),\;\; x(t_0) = x_0,\;\; x(t_f) = x_f$$

  constraints on states and controls

  $$G(x(t)) \le 0$$
  $$|u(t)| \le 1$$

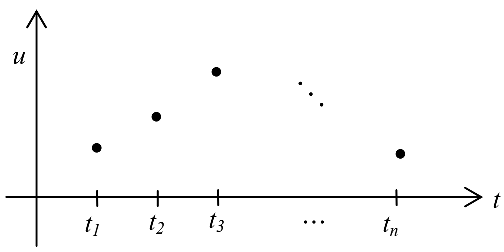
  The control vector $u(t)$ can be parameterized by

  I. Polynomials in time

  $$u = \sum_{i=0}^{N} p_i t^i$$

  II. Splines in time

  III. Discrete control vectors



  Then, the paramerized expressions are

  $J(x,u,t) \Rightarrow J(u_i),\;\; u_i$ are parmeters

  $$x(t_f) \Rightarrow x(t_f, u_i)$$

$|u(t)| \le 1 \Rightarrow |u_i| \le 1$

$G(x) \le 0 \Rightarrow G(u_i) \le 0$

- Polynomial Approximation Method
  Polynomials of sufficiently high degree can approximate a continuous function as high accuracy as arbitrarily possible
  Each degree of polynomial serves as a basis function

The vector $x(t)$ in the TPBVP can be approximated as

$$x(t) = \Phi(t)A \tag{1}$$

where

$$\Phi(t) = \left[\phi_1(t), \phi_2(t), \cdots \phi_{NB}(t)\right]^T$$

$$A = \left[A_1^T, A_2^T, \cdots A_{NB}^T\right]^T$$

$$A_i = \left[a_{i1}, a_{i2}, \cdots a_{iNB}\right]^T$$

- NB is the number of basis function used
- Each of the basis function $\phi_i(t)$ is the $(i-1)^{th}$ degree polynomials in t, and is typically orthogonal polynomials
- The derivatives of $\phi_i(t)$ can be usually obtained through simple recurrence relations
- Each of a_{ij} represents a coefficient such that the i^{th} variable $x_i(t)$ is represented as

$$x_i(t) = \sum_{j=1}^{NB} a_{ij}\phi_j(t), \ \ i = 1, 2, \cdots, N$$

Differentiate (1)

$$\dot{x}(t) = \dot{\Phi}(t)A \tag{2}$$

Consider the linear systems of equations

$$\dot{x}(t) = F(t)x(t) + D(t) \tag{3}$$

with boundary conditions

$$\mathbf{u}x(0) = \alpha \tag{4}$$

$$\mathbf{v}x(t_f) = \beta \tag{5}$$

where

$\mathbf{u}$ and $\mathbf{v}$ are known $N \times N$ matrices

α and β are known $N \times 1$ vectors

Substituting (1) and (2) into (3) leads to

$$\dot{\Phi}A = F\Phi A + D \tag{6}$$

$$\Rightarrow \left[\dot{\Phi} - F\Phi\right]A = D$$

$$\left(\Uparrow\ P_1 \equiv \dot{\Phi} - F\Phi\right)$$

$$\Rightarrow P_1\left(t\right)A = D \tag{7}$$

When P_1 is an $\left[N \times \left(N \times NB\right)\right]$ known time varying matrix, (7) can be evaluated at the number of NS sample points at times t_i on $\left[0, T\right]$ and thereby establish enough equations to determine the coefficient vector A

We must also ensure satisfaction of boundary conditions equation (1), for which (4) and (5) provides two boundary condtions:

$$P_2 A = \alpha \tag{8}$$

$$P_3 A = \beta \tag{9}$$

where

$$P_2 = \mathbf{u}\Phi(0)$$

$$P_3 = \mathbf{v}\Phi(t_f)$$

(7)~(9) merge into a linear algebraic system

$$PA = Q \tag{10}$$

where

$$P = \begin{bmatrix} P_1(t_1) \\ P_1(t_2) \\ P_1(t_3) \\ \vdots \\ P_1(t_{NS}) \\ P_2 \\ P_3 \end{bmatrix} \in R^{\left[N\times\left(NS+1\right)\right]\times\left[N\times NB\right]},\ Q = \begin{bmatrix} D(t_1) \\ D(t_2) \\ D(t_3) \\ \vdots \\ D(t_{NS}) \\ \alpha \\ \beta \end{bmatrix} \in R^{\left[N\times\left(NS+1\right)\right]\times 1}$$

- Collocation Method
  - NS is selected such that P is a square matrix, i.e., $NS = NB - 1$
  - Then, the coefficient a_{ij} in A are uniquely determined by solving the linear system (10)
  - Thus, the solution $x(t)$ can be obtained from equation (1)
  - This solution satisfies the differential equation only at the sample points, however the boundary conditions are satisfied exactly
  - As NB increases sufficiently, the average error at intermediate points typically decreases to zero

5.9Nonlinear Programming Problem (NLP)

- Optimization Problem typically involves constraints
  - Linear Programming Problem (LP): both the performance index and constraints are linear functions of the variables in the optimization problem
  - Nonlinear Programming Problem (NLP): nonlinear terms are present in the performance index and/or constraints in the optimization problem

- Typical NLP:

  Minimize the nonlinear performance index

$$p = f(x) \tag{1}$$

  where $x = [x_1, x_2, \cdots, x_n]^T$ are constrained by the following sets of equations:

$$g_i(x) = C_i, \; i = 1, 2, \cdots, k < n \tag{2}$$

$$g_i(x) \le C_i, \; i = k+1, k+2, \cdots, m \tag{3}$$

  It is assumed that $g_i(x) \in C^1$

- Unconstrained Minimization: Minimize $f(x)$
  - First Order Gradient method

$$x^{new} = x^{old} - \alpha \frac{\partial f}{\partial x}$$

  - Second Order Gradient Method (Gauss-Newton Method)

$$f\left(x^k\right) \simeq f\left(x^*\right) + \underbrace{G^k}_{Jacobian} \Delta x^k + \tfrac{1}{2}\Delta x^k \underbrace{H^k}_{Hessian} \Delta x^k, \quad x^k : \text{current guess}$$

Minimize $\Delta f \simeq f(x^k) - f(x^*)$ with respect to Δx^k

$$\frac{\partial \Delta f}{\partial \Delta x} = 0$$

$$\Rightarrow G^k + H^k \Delta x^k = 0$$

$$\Rightarrow \Delta x^k = -\left[H^k\right]^{-1} G^k$$

$$\Rightarrow x^{k+1} = x^k - \alpha\left[H^k\right]^{-1} G^k$$

It is required that

$$H^k > 0$$

since

$$\Delta f(x) \simeq \frac{\partial f}{\partial x}\Delta x^k = -\alpha G^k \left[H^k\right]^{-1} G^k < 0$$

It is expensive to evaluate $\left[H^k\right]^{-1}$

- Quasi-Newton Method

Approximate the inverse of the Hessian

The search direction is still along the negative gradient

$$\Delta x^k \equiv -A^k G^k$$

where $A^k \simeq \left[H^k\right]^{-1}$ approximates the inverse of the Hessian

$$A^{k+1} = A^k + \Delta A^k, \quad A^0 = I$$

A general update procedure to the above equation is

$$\Delta A^k = \frac{\Delta x^k y^T}{y^T \Delta G^k} - \frac{A^k \Delta G^k z^T}{z^T \Delta G^k} \quad \text{for arbitrary } y \text{ and } z$$

DFP (David, Flecher, Powell) Method:

$$y = \Delta x^k \quad \text{and} \quad z = A^k \Delta G^k$$

Then,

$$A^{k+1} = A^k + \Delta A^k$$

$$\Delta A^k = \frac{\Delta x^k \left(\Delta x^k\right)^T}{\left(\Delta x^k\right)^T \Delta G^k} - \frac{A^k \Delta G^k \left(\Delta G^k\right)^T \left(A^k\right)^T}{\left(\Delta G^k\right)^T \left(A^k\right)^T \Delta G^k}$$

A kep property is that if $A^k > 0$ and α satisfies the so-called Wolfe conditions, then $A^{k+1} > 0$

BFGS Method

Broyden Method

SR1 Method

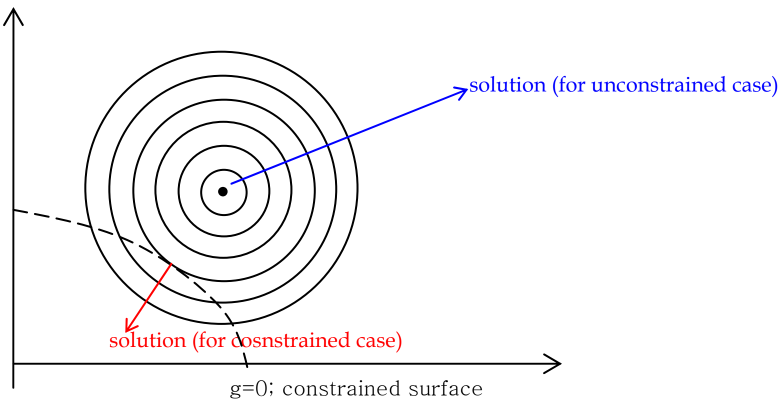
- Constrained (Equality) Minimization
  - Gradient Projection

    Minimize

    $$f(x),\ \ x \in R^N$$

    subject to

    $$g(x) = 0,\ \ g \in R^M,\ \ M < N$$



Algorithm

I. Define Lagrangian as

$$L = f + \lambda^T g$$

II. Start with a feasible solution that satisfies the constraints:

$$\frac{\partial L}{\partial x} = \underbrace{\frac{\partial f}{\partial x} + \left(\frac{\partial g}{\partial x}\right)^T \lambda}_{\text{Projected Gradient}} \simeq 0,$$

$$\left(\Uparrow \; A \equiv \left(\frac{\partial g}{\partial x}\right)^T_{N\times M\ matrix} \right)$$

$$\Rightarrow A\lambda = -\frac{\partial f}{\partial x}$$

$$\Rightarrow \lambda = -\left(A^T A\right)^{-1} A^T \frac{\partial f}{\partial x}$$

$$\Rightarrow \frac{\partial L}{\partial x} = \left[I - A\left(A^T A\right)^{-1} A^T \right] \frac{\partial f}{\partial x}$$

$$\Rightarrow \Delta x^k = -\alpha \frac{\partial L^k}{\partial x} = \left[I - A\left(A^T A\right)^{-1} A^T \right] \frac{\partial f}{\partial \vec{x}}$$

III. Work on Constraints:

$$g(x^{k+1}) \simeq 0$$

$$\Rightarrow g(x^k + \Delta x^k) \simeq g(x^k) + A\Delta x^k \simeq 0$$

$$\Rightarrow A\Delta x^k \simeq -g(x^k)$$

$$\Rightarrow \Delta x^k = -\alpha (A^T A)^{-1} A^T g(x^k)$$

IV. Repeat Step II and III until the convergence criteria meet

$$\left|\frac{\partial L}{\partial x}\right| < \varepsilon_1, \; \left|g\right| < \varepsilon_2$$

■ Multiplier Penalty Problem (or Augmented Lagrangian Problem)

◆ Simple Example for Motivation:

Minimize

$$f(x) = x$$

subject to

$$g(x) = x - 1 = 0$$

Define Lagrangian as

$$L = x + \frac{1}{2}\sigma(x-1)^2$$

where $\sigma(x-1)^2/2$ is a penalty function with a known weight

$(\sigma > 0)$

$$\frac{\partial L}{\partial x} = 0 \Rightarrow 1 + \sigma\left(x - 1\right) = 0 \Rightarrow x = 1 - \frac{1}{\sigma}$$

It turns out that

$$\lim_{\sigma \to \infty} x \to x^* = 1$$

◆ Minimize

$$f(x)$$

subject to

$$g(x) = 0, \quad g(x) \in R^M$$

Construct the Augmented Lagrangian with σ_i and θ_i as parameters

$$\phi_\sigma(x, \theta) = f(x) + \frac{1}{2}\sum_{i=1}^{m} \sigma_i \left[g_i(x) - \theta_i\right]^2$$

$$\Rightarrow \phi_\sigma = f(x) + \frac{1}{2}\sum_{i=1}^{m} \sigma_i g_i^{\,2}(x) - \sum_{i=1}^{m} \sigma_i g_i(x)\theta_i + \tfrac{1}{2}\sum_{i=1}^{m} \sigma_i \theta_i^2$$

Since we are optimizing with respect to x, neglect

$$\sum_{i=1}^{m} \sigma_i \theta_i^2 / 2$$

in the above equation and define

$$\lambda_i = \sigma_i \theta_i$$

θ_i's are parameters that artificially shift the origin

The idea is to increase σ_i (finite) and decrease θ_i so that λ_i is finite

◆ Theorem

If $\vec{x}^*$ and $\vec{\lambda}^*$ satisfy the second order sufficient conditions

$$\frac{\partial \phi}{\partial x} = 0, \frac{\partial \phi}{\partial \lambda} = 0; Y \frac{\partial^2 \phi}{\partial x^2} Y > 0$$

such that

$$\frac{\partial g}{\partial x} y = 0$$

for any $y \neq 0$, then there exists $\bar{\sigma} > 0$, such that $\vec{x}^*$ is strict relative minimum of $\phi_\sigma(x, \lambda^*)$ for any $\sigma > \bar{\sigma}$

- The new problem is to minimize ϕ_σ with respect to x:
  Minimize

$$\phi_\sigma(x,\lambda) = f(x) - \lambda^T g(x) + \frac{1}{2}\sum_{i=1}^{m}\sigma_i g_i^2(x), \quad \sigma_i = \text{constant}$$

$$x^{k+1} = x^k - \frac{\partial \phi_\sigma}{\partial x^k}$$

$$\lambda^{k+1} = \lambda^k - \frac{\partial \phi_\sigma}{\partial \lambda^k} = \lambda + g(x^k)$$

- Quadratic Programming

Minimize

$$f(x) = \frac{1}{2}x^T Q x + c^T x, \quad Q > 0 \qquad \text{(quadratic cost)}$$

subject to

$$g(x) = Ax - b = 0 \qquad \text{(linear equality constraints)}$$

Define Lagrangian as

$$L(x,\lambda) = f(x) + \lambda^T(Ax - b)$$

The first order necessary conditions are

$$\begin{cases} \dfrac{\partial L}{\partial x} = 0 = Qx + c + \lambda^T A \\ \dfrac{\partial L}{\partial \lambda} = 0 = Ax - b \end{cases}$$

$$\Rightarrow \begin{bmatrix} Q & A^T \\ A & 0 \end{bmatrix}\begin{bmatrix} x \\ \lambda \end{bmatrix} = \begin{bmatrix} -c \\ b \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ \lambda \end{bmatrix} = \begin{bmatrix} Q & A^T \\ A & 0 \end{bmatrix}^{-1}\begin{bmatrix} -c \\ b \end{bmatrix}$$

$$\left(\Uparrow \begin{bmatrix} Q & A^T \\ A & 0 \end{bmatrix}^{-1} = \begin{bmatrix} H & P^T \\ P & S \end{bmatrix} \Leftarrow \begin{cases} H = Q^{-1} - Q^{-1}A^T\left(AQ^{-1}A^T\right)^{-1}AQ^{-1} \\ P = \left(AQ^{-1}A^T\right)^{-1}AQ^{-1} \\ S = \left(AQ^{-1}A^T\right)^{-1} \end{cases} \right)$$

$$\Rightarrow \begin{bmatrix} x \\ \lambda \end{bmatrix} = \begin{bmatrix} H & P^T \\ P & S \end{bmatrix} \begin{bmatrix} -c \\ b \end{bmatrix}$$

- Sequential Quadratic Programming (SQP)

  Minimize

  $$f(x)$$

  subject to

  $$g_i(x) = 0, \quad i = 1, 2, \cdots, m$$

  First approximate

  $$f(x^k + \Delta x^k) \simeq f(x^k) + \nabla f(x^k)\Delta x^k + \frac{1}{2}\Delta x^k H^k \Delta x^k$$

  $$g_i(x^k + \Delta x^k) \simeq g_i(x^k) + \nabla g_i(x^k)\Delta x^k$$

  Auxilliary Problem (to be solved by Quadratic Programming):

  Minimize

  $$\nabla f\left(x^k\right)\Delta x^k + \frac{1}{2}\Delta x^k H^k \Delta x^k$$

  subject to

  $$g_i(x^k) + \nabla g_i(x^k)\Delta x^k = 0, \quad i = 1, 2, \cdots, m$$

  Solve for Δx^k and λ^k using Gradient Projection or Quadratic Programming

- Constrained (Inequality) Minimization
  - Augumented Lagrangian

    Minimize

    $$f(x)$$

    subject to

$g(x) \geq 0, \ \ g(x) \in R^M$

Construct the Augmented Lagrangian with σ_i and θ_i as parameters

$$\phi_\sigma(x,\theta) = f(x) + \frac{1}{2}\sum_{i=1}^{m}\sigma_i\left[g_i(x) - \theta_i\right]^2$$

$$\Rightarrow \phi_\sigma = f(x) + \tfrac{1}{2}\sum_{i=1}^{m}\sigma_i g_i^2(x) - \sum_{i=1}^{m}\sigma_i g_i(x)\theta_i + \frac{1}{2}\sum_{i=1}^{m}\sigma_i\theta_i^2$$

Define $\lambda_i = \sigma_i\theta_i > 0$

The new problem is to minimize ϕ_σ with respect to x

$$\phi_\sigma(x,\lambda) = f(x) - \lambda^T g(x) + \frac{1}{2}\sum_{i=1}^{m}\sigma_i g_i^2(x), \ \ \sigma_i = \text{constant}$$

$$x^{k+1} = x^k - \alpha\frac{\partial\phi_\sigma}{\partial x^k}$$

$$\lambda_i^{k+1} = \lambda_i^k - \min\left[\sigma_i g_i(x^k), \lambda_i^k\right]$$

If $g_i > 0$, then $\lambda_i = 0$, i.e., the i-th constraint is inactive

- Kuhn-Tucker Method

Minimize

$f(x)$

subject to

$g(x) \geq 0, \ \ g(x) \in R^M$

Define Lagrangian as

$L = f(x) + \Lambda^T g$

Use $\partial L / \partial x = 0$ to solve for x

$$\begin{cases} \lambda_i > 0, & if \ \ g_i = 0 \\ \lambda_i = 0, & if \ \ g_i > 0 \end{cases}$$

If constraints are violated, the cost increases $(\lambda_i > 0)$

Check the sign of λ_i to determine if the answer is optimal